

Introduction to Structure and Models of Bonding

Intent and Purpose

There are three goals for Chapter 1. The first is to review simple notions of chemical bonding and structure. This review is meant for readers who have a knowledge of atomic and molecular structure equivalent to that given in introductory chemistry and organic chemistry textbooks. In this review, concepts such as quantum numbers, electron configurations, valence-shell electron-pair repulsion (VSEPR) theory, hybridization, electronegativity, polar covalent bonding, and σ and π bonds, are covered in an introductory manner. A large fraction of organic chemistry can be understood and predicted based upon these very simple concepts in structure and bonding. However, the second goal of the chapter is to present a more advanced view of bonding. This is known as qualitative molecular orbital theory (QMOT), and it will lay the foundation for Chapter 14, where computational methods are discussed. This more advanced approach to bonding includes the notion of group orbitals for recurring functional groups, and an extension of molecular orbital theory called perturbational molecular orbital theory that will allow us to make rational predictions as to how bonding schemes arise from orbital mixing. We show these bonding models first with stable molecules, and then apply the lessons to reactive intermediates. By covering stable structures alongside reactive intermediates, it should be clear that our standard models of bonding predict the reactivity and structure of all types of organic structures, stable and otherwise. Showing such a correlation is the third goal of the chapter.

A recurrent theme of this chapter is that organic functional groups—olefins, carbonyls, amides, and even simple alkyl groups such as methylene and methyl—can be viewed as having transferable orbitals, nearly equivalent from one organic structure to another. We will describe several of these molecular orbitals for many common organic functional groups. In all the discussions there is a single unifying theme, that of developing models of bonding that can be used to explain reactivity, structure, and stability, as a preparation for future chapters.

You may be aware that modern computational methods can be used to describe the bonding in organic molecules. Why, then, should we develop simple descriptive theories of bonding? With the advent of universally available, very powerful computers, why not just use quantum mechanics and computers to describe the bonding of any molecule of interest? In the early twenty-first century, it is true that any desktop computer can perform sophisticated calculations on molecules of interest to organic chemists. We will discuss the methodology of these calculations in detail in Chapter 14, and we will often refer to their results during our discussions in this and other chapters. However, for all their power, such calculations do not necessarily produce *insight* into the nature of molecules. A string of computer-generated numbers is just no substitute for a well-developed *feeling* for the nature of bonding in organic molecules. Furthermore, in a typical working scenario at the bench or in a scientific discussion, we must be able to rapidly assess the probability of a reaction occurring without constantly referring to the results of a quantum mechanical calculation. Moreover, practically speaking, we do not need high level calculations and full molecular orbital theory to understand most common reactions, molecular conformations and structures, or kinetics and thermodynamics. Hence, we defer detailed discussions of sophisticated calculations and full molecular orbital theory until just before the chapters where these methods are

essential. Also, as powerful as they are, calculations are still severely limited in their ability to address large systems such as proteins, nucleic acids, or conducting polymers. This limitation is even more severe when solvation or solid state issues become critical. Therefore, it is still true—and will be true for some time—that descriptive models of bonding that are readily applicable to a wide range of situations are the best way to attack complex problems. The models must be firmly rooted in rigorous theory, and must stand up to quantitative computational tests. Two such models are developed in this chapter.

1.1 A Review of Basic Bonding Concepts

In this section we present a number of basic concepts associated with chemical bonding and organic structure. Most of this material should be quite familiar to you. We use this section to collect the terminology all in one place, and to be sure you recall the essentials we will need for the more advanced model of bonding given in Sections 1.2 and 1.3. For most students, a quick read of this first section will provide an adequate refresher.

1.1.1 Quantum Numbers and Atomic Orbitals

Every molecule is made up from the nuclei and electrons of two or more atoms via bonds that result from the overlap of atomic orbitals. Hence, the shapes and properties of atomic orbitals are of paramount importance in dictating the bonding in and properties of molecules. The **Bohr model** of atoms had electrons moving in specific orbits (hence the term **orbitals**) around the nucleus. We now view the shapes and properties of atomic orbitals as they are obtained from basic quantum mechanics via solution of the Schrödinger equation. The solutions to the Schrödinger equation are termed **wavefunctions**, and in their most common implementation these wavefunctions correspond to atomic or molecular orbitals.

The atomic orbital wavefunctions come in sets that are associated with four different quantum numbers. The first is the **principal quantum number**, which takes on positive integer values starting with 1 ($n = 1, 2, 3, \dots$). An atom's highest principal quantum number determines the **valence shell** of the atom, and it is typically only the electrons and orbitals of the valence shell that are involved in bonding. Each row in the periodic table indicates a different principal quantum number (with the exception of *d* and *f* orbitals, which are displaced down one row from their respective principal shells). In addition, each row is further split into **azimuthal quantum numbers** ($m = 0, 1, 2, 3, \dots$; alternatively described as *s*, *p*, *d*, *f*, $\dots$). This number indicates the angular momentum of the orbital, and it defines the spatial distribution of the orbital with respect to the nucleus. These orbitals are shown in Figure 1.1 for $n = 2$ (as with carbon) as a function of one of the three Cartesian coordinates.

The shapes given in Figure 1.1 are a schematic representation of the orbitals in regions of space around the nucleus. For $n = 1$, only a 1*s* atomic orbital is allowed. The highest electron density is at the atomic nucleus, with decreasing density in all directions in space at increas-

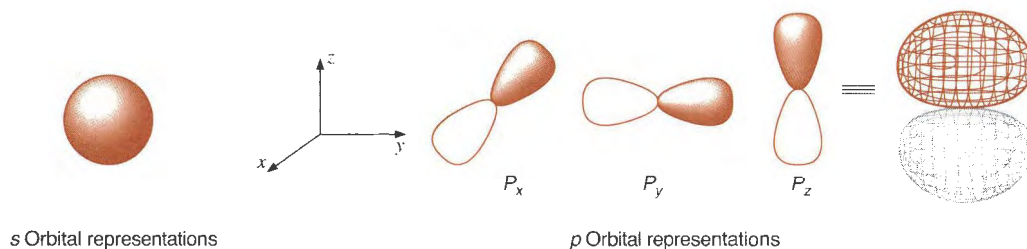


Figure 1.1

The general shape of *s* and *p* atomic orbitals for carbon. These cartoons are the schematics that chemists typically sketch. Shown also is a more realistic representation for the *p* orbital produced by quantum mechanical calculations.

ing distances from the nucleus. We pictorially represent such a population density as a sphere.

The principal quantum number 2 has *s* and *p* orbitals. The 2*s* orbital is similar to the 1*s* orbital, but has a spherical surface in three-dimensional space where the electron density goes to zero, called a **node**. A node is a surface (a sphere for *s* orbitals, a plane for *p* orbitals) that separates the positive and negative regions of a wavefunction. There is zero probability of finding an electron at an orbital node. The spherical node of a 2*s* orbital cannot be seen in the representation of Figure 1.1. In reality, this nodal surface in the 2*s* orbital has little impact on bonding models, and again, we pictorially represent this orbital as a sphere, just as with a 1*s* orbital.

A *p* orbital can orient along three perpendicular directions in space, defined to be the *x*, *y*, and *z* axes. The 2*p* orbitals have a **nodal plane** that contains the nucleus and is perpendicular to the orbital axis. As such, the electron density is zero at the nucleus. The population density of a *p* orbital reaches a maximum along its axis in both the negative and positive spatial directions, and then drops off. This population density is shown as a dumbbell-like shape.

The directionality of an orbital in space is associated with a third set of quantum numbers called **magnetic**. For the *p* orbitals the magnetic quantum numbers are $-1, 0$, and 1 , each representing one of the three different orthogonal directions in space (see the three 2*p* orbitals in Figure 1.1). The 2*s* and 2*p* orbitals make up the valence shell for carbon. Later in this chapter, we will examine metals, which contain *d* orbitals. The magnetic quantum numbers for *d* orbitals are $-2, -1, 0, 1$, and 2 .

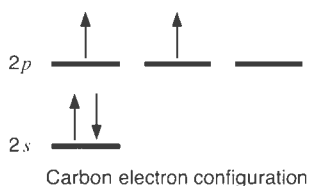
The **phasing** of the atomic orbitals shown in Figure 1.1 (color and gray / clear) is solely a result of the mathematical functions describing the orbitals. One color indicates that the function is positive in this region of space, and the other color indicates that the function is negative. It does not matter which color is defined as positive or negative, only that the two regions are opposite. There is no other meaning to be given to these phases. For instance, the probability of finding an electron in the differently phased regions is the same. The probability is defined as the **electron density** or **electron distribution**. It is specifically related to the square of the mathematical function that represents the orbitals.

The fourth and final quantum number, m_s , is associated with the spin of an electron. Its value can be $+\frac{1}{2}$ or $-\frac{1}{2}$. An orbital can only contain two electrons, and their spin quantum numbers must be oppositely signed (termed **spin paired**) if the electrons reside in the same orbital. Because electrons have wave-like properties, these waves are overlapping in space when the electrons are in the same orbital. However, because the electrons are negatively charged and have particle character also, they tend to repel each other. As a result, their movements are actually correlated, so as to keep the like charges apart. **Correlation** is the ability of an electron to feel the trajectory of another electron and therefore alter its own course so as to minimize Coulombic repulsions and keep the energy of the system to a minimum.

1.1.2 Electron Configurations and Electronic Diagrams

The **electron configuration** of an atom describes all the atomic orbitals that are populated with electrons, with the number of electrons in each orbital designated by a superscript. For example, carbon has its 1*s*, 2*s*, and 2*p* orbitals each populated with two electrons. Hence, the electron configuration of carbon is $1s^2 2s^2 2p^2$. This is the **ground state** of carbon, the most stable form. Promotion of an electron from an atomic orbital to a higher-lying atomic orbital produces a higher energy **excited state**, such as $1s^2 2s^1 2p^3$.

In an **electronic diagram** the atomic orbitals are represented by horizontal lines at different energy levels, where the higher the line on the page the higher the energy. Symbols are placed near the lines to indicate which orbitals the lines are meant to represent. The arrows represent electrons, and their direction indicates the relative spin of the individual electrons. Several rules are used to decide how these lines (orbitals) are populated (filled) with electrons. The **aufbau principle** (from German for “building-up”) states that one populates the lower energy orbitals with electrons first. Furthermore, only two electrons can be in each or-



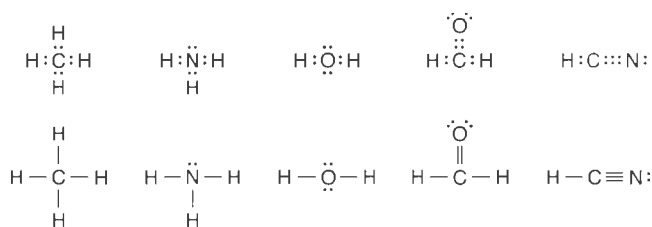
bital, and when they are in the same orbital they must be spin paired (a result of the **Pauli principle**). **Hund's rule** tells us how to handle the population of **degenerate orbitals**, which are orbitals that have the same energy. We singly populate such orbitals sequentially, and all electrons in singly-occupied orbitals have their spins aligned.

Carbon has six electrons, two in the 1s orbital, and four valence electrons that occupy the 2s and 2p valence orbitals. Based on the rules briefly reviewed here, the lowest energy electronic diagram of the valence shell of carbon is as shown in the margin.

The familiar **octet rule**, which states that atoms are most stable when their valence shell is full, suggests that carbon in a molecule will take on four more electrons from other atoms so as to possess an octet of electrons and thereby attain a noble gas configuration. The number of bonds that an atom can make is called its **valence number**. If each bond that carbon makes is created by the donation of a single electron from an adjacent atom's atomic orbitals, carbon will make four bonds. Carbon is said to have a valence of four. This valence is by far the most common bonding arrangement for C. When carbon has fewer than four bonds it is in a reactive form, namely a carbocation, radical, carbanion, or carbene. When a similar analysis is done for N, O, and F, it is found that these atoms prefer three, two, and one bond(s), respectively.

1.1.3 Lewis Structures

G. N. Lewis developed a notation that allows us to use the valence electrons of atoms in a molecule to predict the bonding in that molecule. In this method, the electrons in the valence shell of each atom are drawn as dots for all atoms in the molecule (see examples below). Bonds are formed by sharing of one or more pairs of electrons between the atoms, such that each atom achieves an octet of electrons. In an alternative to the electron dot symbolism, we can draw a line to represent a bond. A single bond is the sharing of two electrons, while double and triple bonds involve the sharing of four and six electrons, respectively. Despite its simplicity, this notation can be used to accurately predict the number of lone pairs that an atom will have and whether that atom will use single, double, or triple bonds when incorporated into specific molecules.



A few examples of Lewis structures

The problem with Lewis dot structures is that they provide no insight into molecular shapes, orbitals, or distributions of electrons within molecules. Instead, they are only useful for predicting the number of bonds an atom forms; whether the atom has lone pairs; and whether single, double, or triple bonds are used. Once an atom is found to have an octet using a Lewis analysis, no further insight into the structure or reactivity can be obtained from the Lewis structure. We have to turn to more sophisticated molecular structure and bonding concepts to understand structure and reactivity.

1.1.4 Formal Charge

Often it is convenient to associate full charges with certain atoms, even though the charges are in fact delocalized among the atoms in the molecule, and the overall molecule may be neutral. Such charges derive from the Lewis structure, and these full charges on atoms are called **formal charges**, denoting they are more of a formality than a reality.

The formula generally given in introductory chemistry textbooks for calculating the formal charge is $\text{formal charge} = \text{number of valence electrons} - \text{number of unshared electrons}$

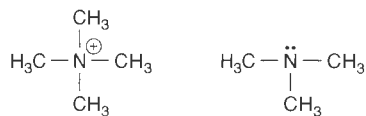
$-\frac{1}{2}$ the number of shared electrons. In organic chemistry, it is easier to just remember a few simple structures. For example, the oxygen in water has two bonds to hydrogen and is neutral. In contrast, the oxygen in the hydronium ion (H_3O^+) bonds to three hydrogens and is positive; and the oxygen of hydroxide (OH^-) has only one bond and has a formal negative charge. This series can be generalized. Whenever oxygen has an octet of electrons and has one, two, or three bonds it is negative, neutral, or positive, respectively. More generally, whenever an atom has an octet and has one bond more than its neutral state it is positive; when it has one bond fewer it is negative. Hence, a nitrogen atom having two, three, or four bonds is negative, neutral, or positive, respectively; similarly, a carbon having an octet and three, four, or five bonds is negative, neutral, or positive, respectively. Although formal charge can be rapidly evaluated in this manner, you should not take the charge on a particular atom too literally, as demonstrated in the following Going Deeper highlight.

Going Deeper

How Realistic are Formal Charges?

Formal charge is more or less a bookkeeping tool. For the tetramethylammonium ion, for example, we draw a positive charge on the nitrogen because it is tetravalent. However, it is now possible to develop very accurate descriptions of the electron distributions in molecules using sophisticated computational techniques (Chapter 14). Such calculations indicate that a much more reasonable model for the tetramethylammonium ion describes the N as essentially neutral. The positive charge resides on the methyls, each carrying one-fourth of a charge. What is going on here? Looking ahead to Section 1.1.8, we know that N is more electronegative than C, so it should have more negative charge (less positive charge) than C. Indeed, in trimethylamine there is a substantial negative

charge on the N. On going from trimethylamine to tetramethylammonium the N does become more positive than in a neutral molecule. It is just that it goes from partial negative to essentially neutral, rather than from neutral to positive, as implied by the formal charge symbolism. Beyond bookkeeping, formal charge is really only useful for indicating the charge on the *molecule*, not on individual atoms.



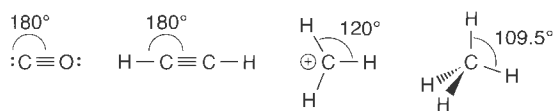
Formal charge on
quaternary ammonium

1.1.5 VSEPR

Once we have a basic idea of the bonds to expect for organic structures, the next key issue is the three-dimensional shape of such structures. We now introduce two important concepts for rationalizing the diverse possibilities for shapes of organic molecules: VSEPR and hybridization.

The **valence-shell electron-pair repulsion (VSEPR) rule** states that all groups emanating from an atom—whether single, double, or triple bonds, or lone pairs—will be in spatial positions that are as far apart from one another as possible. The VSEPR method does not consider singly occupied orbitals to be groups (see below for the reason). VSEPR is purely a theory based upon the notion that the electrostatic repulsions between entities consisting of two or more electrons dictate molecular geometries.

This rule can be applied to carbon when it is bonding to either four, three, or two other atoms. Acetylene has a **linear** arrangement of the C–C triple bond and the C–H bond, because a 180° angle places these two groups as far apart as possible. When three groups are attached to an atom, such as the three hydrogens of CH_3^+ , the geometry is **trigonal planar**



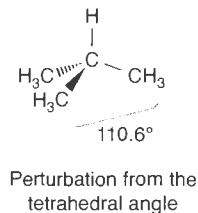
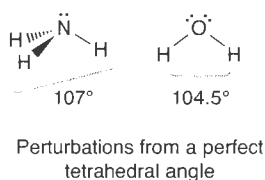
Geometries based upon VSEPR

with 120° H–C–H bond angles. Finally, in a molecule such as methane, a **tetrahedral** arrangement of the four bonds places them as far apart in space as possible (H–C–H angles of 109.5°).

The geometries for acetylene, methyl cation, and methane correspond to the bond angles for idealized linear, trigonal, and tetrahedral systems. While these geometrically perfect angles do appear in simple molecules like these, in most organic molecules where different groups are attached to the various atoms, measurable deviations from these ideals are observed. However, we will still loosely refer to the carbons as tetrahedral or trigonal, even though we don't expect angles of exactly 109.5° or 120° , respectively.

The VSEPR model provides a simple way to understand such deviations from perfection. Since the geometries derived from VSEPR are based solely upon maximizing the distance between electron pairs, it makes sense that the geometries would also depend upon the “sizes” of the electron pairs. A central tenant of VSEPR is that lone pairs *behave as if* they are larger than bonded pairs. Always keep in mind that VSEPR is not based on any first principles analysis of electronic structure theory. It is a simple way to rationalize observed trends. It is debatable whether a lone pair of electrons actually is larger than a bonded pair of electrons plus the associated atoms. In fact, it is not even clear that size is a well defined concept for a lone pair. The point is, in VSEPR we consider lone pairs to be larger than bonded pairs because that approach leads to the right conclusions. This view allows us to rationalize the fact that the H–X–H angles in ammonia and water are smaller than 109.5° . Both systems are considered to have four groups attached to the central atom because, as stated earlier, lone pairs count as groups in VSEPR. Since a lone pair is larger than a bonding pair, the N–H bonds of ammonia want to get away from the lone pair, causing contractions of the H–N–H angles. The effect is larger in water, with two lone pairs.

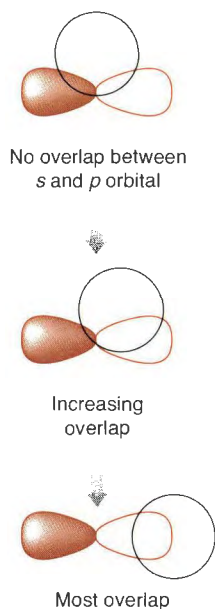
The VSEPR rule uses a common principle in organic chemistry to predict geometry, that of **sterics**, a notion associated with the through-space repulsion between two groups. **Steric repulsion** arises from the buttressing of filled orbitals that cannot participate in bonding, where the negative electrostatic field of the electrons in the orbitals is repulsive. The reason that singly occupied orbitals are not considered to be groups in VSEPR is that they can participate in bonding with doubly occupied orbitals. Intuitively, we expect larger groups to be more repulsive than smaller groups, and this is the reasoning applied to the lone pairs in ammonia and water. Likewise, due to sterics, we may expect the central carbon in 2-methylpropane to have an angle larger than 109.5° , and indeed the angle is larger than this value (see margin).

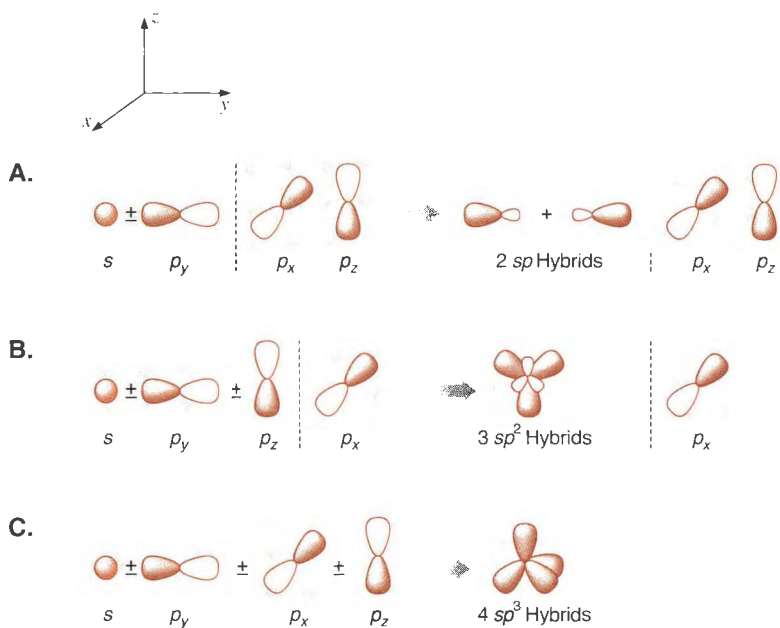


1.1.6 Hybridization

It was stated earlier that CH_3^+ prefers bond angles of 120° , and methane prefers bond angles of 109.5° . How do we achieve such bond angles when the s and p atomic orbitals are not oriented at these angles? The s orbitals are spherical and so have no directionality in space, and the p orbitals are oriented at 90° angles with respect to each other. We need a conceptual approach to understand how s and p atomic orbitals can accommodate these experimentally determined molecular bond angles. The most common approach is the idea of hybridization, first introduced by Pauling.

Pauling's assumption was that bonds arise from the overlap of atomic orbitals on adjacent atoms, and that the better the overlap the stronger the bond. **Orbital overlap** has a quantitative quantum mechanical definition (given in Chapter 14). In a qualitative sense, overlap can be thought of as the extent to which the orbitals occupy the same space. However, if there are regions of overlap with matched and mismatched phasing, the contributions to the overlap have opposite signs and will cancel. The more space occupied where the phasing reinforces, the larger the overlap. When the opposite phasing in the various areas completely cancels, there is no overlap. For example, consider the arrangements of the s and p orbitals shown in the margin. The top shows how the s and p occupy some of the same space, but the phasing completely cancels: zero overlap is the result. Any movement of the s orbital to the side increases overlap, until the greatest overlap, shown for the bottom arrangement, takes advantage of the directionality of a p orbital.



**Figure 1.2**

Forming hybrid orbitals. Combining an s orbital with one, two, or three p orbitals produces the familiar **A.** sp , **B.** sp^2 , and **C.** sp^3 hybrid orbitals.

Pauling also argued that orbitals with directionality would give stronger bonds because the overlap would be higher. Pauling suggested that to achieve orbitals with directionality, mixtures of atomic orbitals on the same atom are formed in a process known as hybridization. **Hybridization** is the method of adding and subtracting atomic orbitals on the same atom. Remember that orbitals are mathematical solutions to the Schrödinger equation, and that the addition and subtraction of mathematical equations is just an exercise in algebra. It is a perfectly valid operation to add orbitals as long as one also does the corresponding subtraction. Qualitatively, we use the positive and negative phasing along with the three-dimensional shapes of the orbitals to visualize what the result of adding and subtracting the orbitals would be. For example, Figure 1.2 **A** shows the result of combining a $2s$ and one of the $2p$ orbitals, in this case the $2p_y$ orbital. Each of the resultant orbitals has a large lobe on one side of the atom and a small lobe on the other side, and therefore has greater directionality than the original orbitals. The addition leads to an orbital with directionality along the negative y axis, and the subtraction leads to an orbital with directionality along the positive y axis. The two lobes have different phasing. The combination of an s orbital with a single p orbital creates what are called **sp hybrid orbitals**. Note that these two new orbitals point 180° apart, as is found in acetylene. Hence, the carbons in acetylene are considered to be sp hybridized in order to accommodate the experimentally determined geometry. This leaves two pure p orbitals: p_x and p_z .

This addition and subtraction can be carried further to give **sp^2** and **sp^3 orbitals**. Figure 1.2 **B** shows this addition and subtraction for sp^2 . Note that the new orbitals are now all 120° apart. The remaining p orbital is perpendicular to the sp^2 hybrid orbitals. It makes good sense that the mixing of this last p orbital with the sp^2 hybrids would lead to new hybrids that are above and below the plane formed by the sp^2 hybrids. In this case four identical orbitals called **sp^3 hybrids** are the result (Figure 1.2 **C**). Each points toward the corners of a tetrahedron. In organic molecules one of these hybridization states— sp , sp^2 , or sp^3 —is invoked as appropriate when explaining the linear, trigonal planar, or tetrahedral geometry of an atom, respectively.

The geometries for acetylene, methyl cation, and methane correspond to the bond angles for the different hybridization states sp , sp^2 , and sp^3 , respectively. Again, most organic molecules display measurable deviations from these ideals, but we still loosely refer to the atoms as sp , sp^2 , or sp^3 hybridized, even though we don't expect angles of exactly 180° , 120° or 109.5° .

Hybridization provides an alternative "explanation" to VSEPR for such deviations from ideal angles. In going from pure sp to sp^2 , sp^3 , and pure p , the angles go from 180° to 120° , 109.5° , and 90° . Thus, decreasing s character leads to decreasing bond angles. We could say that in ammonia the N–H bonds have lost s character from N relative to a pure sp^3 N, because the angle is smaller than the perfect tetrahedral angle. In fact, we can quantify this analysis with a simple relationship. We define a **hybridization index**, i (Eq. 1.1). Here, the observed bond angle θ is used in the equation to solve for i .

$$1 + i \cos \theta = 0 \quad (\text{Eq. 1.1})$$

We then define the hybridization as sp^i . For example, since by definition the tetrahedral angle is the arc $\cos(-1/3)$ ($\sim 109.5^\circ$), perfect tetrahedral angles imply $i = 3$. For ammonia, we conclude that the N hybrids that bond to H are $sp^{3.4}$, and in water the bonds to H are formed by sp^4 hybrids. That is, in water the orbitals that make up the O–H bonds are 80% p in character and 20% s , versus the 75:25 mixture implied by sp^3 . The lone pairs must compensate, and they take on extra s character in NH_3 and H_2O . We will see that this notion of non-integral hybridizations is more than just an after-the-fact rationalization, and has experimental support (see the following Connections highlight). It can have predictive power. However, we must first introduce another important bonding concept: electronegativity.

Connections

NMR Coupling Constants

The view of variable hybridization has some experimental support. The magnitude of ^{13}C – ^1H NMR coupling constants is expected to be proportional to the amount of carbon s character in the bond, because only s orbitals have density at the carbon nucleus and can affect neighboring nuclear spin states. In several systems, a clear correlation has been observed between NMR coupling constants and percent s character, as predicted from the geometry and the associated hybridization index.

For example, in cyclic alkanes, the smaller the ring, the larger the p character that would be expected in the hybrid orbitals used to form the C–C bonds, because p orbitals better accommodate smaller bond angles. Correspondingly, the C–H bonds would have higher s character.

This correspondence is indeed seen from an analysis of the C–H coupling constants given below. The smaller rings have the larger coupling constants.

Ring system	$J_{\text{C-H}}$
Cyclopropane	161
Cyclobutane	134
Cyclopentane	128
Cyclohexane	124
Cycloheptane	123
Cyclooctane	122

Ferguson, L. N. (1973). *Highlights of Alicyclic Chemistry, Part 1*. Franklin Publishing Company, Inc., Palisade, MI.

1.1.7 A Hybrid Valence Bond/Molecular Orbital Model of Bonding

There are two dominant models for considering bonding in organic molecules: valence bond theory (VBT) and molecular orbital theory (MOT). While often viewed as competing theories, VBT and MOT actually complement each other well, and our ultimate model for bonding will borrow from both theories. VBT was developed first. The idea, as originally put forth by Heitler and London and expanded by Pauling, was that the binding energy between two atoms arises primarily from exchange (resonance) of electrons between the two atoms in a bond. The starting point for VBT has one electron on each atom that contributes to an **electron pair bond**. It is this assignment of electrons primarily to individual atoms—or, more precisely, to individual orbitals on atoms—that is the hallmark of VBT. Bonding is, in

effect, viewed as a perturbation of this arrangement. That is, when two atoms are brought together, each electron is permitted to interact with either nucleus, and this produces bond energies in adequate agreement with the experimental values. Hence, the conclusion was that bonds consist of two electrons in the region between two nuclei.

In VBT a molecule is formed by adjacent atoms sharing electrons. As suggested by the name, the electrons that are involved in bonding are those from the atoms' valence shells. Each atom donates one electron to the bond, and the resulting electron pair is considered to be mostly localized between the two adjacent atoms. This localization of the electrons is exactly the impression of bonding that is given by a Lewis structure. Furthermore, localization of the electrons between the atoms would require orbitals that point in the appropriate directions in space. It is this kind of reasoning that led Pauling to develop hybrid orbitals, an essentially valence bond concept. In essence, VBT nicely encompasses the topics discussed to this point in the chapter. However, one other notion is required by VBT—that of resonance. As discussed in Section 1.1.10, if more than one Lewis dot structure can be drawn for a molecule, then VBT states that the actual molecule is a hybrid of these “canonical forms”.

Creating Localized σ and π Bonds

The most common model for bonding in organic compounds derives from VBT and the hybridization procedure given previously. **Sigma bonds** (σ bonds) are created by the overlap of a hybrid orbital on one atom with a hybrid orbital on another atom or an s orbital on hydrogen (Figures 1.3 A and B, respectively). **Pi bonds** (π bonds) are created by the overlap of two p orbitals on adjacent atoms (Figure 1.3 C). Specifically, σ bonds are defined as having their electron density along the bond axis, while π bonds have their electron density above and below the bond axis. The combination of the two orbitals on adjacent atoms that creates **in-phase** interactions (signs of the orbitals are the same) between the two atoms is called the **bonding orbital**. The combination that results in **out-of-phase** interactions (signs of the orbitals are opposite) is called the **antibonding orbital**. The bonding orbital is lower in energy than the antibonding orbital. There are also orbitals that contain lone pairs of electrons, which are not bonding or antibonding. These are called **nonbonding orbitals**. In standard neutral organic structures, only the bonding orbitals and nonbonding orbitals are occupied with electrons. Recall that an alkene functional group has a single σ and a single π bond between the adjacent carbons, whereas an alkyne has a single σ and two π bonds between the carbons. The number of bonds between two atoms is called the **bond order**.

The creation of bonding and antibonding orbitals is actually a molecular orbital theory notion. Therefore, the orbitals of Figure 1.3 are in effect **molecular orbitals**. We will have much more to say in Section 1.2 about how to linearly mix orbitals to create bonding and antibonding molecular orbitals. However, you may recall molecular orbital mixing diagrams from introductory organic chemistry, such as that shown in the margin for the π bond in ethylene. These diagrams give a picture of how chemists visually create bonding and antibonding orbitals via mixing. The mixing to derive the molecular orbitals gives both a plus

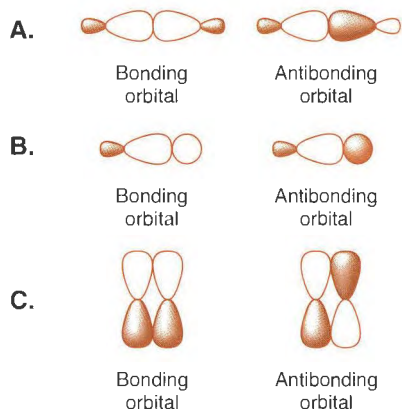
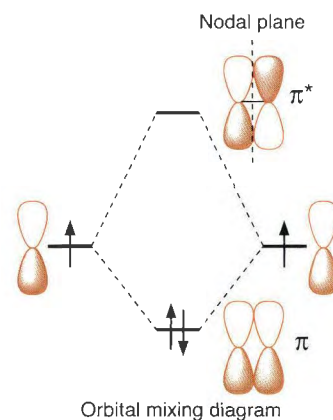


Figure 1.3

A. Combination of two hybrid orbitals on adjacent atoms gives bonding and antibonding orbitals. Population of the bonding orbital with two electrons creates a σ bond. **B.** Combination of a hybrid and a $1s$ orbital on hydrogen gives a bonding and antibonding pair. Population of the bonding orbital with two electrons creates a σ bond. **C.** Combination of two p orbitals on adjacent atoms also gives a bonding and antibonding set. Population of the bonding orbital with two electrons creates a π bond.

and minus combination of the starting atomic orbitals. Note, therefore, that the orbitals given in Figure 1.3 are actually derived from a hybrid VBT/MOT approach to bonding. One creates discrete, localized bonds between adjacent atoms as pictured with VBT, but it is done using the linear combination ideas of MOT.

The simple molecular orbital mixing diagram given above serves to illustrate many concepts and terms used with molecular orbital theory. The bonding molecular orbital (MO) is **symmetric** with respect to a mirror plane that resides between the two carbons making the bond, while the antibonding MO is **antisymmetric**. Furthermore, a nodal plane exists in the antibonding MO between the two atoms making the π bond, which means that populating this orbital with electrons leads to a repulsive interaction between the atoms.

The picture of σ and π bonds that consist of bonding and antibonding molecular orbitals that reside primarily between adjacent atoms is the standard that organic chemists most commonly use. The reactivity of the vast majority of organic compounds can be nicely modeled using this picture, and it forms the starting point for the electron pushing method of presenting organic reaction mechanisms (see Appendix 5). Hence, this theory of bonding is extremely important in organic chemistry.

1.1.8 Polar Covalent Bonding

Once the geometry of a molecule has been established, the next crucial feature for predicting the reactivity is its charge distribution. Notions such as VSEPR and hybridization control shape and structure. Here we discuss how electronegativity is the primary determinant of the charge distribution in a molecule, with hybridization playing a secondary but still important role.

Covalent bonds predominate in organic chemistry. In our simple theory of bonding, the two electrons in the bond are shared between the two adjacent atoms, as implied by a Lewis dot structure and the σ and π bonds discussed previously. Very few, if any, organic structures can be considered to have ionic bonds. However, whenever a carbon forms a bond to any atom or group not identical to itself, the bond develops some polar character; there is a positive end and a negative end to the bond. This charge separation means the sharing of electrons is unequal. A covalent bond that has an unequal sharing of the bonding pair of electrons is called **polar covalent**. Pauling argued that introducing polarity into a bond strengthens it, and we will see in Chapter 2 that trends in bond strengths generally support this view.

Electronegativity

To predict the charge distribution in an organic molecule, we need to examine the **electronegativity** of the atoms in the molecule. Pauling originally developed this important concept and described it as “the power of an atom in a molecule to attract electrons to itself”. Pauling assigned values to various atom types by examining bond dissociation energies of molecules. As such, the Pauling electronegativity scale depends upon *molecular* properties, and is not an intrinsic property of the atoms. The Pauling scale is most commonly used, and is given in all introductory chemistry textbooks.

Mulliken defined an electronegativity scale that is derived from the average of the ionization potential and electron affinity of an atom, and therefore is solely an *atomic* property. The **ionization potential** is the energy required to remove an electron from an atom or molecule. Hence, this number reflects the affinity of an atom for the electrons it already has. The **electron affinity** is the amount of energy released or required to attach another electron to an atom or molecule. Hence, this number reflects the affinity of the atom for an additional electron. Using these values is a logical basis for determining the ability of an atom to attract electrons toward itself. Along with the electronegativity scales of Pauling and Mulliken, comparable scales have been developed by Nagle, Allen, Sanderson, Allred-Rochow, Gordy, Yuan, and Parr. Suffice it to say that the electronegativity of an atom is a difficult concept to put a precise number on, and that the use of different scales is appropriate for different applications. Table 1.1 compares the Pauling and Mulliken electronegativity scales, showing that the two are similar. We should always remember that the key issue is the *rela-*

Table 1.1
Electronegativities of Atoms According to
the Scales of Pauling and Mulliken*

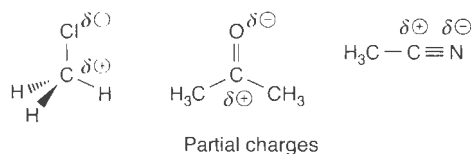
Atom	Pauling	Mulliken
H	2.1	3.01
B	2.0	1.83
C	2.5	2.67
N	3.0	3.08
O	3.5	3.22
F	4.0	4.44
Cl	3.0	3.54
Br	2.8	3.24
I	2.5	2.88
Li	1.0	1.28
Na	0.9	1.21
K	0.8	1.03
Mg	1.2	1.63
Ca	1.0	1.30
Al	1.5	1.37
Si	1.8	2.03
P	2.1	2.39
S	2.5	2.65

*Pauling, L. (1960). *The Nature of the Chemical Bond and the Structure of Molecules and Crystals; an Introduction to Modern Structural Chemistry*, 3d ed., Cornell University Press, Ithaca, NY, and Allen, L. C. "Electronegativity is the Average One-Electron Energy of the Valence-Shell Electrons in Ground-State Free Atoms." *J. Am. Chem. Soc.*, **111**, 9003 (1989).

five electronegativities of two moieties—we mainly want to know which is the more electronegative of the two, and whether the difference is relatively large or small. In most situations, all the various electronegativity scales lead to the same predictions.

The major factor influencing electronegativity is the energy of the orbitals that the atom uses to accept electrons. As one moves left-to-right across the periodic table, the valence orbitals become lower in energy within the same row. Going down a column, the atoms get bigger and the valence orbitals are higher in energy. This would mean that He has the lowest energy valence orbitals, but He cannot accept any more electrons because it is a noble gas. The atom with the lowest energy valence orbitals that is not a noble gas is F, which is the most electronegative element. In fact, a useful way to estimate electronegativities if Table 1.1 is unavailable is simply to recall that F is the most electronegative element, and moving left or down in the periodic table progressively diminishes electronegativity.

When an atom with a higher electronegativity than carbon forms a bond to carbon, the electrons in the bond will reside more toward the electronegative atom, producing a partial negative charge on this atom and a partial positive charge on carbon. Conversely, when the atom has a lower electronegativity, the carbon will possess a partial negative charge. These partial charges are denoted $\delta+$ and $\delta-$ (see a few examples below). The larger the difference between the electronegativities for atoms in a bond, the more the bond is polarized. The magnitude of the polarization can be gauged using the values of Table 1.1. For example, us-

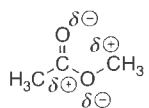


ing the Pauling scale, a C–F bond is more polar than a C–Cl bond, because the electronegativity differences for the atoms in these bonds are 1.43 and 0.61, respectively. A bond whose electronegativity difference is 1.7 is considered to be 50% ionic and 50% covalent, and so bonds with differences greater than this are considered to be ionic. In this view, bonds of Li, Na, or K to F, Cl, Br, or I are all ionic.

Some conclusions that can be drawn from the data of Table 1.1 may seem a bit counterintuitive. For example, the electronegativity difference between C and I in both scales is actually smaller than the difference between C and H. Therefore, a C–I bond is predicted to have a smaller charge polarization than a C–H bond, based solely upon electronegativities. This polarization may come as a surprise, since iodide is a good leaving group in S_N2 reactions. Often the electronegativity of iodine is erroneously invoked to explain such reactions. As we will see below, electronegativity is not the whole story. Polarizability is also important, especially when rationalizing reactivity trends. Also note that the electronegativity difference between C and O is smaller than between P and S, or P and O, on the Mulliken scale. Hence, S–O and P–O bonds are more polarized than C–O bonds.

Electrostatic Potential Surfaces

In a polar bond, one end is designated as δ^+ and the other as δ^- , and this designation is often adequate for discussing simple molecules such as methyl chloride. However, in more complex molecules there will be many different types of bonds with differing degrees of polarity, and the overall molecule will reflect the sum of these and any interactions they experience. The simple δ^+ / δ^- symbolism is no longer adequate, so we need an alternative way to view the charge distribution in complex organic molecules. In recent years, many scientists have found that plots of electrostatic potential surfaces are quite useful in this regard. Such plots are given in Appendix 2, which contains a gallery of representative electrostatic potential surfaces for prototype organic structures. In these pictures, red represents negative electrostatic potential whereas blue represents positive electrostatic potential. A green color is a region that is essentially neutral.



Let's examine for a moment the electrostatic potential surface for methyl acetate (see Appendix 2). In our δ^+ / δ^- method, we would denote methyl acetate as shown in the margin. The carbonyl carbon is partially positive, and so is the methyl group. However, both oxygens would be denoted as negative. This notation is less than optimum, however, because we have no idea which oxygen is more negative. Yet, a quick glance at the electrostatic surface shows that the carbonyl oxygen is more negative (you may predict this fact from resonance; see Section 1.1.10). Thus, an electrostatic potential surface can provide deeper insight into the electronic distribution in a molecule than a simple δ^+ / δ^- picture.

What exactly are these surfaces? First, they result from a full quantum mechanical calculation of the electronic structure of the molecule. Note that we do not use electronegativities or hybridizations or bond dipoles or any of the descriptive features of our bonding model in such calculations. These are *a priori* quantum mechanical calculations, and their output enables the charge distribution in a molecule to be computed.

What are we actually showing in such plots? First, we give a surface to the molecule. The surface is very similar to a **van der Waals surface**, the surface that would be obtained by considering each atom to be a sphere with a radius equal to its van der Waals radius (Table 1.5). For technical reasons, though, the surface is more typically an **isodensity surface**, meaning a surface with a constant electron density, such as 0.002 electrons/ $\text{\AA}^2$. The distinction is small. Next, we color the surface according to **electrostatic potential** (i.e., red for negative and blue for positive).

What is electrostatic potential and how is it determined? Imagine taking a very small sphere with a charge of +1 and rolling it around the isodensity surface. At each point, we ask whether the sphere is attracted to or repulsed by the surface and what the energetic magnitude of the interaction is. The magnitude of the interaction is the electrostatic potential. Thus, the plots have units not of charge but of energy (we will use kcal/mol). We are not plotting partial charge; we are plotting electrostatic potential, although the two will track

each other. As a result, these plots are extremely useful for visualizing the charge distribution in organic molecules. We encourage the student to consult Appendix 2 frequently while reading this text.

There are definitely some caveats to the interpretation of electrostatic surface potentials, and some are given in the next Going Deeper highlight. An important caveat is to appreciate that these are electrostatic potential surfaces for the *ground states* of the molecules. They show the charge distribution absent any external perturbation. When a chemical reaction occurs, we expect a substantial reorganization of charge. For example, when an anionic nucleophile adds to the carbonyl of acetone, we expect a very different electrostatic potential surface for the transition state than in the picture of acetone shown in Appendix 2. Since transition states control reactivity, it is risky to use these electrostatic potential surfaces to predict or rationalize reactivity. They can be helpful in this regard, but caution is in order.

Going Deeper

Scaling Electrostatic Surface Potentials

Electrostatic potential surfaces are very useful guides to charge distributions, and they are now commonly shown in introductory organic texts. However, some caution is warranted when interpreting them. Most important is to pay attention to the energy scale associated with any particular structure. That is, what value of positive electrostatic potential does it take to achieve the maximum in blue, and what negative electrostatic potential will maximize the red? Realize that in the analysis of any electrostatic potential surface, positive or negative potentials larger in magnitude than the arbitrarily set range will simply be the most intense blue or red, and will not be distinguished from any other value over the limit.

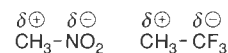
There are two ways to scale these plots. Some electrostatic potential surface presentations take the direct results of the calculation and use as the maxima the maximum values for plus and minus electrostatic potential in the molecule. This appears to be the most common approach in introductory texts. In this approach, we could end up with a range such as +57.29 kcal/mol to -36.43

kcal/mol. A potential problem with this approach is that the color scale is “linear” from plus to minus, so the zero electrostatic potential color—an important benchmark—will be different for this structure than for some other structure with a range such as +27.22 to -49.83. For this reason, we avoid such presentations in Appendix 2 and present electrostatic potential surfaces with a symmetrical range of electrostatic potentials, such as ± 25 kcal/mol or ± 50 kcal/mol. In this way, zero electrostatic potential is always the same green color. However, *it is also very important to be aware of the range of electrostatic potentials being plotted.* A plot of the benzene electrostatic potential surface with a ± 25 kcal/mol range will look different from a plot with a ± 50 kcal/mol range. It is especially risky to compare two structures with different ranges plotted. Whenever possible, we will provide comparisons with the same range, and we will always make it clear what range of electrostatic potentials is used for any figure. The student always needs to be aware of electrostatic potential range when interpreting and comparing such plots.

Inductive Effects

We have seen that when carbon bonds to an electronegative element like O, N, Cl, or F, a bond polarization develops, making the C δ^+ and the heteroatom or halogen δ^- . We might expect a functional group containing electronegative atoms to also be electron withdrawing (see examples in the margin). The phenomenon of withdrawing electrons through σ bonds to the more electronegative atom or group is called an **inductive effect**. It is an effect that we will often cite to explain trends in thermodynamics or reactivity throughout this book (an interesting twist to induction is given in the next Going Deeper highlight). The inductive effect is what gives rise to bond polarizations, polarizations within molecules, and bond and molecular dipole moments.

A similar but separate phenomenon is a **field effect**. This is a polarization in a molecule that results from charges that interact through space, rather than through σ bonds, and it can influence the structure and reactivity of other parts of the molecule. We will see systems later in which both field and inductive effects seem to be operative.



Partial charges due to induction

Going Deeper

1-Fluorobutane

Consider the specific case of 1-fluorobutane. We expect a large bond dipole, with C1 positive, and the F negative. Moving along the σ chain we would expect the magnitude of the charges to progressively diminish, with some polarization at C2, and less at C3, etc.

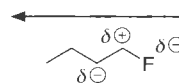
Inductive effects have been extensively investigated experimentally, and some conflicting trends are seen. Advanced quantum mechanical calculations of the sort described in Chapter 14 are able to assign partial charges to atoms or groups of atoms, and while there is some debate as to the best way to do this, all methods produce similar trends. In the case of 1-fluorobutane the F atom carries a large partial negative charge, on the order of -0.44 . As expected, the CH_2 group at C1, as a unit, carries a comparable positive charge, creating a very large C–F bond dipole. What about the CH_2 at C2? Instead of being just part of a simple C–C bond, it is paired to a carbon with a substantial positive charge. At the same time, it is two atoms away from a very electronegative element—fluorine.

Perhaps surprisingly, the C2 CH_2 shows a small

negative charge, on the order of -0.03 . Rather than a progressively diminishing positive charge as we move down the chain, a charge alternation is seen. At some levels of theory, this charge alternation continues down the chain, but the magnitude of the charges at C3 and C4 are so small as to be inconsequential.

This result is fairly general in computational studies, but it is not in line with experimental observations of inductive effects. As we will see in Chapter 5 and other places, inductive effects on thermodynamics and kinetics do not usually show an alternation pattern. For example, for a linear alkanoic acid, adding a strongly electronegative element like F to the alkyl chain always increases the acidity of the carboxylic acid functional group, and the effect is always stronger the closer the F is to the incipient carboxylate. This trend is an example of the danger of directly extrapolating ground state electronic structure features to reactivity patterns.

Magnitude of the delta charges
diminishes but alternates



Group Electronegativities

It is often convenient to consider groups that make up particular portions of a molecule as having their own electronegativity. For example, we would expect a CF_3 group to affect the charge distribution in a molecule via induction much more than a CH_3 group, but if we consider only carbon electronegativities, the two are the same. Table 1.2 lists some group electronegativity values that were derived to be comparable to the Pauling scale for atoms. We find that a methyl group is essentially the same as a C, whereas the CF_3 group has an electronegativity similar to that for O. Alkenyl and alkynyl groups are quite electronegative, as are nitro and cyano groups. Finally, a full positive charge, such as that associated with a protonated amine, has the highest group electronegativity.

Table 1.2
Group Electronegativities, Scaled to be
Compatible with the Pauling Scale*

Group	Electronegativity
CH_3	2.3
CH_2Cl	2.8
CHCl_2	3.0
CCl_3	3.0
CF_3	3.4
Ph	3.0
$\text{CH}=\text{CH}_2$	3.0
$\text{C}\equiv\text{CH}$	3.3
$\text{C}\equiv\text{N}$	3.3
NH_2	3.4
NH_3^+	3.8
NO_2	3.4
OH	3.7

*Wells, P. R. "Group Electronegativities." *Prog. Phys. Org. Chem.*, **6**, 111 (1968).

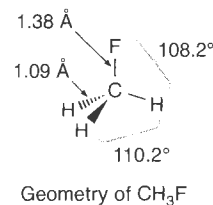
Hybridization Effects

The relative electronegativities of C and H, a critical issue in organic chemistry, has in fact been the topic of some debate. In Table 1.1 we see that the Pauling scale describes C as more electronegative than H, while the Mulliken scale gives the opposite ordering. It is now believed that the cause of this discrepancy is a hybridization effect. Since s orbitals have substantial density at the nucleus while p orbitals have a node at the nucleus, the more s character in a hybrid orbital, the closer to the nucleus the electrons in that hybrid tend to be. Because electronegativity describes an atom's ability to attract electrons to itself, sp^2 hybrids should be more electronegative than sp^3 hybrids, and this is indeed the case. The data in Table 1.2 are completely consistent with this view. The electronegativities are $C\equiv CH > CH=CH_2 > CH_3$; $sp > sp^2 > sp^3$.

What does this mean, then, about the relative electronegativities of C and H? A good deal of evidence, including molecular quadrupole moments discussed below, points to the conclusion that an sp^2 C is more electronegative than H, while an sp^3 C and H have very similar electronegativities. In a sense, both Pauling and Mulliken were right. A great many observations, especially those involving noncovalent interactions (Chapters 3 and 4), can be understood from this simple statement.

Taken together, electronegativity and hybridization provide an appealing rationalization of many structural trends. For example, the smaller bond angles in ammonia and water vs. methane discussed previously are nicely explained. There is a competition as to whether the central atom (O or N) should place more s character in the hybrid orbital that contains the lone pair(s) or the hybrid orbital used to make bonds to the hydrogen atoms. Since the lone pair electrons are not shared with another atom, an electronegative element prefers greater s character in its own lone pair orbitals, because it can better keep these electrons to itself. This effect places more p orbital character in the bonds to hydrogen, which in turn reduces the H–N–H or H–O–H bond angles relative to methane. The effect is more pronounced for O because it is more electronegative.

As another example, let's consider methyl fluoride. The H–C–F angle is contracted, and as a result the H–C–H bonds are slightly expanded. The H–C–F contraction is due to the fact that F is the more electronegative substituent. The F prefers to bond to a carbon hybrid that has more p character, because it is easier to withdraw electrons from a p orbital on carbon than an s orbital on carbon. It is often said that s orbitals have better **electron penetration** to the nucleus than p orbitals, suggesting again that it is harder to withdraw electrons from s orbitals. If the carbon uses more p character in a hybrid to bond to F, more s character will be devoted to the hybrids that comprise the H–C–H bonds. It is difficult to imagine a rationalization of this result using VSEPR, because F is larger than H, and may be expected by VSEPR to open up the H–C–F angle. For the most part, organic structures are better rationalized using hybridization and electronegativity arguments than VSEPR.



Geometry of CH_3F

1.1.9 Bond Dipoles, Molecular Dipoles, and Quadrupoles

One goal of our discussion of electronegativity was to delineate the relative electron withdrawing nature of an atom, group, or orbital. The term “relative” is important, because the exact numbers associated with atom and group electronegativities are not used on a day to day basis when practicing organic chemistry. Instead, the trends and relative electron donating and accepting abilities are of paramount importance. Now, however, we will be considering bond dipoles, and we need to get more quantitative. The exact charges on the atoms in the bonds need to be known, or the electronegativity numbers associated with atoms need to be used.

Bond Dipoles

When two atoms of differing electronegativities are bonded, one end of the bond will be $\delta+$ and the other will be $\delta-$. This analysis leads to the notion of a **bond dipole** as the local moment that is associated with a polar covalent bond. A **moment** reflects the electrostatic force that would be exerted by a charge on a neighboring charge. The dipole moment pro-

vides a means of comparing which bonds are more polar and evaluating the relative force that a dipole exerts on neighboring charges or dipoles. Certainly, examination of the electrostatic potential surfaces of Appendix 2 reveals bond dipoles. If we know the partial charges on the atoms of the bond, we can calculate a bond dipole.

A dipole moment (μ) is given in units of electrical charge times distance (Eq. 1.2), where q is charge and r is distance. It is usually expressed in units of Debye (D, where $1\text{D} = 10^{-18}\text{esu cm}$). “Esu” stands for “electrostatic unit”, and the charge of an electron or proton is negative or positive $4.80 \times 10^{-10}\text{esu}$, respectively.

$$\mu = q \times r \quad (\text{Eq. 1.2})$$

For example, a bond that has a 0.2 positive and negative charge on the opposite ends with a separation of $1.54\text{ \AA}$ ($1\text{ \AA} = 10^{-8}\text{ cm}$) would have an associated bond dipole of $(0.2)(4.80 \times 10^{-10}\text{ esu})(1.54 \times 10^{-8}\text{ cm}) = 1.47 \times 10^{-18}\text{ esu cm} = 1.47\text{ D}$. Chemists consistently use a symbolism in which the positive end of the dipole is represented by a cross, along with an arrow that points in the direction of the negative end of the dipole.

Recognizing polar covalent bonds and bond dipoles in organic molecules is a great aid to predicting chemical reactivity. Species with partial or full negative charges should be attracted to the $\delta+$ region in a molecule or the positive end of the bond dipole. Conversely, positively charged species would be attracted to the $\delta-$ region of the molecule or the negative end of the bond dipole (see Chapter 10 for the use of these guidelines in predicting reactivity). Such attractions are crucial in controlling weak, noncovalent interactions such as solvation and molecular recognition (see Chapters 3 and 4). Just like electrostatic potential surfaces, molecular and bond dipoles reflect ground states. While the polarization patterns described here can provide valuable clues to reactivity, it is also crucial to consider how bond polarity affects transition states when discussing reactivity.

Molecular Dipole Moments

While the bond dipoles we have just described are, in a sense, conceptualizations, related to our notions of electronegativity, the **molecular dipole** is a well-defined, intrinsic property of a molecule. A molecule has a dipole moment whenever the center of positive charge in the molecule is not coincident with the center of negative charge. This separation of charged centers feeds into Eq. 1.2 also, making it possible to calculate the molecular dipole moment.

Table 1.3 lists experimentally determined dipole moments for select molecules. The numbers tell the relative separation of charges within the molecules, thereby giving an idea of the intensity of the electric field around the molecule. They also give a sense as to how

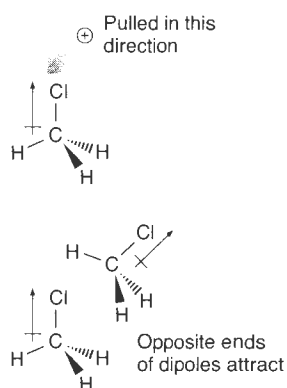
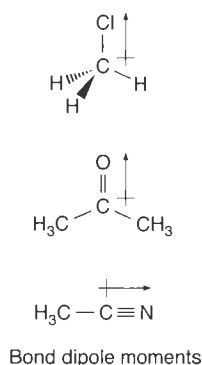


Table 1.3
Molecular Dipole Values*

Compound	Molecular dipole (D)	Compound	Molecular dipole (D)
CCl_4	0.0	CH_3COCH_3	2.9
CHCl_3	1.0	CH_3COOH	1.7
CH_2Cl_2	1.6	CH_3COCl	2.7
CH_3Cl	1.9	$\text{CH}_3\text{COOCH}_3$	1.7
CH_3F	1.8	$\text{C}_6\text{H}_5\text{Cl}$	1.8
CH_3Br	1.8	$\text{C}_6\text{H}_5\text{NO}_2$	4.0
CH_3I	1.6	1-Butene	0.34
CH_3OH	1.7	1-Propyne	0.80
CH_3OCH_3	1.3	<i>cis</i> -2-Butene	0.25
CH_3CN	4.0	<i>cis</i> -1,2-Dichloroethene	1.9
CH_3NO_2	3.4	Tetrahydrofuran	1.6
CH_3NH_2	1.3	Water	1.8

*Handbook of Chemistry and Physics, CRC Press, Inc., Boca Raton, FL (1979).

strongly an approaching molecule or charge can differentiate one end of the molecule from the other or, alternatively, how favorable a potential electrostatic interaction can be. For example, all approaches to a molecule with a molecular dipole of zero, such as tetrachloromethane, encounter essentially the same electric field. This argument ignores both polarization effects and higher moments such as quadrupoles. In contrast, the electric field felt by a molecule approaching a structure with a dipole of 4.0, such as acetonitrile, is quite different, depending upon the direction of approach. Note that the electrostatic potential surfaces of several small molecules in Appendix 2 provide a clear way to visualize molecular dipole moments.

An often informative exercise is to analyze a molecular dipole as a vector sum of bond dipoles. Examples of this analysis are shown in Figure 1.4. Note that in high symmetry cases all the local bond dipoles cancel and the overall molecule has no molecular dipole. Thus, the absence of a molecular dipole does not rule out the existence of bond dipoles, and the presence of bond dipoles does not guarantee the existence of a molecular dipole.

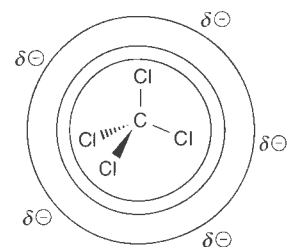
Several trends emerge from examining Table 1.3. The more chlorines attached to methane, from CH_3Cl to CCl_4 , the lower the dipole. This trend might at first seem counterintuitive, because we are progressively adding polar bonds to the system. However, it can be understood as a consequence of vector mathematics, in which the individual bond dipoles increasingly cancel as the number of chlorines increases. The incorporation of nitro or cyano groups into molecules results in very large molecular dipoles when there are no other bond dipoles to cancel them. An important feature of dipole moments is illustrated by the fact that the dipole moments of CH_3Br and CH_3F are the same. We would expect a much larger charge polarization in the C–F bond compared to the C–Br bond, and this is so. However, the C–Br bond is longer than the C–F bond, and even though the charge separation is smaller, the distance is larger. The two phenomena both affect the molecular dipole, and coincidentally lead to the same dipole moment for the two compounds.

Molecular Quadrupole Moments

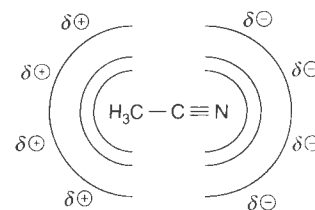
In a complete description of a molecule's charge distribution, the dipole moment is just one term in a series: monopole, dipole, quadrupole, octupole, hexadecapole, etc. A monopole is just a point charge—the dominant term for ions. For neutral molecules organic chemists usually truncate the series after the dipole. However, the quadrupole moment of a molecule can often be quite important. As such, we take a moment here to remind you about some basic electrostatics.

A **quadrupole** is simply two dipoles aligned in such a way that there is no net dipole (if there was a dipole, we'd have a dipole, not a quadrupole). Interestingly, the multipole expansion follows a familiar topological pattern. Monopoles look like *s* orbitals (spheres); dipoles look like *p* orbitals (a + end and a – end); quadrupoles look like *d* orbitals; octupoles like *f* orbitals, etc. The analogy between multipoles and orbitals is given just to illustrate phasing properties; orbitals do not have polar character.

Figure 1.5 illustrates this point. The most common quadrupole has the two dipoles side-by-side pointing in opposite directions, giving four charge regions (two + and two –) and the topology of a d_{xy} orbital. This topology is also the arrangement in a quadrupole mass spectrometer. However, there is an alternative arrangement—two end-to-end dipoles point-



Electric field around CCl_4



Electric field around acetonitrile

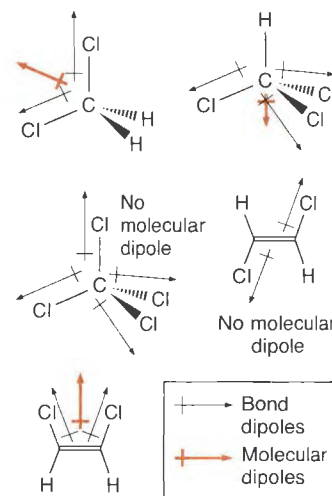


Figure 1.4
Examples of analyzing molecular dipoles as a sum of bond dipoles.

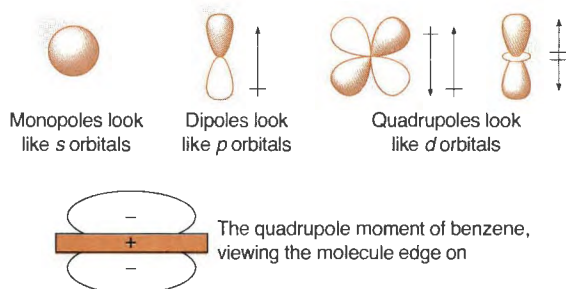


Figure 1.5
The topological relationships between atomic orbitals and electrostatic moments, and the quadrupole moment of benzene.

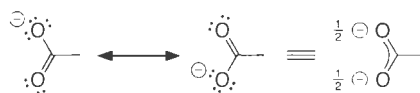
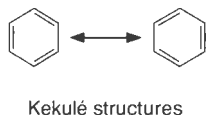
ing in opposite directions—the topology of a d_{z^2} orbital. Actually, this arrangement is more important in organic chemistry, because it is present in benzene (see below). The multipole expansion series—monopole, dipole, quadrupole, etc.—is *not* a perturbation series. It is not true that quadrupoles are somehow intrinsically weaker than dipoles in electrostatic interactions. In fact, in some important organic molecular recognition phenomena, quadrupoles prove to be stronger than dipoles (see Chapter 3).

The most common and important quadrupole moment in organic chemistry is that of benzene. Experimental measurements have determined that benzene has a large quadrupole moment, with a charge distribution as in Figure 1.5 (see Appendix 2). Just as with molecular dipole moments, we can rationalize a molecular quadrupole moment as a sum of bond dipoles. In this case, we add six $C^{\delta-}-H^{\delta+}$ dipoles to get the molecular quadrupole. The existence of a large, permanent quadrupole moment in benzene is *unambiguous proof* that an sp^2 C is more electronegative than H. We must have six $C^{\delta-}-H^{\delta+}$ dipoles to explain the effect. Note that cyclohexane has a negligible quadrupole moment, indicating that an sp^3 C and H have similar electronegativities.

1.1.10 Resonance

The bonding model we have developed thus far is quite “classical”, relying on fairly simple notions, such as Lewis structures. Some structures, however, cannot be adequately described by a single Lewis structure. In these cases, two or more Lewis structures are drawn, and the actual molecule is a hybrid or mixture of these **resonance structures**. The superposition of two or more Lewis structures to describe the bonding in a molecule is called **resonance** (also known as **mesomerism** in very old literature).

A classic example of resonance occurs for acetate. Two structures showing different positions for the double bond and the negative charge are possible. In this case the two structures are identical, and the charge on each oxygen is $-\frac{1}{2}$. Another familiar case is benzene, which also involves two equivalent structures, so that the C–C bond length is appropriate for a bond order of 1.5. All six bonds are equivalent and are represented by two equivalent resonance structures (called **Kekulé structures**).

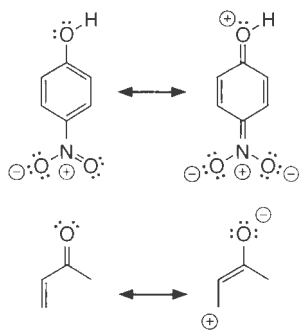


Acetate resonance structures

Resonance structures are not separate molecules that are interconverting. There is really only one structure, which is best thought of as a hybrid of the various resonance structures. The two C–O bonds of acetate are equivalent and the negative charge is distributed equally between the two oxygens. Often, one symbolizes a combination of the resonance structures by a single structure meant to describe the hybrid.

Although the examples of acetate and benzene are ones in which the resonance structures are equivalent, this is not usually the case. For example, *p*-nitrophenol and methylvinylketone also have reasonable resonance structures, but they are significantly different in the arrangement of the bonds, lone pairs, and charges.

The picture of resonance implies that the electrons are covering a larger number of atoms than given by any one resonance structure, and this is defined as **delocalization**. Generally speaking, the more resonance structures that a molecule has and the more reasonable these structures are, the more stable the molecule. The energy of stabilization imparted by resonance is called the **resonance energy** or **delocalization energy**. The reason that a molecule with resonance structures is considered more stable is the effect of delocalization. As we will see in Chapter 14, the more spread out the orbital that electrons occupy, the lower the energy of those electrons. This is related to a calculation often covered in physical chemistry classes, called the “particle-in-a-box” calculation, which we briefly review in the following Going Deeper highlight.



Acceptable resonance structures

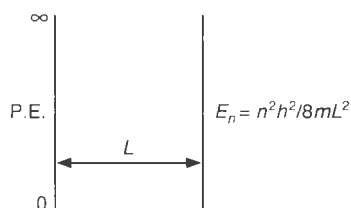
Going Deeper

Particle in a Box

The manner in which one finds the energy of electrons is to solve the Schrödinger equation ($H\Psi = E\Psi$). As a very simple example, imagine an electron in a one-dimensional “box”. Here, the potential energy of the electron is zero if the electron is within the box, but is infinite at the edges and beyond the edges of the box. The potential energy cannot be infinite, so the electron is confined to the region within the box. The solutions (E_n) to the Schrödinger equation for this scenario are very simple, and take the form shown (consult any undergraduate physical chemistry textbook to see how the solutions are derived). The possible energies are quantized (n is an integer, 1, 2, 3, . . .), with the length of the box (L) in the denominator (m is the mass of the electron and h is Planck’s constant). As the box gets bigger, the energy decreases.

The “box” is an analogy to an orbital. With an orbital the electrons have their greatest probability in certain

regions of space. The lesson is that if the electrons are allowed to occupy a larger amount of space, their energy decreases. Specifically, the kinetic energy of the electrons drops, which will be a key issue we discuss in Chapter 14. Because resonance yields a picture of bonding that spreads the electrons out in space, it is a stabilizing concept.



Parameters for the particle in a box

In order to consider what are appropriate resonance structures for a molecule, we first draw all the possible Lewis structures. In these structures only electrons are allowed to “move around”. *The positions of the nuclei never change.* The Lewis structures can have the maximum number of electrons appropriate for each atom (for example, eight for second-row atoms) or fewer electrons. Next, judgment must be made about which resonance structures are reasonable, and Figure 1.6 gives some guidance. Factors that contribute to making a particular resonance structure acceptable include having a noble gas configuration for the atoms, a maximum number of covalent bonds, a minimum number of like charges, close proximity of unlike charges, and placement of negative charges on electronegative atoms. Not all these guidelines need to be met to make a reasonable resonance structure. Many reasonable structures may contribute only a little to the true electronic structure of the molecule, depending upon just how reasonable they are. Yet, the identification of all reasonable resonance structures imparts information about polarity and polarizations in a molecule. As already mentioned, it is generally true that the larger the number of reasonable resonance structures associated with a molecule, the more stable it is. In addition, resonance is especially favorable when it involves two or more equivalent resonance structures (as, for example, with acetate and benzene), and when all second-row atoms have a full complement of eight valence electrons.

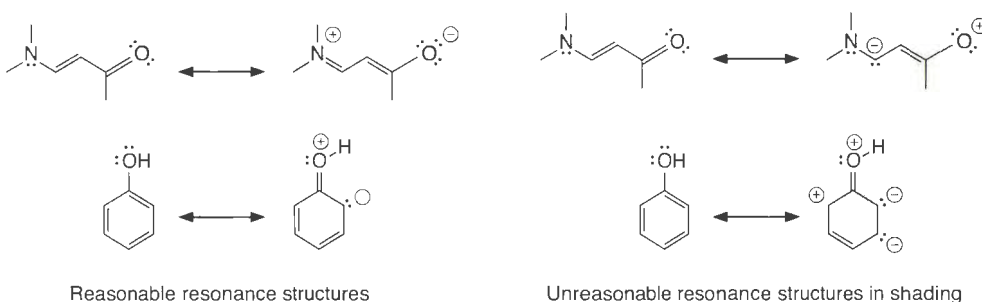
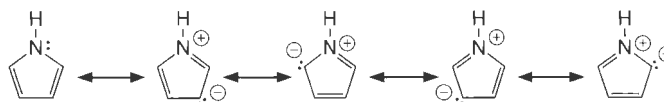


Figure 1.6
Reasonable and unreasonable resonance structures.



Resonance structures of pyrrole

Often we will see an atom adopt a nonstandard hybridization in order to maximize resonance. This most commonly arises when an atom has a lone pair of electrons that is in conjugation with (directly bonded to) a π system. For example, consider pyrrole, shown above. The nitrogen atom would seem at first glance to best be described as an sp^3 hybrid, because four groups are attached to it: two N–C bonds, an N–H bond, and a lone pair. However, in order to accommodate the resonance structures shown, four of which have a double bond between N and a C, the lone pair must be in a p orbital, not an sp^3 hybrid orbital. This requirement makes the nitrogen atom sp^2 hybridized, which is experimentally supported by the fact that the N is trigonal planar. Such resonance effects on hybridization are common and should be routinely looked for when assigning hybridization to various atoms. Another example is the hybridization of N in an amide, which should be classified as sp^2 . However, as shown in the following Connections highlight, the validity of resonance in an amide has recently come into question.

1.1.11 Bond Lengths

The simple bonding model we have developed thus far can rationalize many geometrical features. We have already discussed bond angles, and most organic chemists know what standard angles are for organic structures. It is generally true that it is more difficult to distort bond lengths from standard values than it is to bend angles. Nevertheless, it is worthwhile to know some standard bond lengths, as a significant deviation from these values is a clear indication of substantial strain in a molecule or some non-standard bonding situation (see Chapter 2). In addition, we need to know bond lengths to evaluate bond and molecular dipoles.

Several trends can be gleaned from the series of average bond lengths listed in Table 1.4.

Table 1.4
Typical Bond Lengths of Some Covalent Bonds*

Bond	Length (Å)	Bond	Length (Å)
Single bonds		Double bonds	
$C(sp^3)-C(sp^3)$	1.53–1.55	$C(sp^2)-C(sp^2)$ alkenes	1.31–1.34
$C(sp^3)-C(sp^2)$	1.49–1.52	$C(sp^2)-C(sp^2)$ arenes	1.38–1.40
$C(sp^2)-C(sp^2)$ conjugated	1.45–1.46	$C(sp^2)-O(sp^2)$ aldehydes and ketones	1.19–1.22
$C(sp^2)-C(sp^2)$ nonconjugated	1.47–1.48	$C(sp^2)-O(sp^2)$ esters	1.19–1.20
$C(sp)-C(sp)$	1.37–1.38	$C(sp^2)-O(sp^2)$ amides	1.225–1.24
$C(sp^3)-O(sp^3)$ ethers	1.42–1.44	$C(sp^2)-N(sp^2)$ imines	1.35
$C(sp^3)-N(sp^3)$ amines	1.46–1.48	Triple bonds	
C–F	1.39–1.43	$C(sp)-C(sp)$ alkynes	1.17–1.20
C–Cl	1.78–1.85		
C–Br	1.95–1.98		
C–I	2.15–2.18		
$C(sp^3)-H$	1.09–1.10		
$C(sp^2)-H$	1.075–1.085		
$C(sp)-H$	1.06		
N–H	1.00–1.02		
O–H	0.96–0.97		

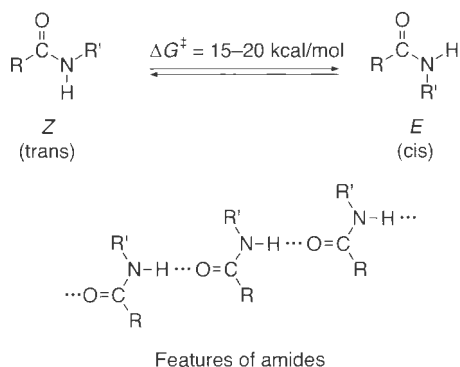
*Allen, F. H., et al. "Tables of Bond Lengths Determined by X-ray and Neutron Diffraction. Part 1. Bond Lengths in Organic Compounds." *J. Chem. Soc., Perkin Trans. II*, S1–S19 (1987).

Connections

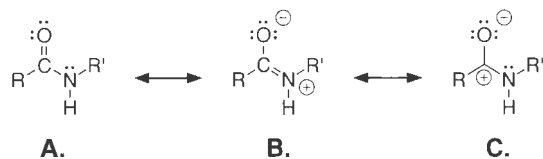
Resonance in the Peptide Amide Bond?

A foundation of structural biology and bioorganic chemistry is the peptide bond, the link between consecutive amino acids in a protein. Interestingly, the modern view of this prototype structure is evolving, and as such it provides an excellent example of various rationalizations of structure and bonding.

The **peptide bond** is an amide formed between a primary amine and a carboxylic acid. The most important structural features are (a) the amide group is planar, with a substantial barrier to rotation (typically in the range of $\Delta G^\ddagger = 15\text{--}20$ kcal/mol); and (b) there is a significant preference for the *Z* conformer (termed “trans” in the protein literature), which places the N–H and C=O bonds trans to each other. Another key feature of amides is their strong propensity toward hydrogen bonding. Two key protein secondary structural elements—the α -helix and the β -sheet—depend on the amide group to act as both a hydrogen bond donor and a hydrogen bond acceptor.

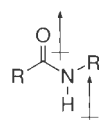


In the traditional model of an amide, resonance is the key concept. As shown below, one can write a reasonable resonance structure for an amide that places a double bond between the C and the N (structure **B**). This “double-bond character” leads to a planar structure, and hindered rotation about the C–N bond. To the extent that electrostatic interactions control hydrogen bond strengths (see Chapter 3), the charges implied by resonance structure **B** suggest strong hydrogen bonding in amides, as is observed.



More recent work, however, has suggested a refinement of the traditional model. Modern computational methods allow detailed analyses of molecular and electronic structure. These tools were used to look in detail at amide rotation. The classical resonance model predicts C–N double-bond character and C–O single-bond character in a planar amide. Since resonance is destroyed on rotating 90° about the C–N bond, we would expect the C–N bond to lengthen and the C–O bond to shorten in the perpendicular form. Calculations do show that the C–N bond lengthens upon rotation, but the C–O bond length is essentially unchanged. Also, examination of the charges calculated on the atoms of a typical amide do not support structure **B**. The N of an amide is not significantly more positive than the N of an amine, for which such resonance is not possible. Also, the O of an amide is not significantly more negative than the O of a ketone. Finally, the C of an amide is quite positive, just like the C of a ketone.

These observations suggested that the simple two-structure resonance model was inadequate. Instead three structures seem necessary, as shown. Both resonance structures **B** and **C** are considered to be of major importance for the rotation barrier. In this view, the role of the oxygen is to polarize the C–O bond, introducing a large δ^+ on carbon (resonance structure **C**). The N does not develop a significant δ^+ , despite the implications of resonance structure **B**. Instead, this model emphasizes dipole interactions—the C–O dipole is aligned for a favorable interaction with the N–H bond dipole. Note that a δ^+ on N would not be consistent with this picture.



Amide bond dipoles

This refined resonance model was inspired by high-level calculations, and it reflects changing views on the importance of bond dipoles (i.e., $C^{\delta+}-O^{\delta-}$) in structure and reactivity. Bond dipoles are also important in rationalizing the *Z*–*E* preference. There are two strong bond dipoles in a secondary amide: $C^{\delta+}-O^{\delta-}$ and $H^{\delta+}-N^{\delta-}$. As shown, the *Z* form aligns these two dipoles favorably (the positive end near the negative end), but the *E* form would produce an unfavorable alignment. Finally, the strong propensity of the peptide bond toward hydrogen bonding is readily understood based on the charge distribution implied by the resonance and bond dipole arguments.

Wiberg, K. B. “The Interaction of Carbonyl Groups with Substituents.” *Acc. Chem. Res.*, **32**, 922–929 (1999).

One obvious trend is that multiple bonds tend to be shorter than single bonds. The bond lengths in arenes are between C–C and C=C bond lengths, just as predicted by resonance theory. Bonds involving atoms with larger atomic radii (further down in a column and / or to the left within a row in the periodic table) are longer.



The state of hybridization affects bond lengths; more *s* character in the hybrid decreases the bond length. This effect is another manifestation of the fact that *s* orbitals lie closer to the nucleus than *p* orbitals. This last effect is sometimes under appreciated. For example, the fact that the C2–C3 bond of 1,3-butadiene is much shorter than a typical single bond in an alkane, 1.48 Å vs. 1.54 Å, might lead to the conclusion that there is some double-bond character in C2–C3 (see the resonance structure in the margin, which is sometimes taught in introductory organic chemistry). However, Table 1.4 shows that this is primarily a hybridization effect, as systems that are not planar (nonconjugated and hence have no double-bond character) but are *sp*²–*sp*² single bonds, show comparably short bond lengths. Further support that the conjugation in butadiene has only a small effect on the C2–C3 bond length comes from the rotation barrier about this bond, which is only on the order of 4–5 kcal/mol (see Chapter 2 for an expanded discussion of rotational barriers).

Bond lengths can also be understood with reference to the radii of the constituent atoms. There are actually three different kinds of radii that we use to understand molecular dimensions: **covalent**, **ionic**, and **van der Waals**. The covalent radius is half the distance between two identical atoms bonded together. For example, the C–C bond length in ethane is 1.54 Å, and therefore the covalent radius of C is 0.77 Å. The ionic radius of an atom is its size as determined in the crystal lattice of various salts. The radius depends upon the charge on the ion, and is smaller as the positive charge increases and larger as the negative charge increases. Finally, the van der Waals radius of an atom is the effective size of the electron cloud around an atom when in a covalent bond, as perceived by an atom to which it is *not* attached. The measurement comes from an analysis of crystal packing, and effectively sets the steric size of an atom or a group. Table 1.5 shows just a few covalent, ionic, and van der Waals radii. There is some debate as to the exact values of some of these, especially the van der Waals radii, but the basic trends are clear. Using these values molecular surface areas and molecular volumes can be calculated.

Table 1.5
Covalent, Ionic, and van der Waals Radii of Select Atoms (Å)*

Atom	Covalent	VDW	Ion	Ionic
C	0.77	1.68		
H	0.30	1.11	H	2.08
N	0.70	1.53		
O	0.66	1.50		
F	0.64	1.51	F [−]	1.36
Cl	0.99	1.84	Cl	1.81
Br	1.14	1.96	Br [−]	1.95
I	1.33	2.13	I	2.16

*Pauling, L. (1960). *The Nature of the Chemical Bond and the Structure of Molecules and Crystals; an Introduction to Modern Structural Chemistry*, 3d ed., Cornell University Press, Ithaca, NY.

1.1.12 Polarizability

An important property of molecules that we have yet to discuss is polarizability. Since electrons are charged particles they respond to electric fields. In particular, electrons in molecules are mobile to varying degrees, and so their positions can shift to differing extents in response to an applied electric field. The electron cloud is said to become polarized in response to the electric field. The ability of the electron cloud to distort in response to an external field is known as its **polarizability**. Upon distortion, a dipole is typically induced in the molecule, adding to any permanent dipole already present.

Table 1.6
Atomic and Molecular Polarizabilities (α , $\text{cm}^3/10^{-24}$)*

Atomic polarizabilities							
H	0.6668						He 0.205
	C 1.76	N 1.10	O 0.802	F 0.557			
		P 3.13	S 2.90	Cl 2.18			
				Br 3.05			
				I 4.7 (or 5.35)			
Selected molecular polarizabilities							
CH ₄	2.6	NH ₃	2.21	H ₂ O	1.45	H ₂ S	3.8
CO ₂	2.91	CS ₂	8.8	CF ₄	3.84	CCl ₄	11.2
C ₂ H ₂	3.6	C ₂ H ₄	4.25	C ₂ H ₆	4.45	CH ₃ OH	3.23
Benzene	10.32	Cyclohexene	10.7	Cyclohexane	11.0		

*CRC Handbook of Chemistry and Physics, D. R. Lide (ed.), CRC Press, Inc., Boca Raton, FL (1990–1), pp. 10-193–10-209.

We define the polarizability of a molecule as the magnitude of the dipole induced by one unit of field gradient, which works out to be in units of volume. Often, the larger the volume occupied by the electrons, the more polarizable those electrons. Since an electric field gradient is directional, it can encounter a molecule or bond in different ways depending upon the relative orientation between the field and the molecule or bond. Hence, polarizabilities are often broken down into one longitudinal (along the bond or molecular axis) and two transverse (perpendicular to the axis) values. Table 1.6 lists some atomic and molecular polarizabilities, for which the directional components have been averaged. Just as with permanent dipoles, the induced dipoles that arise from such polarizations can be analyzed as molecular dipoles or as bond dipoles. We can thus speak of molecular / atomic polarizabilities, as well as of bond polarizabilities.

Several trends are evident in Table 1.6. First, as we move left to right across a row of the periodic table, polarizability decreases. This trend is clear in atomic polarizabilities ($\text{C} > \text{N} > \text{O} > \text{F}$) and molecular polarizabilities ($\text{CH}_4 > \text{NH}_3 > \text{H}_2\text{O}$). Electronegativity plays an important role. Atoms that hold on to their electrons tightly are not polarizable.

Polarizability has profound consequences. Water is a very polar molecule—it has a large dipole moment. However, methane is much more polarizable than water, and alkanes in general are among the most polarizable of molecules. Thus, while aqueous media provide a very polar environment, alkanes and other hydrophobic environments are more polarizable.

The second clear trend in Table 1.6 is that as we move down a column in the periodic table, polarizability increases substantially ($\text{S} > \text{O}$, $\text{P} > \text{N}$, and $\text{H}_2\text{S} > \text{H}_2\text{O}$). This observation explains an earlier apparent contradiction. According to Table 1.1, I is not significantly more electronegative than C, yet C–I bonds are much more reactive than C–Cl bonds in reactions like $\text{S}_\text{N}2$ and E2. In such instances, $\text{C}^\delta- \text{X}^\delta$ polar bonds are often invoked, but C–I bonds are not very polar. However, dramatic changes in molecular and electronic structure occur in the course of a reaction. A model for just the ground state of the reactant is often inadequate. When an anionic nucleophile approaches a C–X bond for an $\text{S}_\text{N}2$ reaction, it can induce a large bond dipole, especially if X is highly polarizable. Thus, the large polarizability of I makes up for the lower electronegativity, and the C–I bond is more reactive than C–Cl.

Another perhaps surprising point is revealed when considering the polarizabilities of the simple hydrocarbons given in Table 1.6. Because of their greater reactivity, most chemists might assume that alkenes are more polarizable than alkanes, but the opposite is true. Once again, alkanes are among the most polarizable of molecules. In particular, ethane is significantly more polarizable than ethylene. Perhaps even more surprisingly, cyclohexane is more polarizable than benzene. While perhaps counterintuitive, these trends are consistent with electronegativity arguments. Alkenes and arenes, with the more electronegative sp^2 carbons, are less polarizable than alkanes with only sp^3 carbons.

1.1.13 Summary of Concepts Used for the Simplest Model of Bonding in Organic Structures

The material presented to this point has been mostly a review of concepts presented in introductory organic chemistry classes. Only in a few cases was the discussion extended further. This review sets the stage for us to go deeper into bonding and its effects on structure, and a deeper analysis is what we present in the remainder of this chapter. First, however, it is instructive to summarize the concepts used in the simplest picture of bonding in organic structures; many of these concepts extend nicely into our second approach to bonding.

Hybridization. Atoms contribute consistent sets of hybrid atomic orbitals to the bonding in a molecule. A carbon with four ligands is sp^3 hybridized, a C with three ligands is sp^2 hybridized, and a C with two ligands is sp hybridized.

σ and π Bonds. Hybrid orbitals on two separate atoms overlap in the region between the adjacent atoms to create σ bonds. The overlap of p orbitals on adjacent atoms creates π bonds. The localized bonds consist of bonding and antibonding molecular orbitals, of which only the bonding orbitals are populated with electrons. This arrangement of bonds is a VBT notion, with discrete and localized bonds between adjacent atoms, but the orbitals are created using MOT mixing notions.

Resonance. For certain molecules, a single valence bond structure cannot properly describe the bonding, so resonance is involved. Resonance structures show various arrangements of electrons within a structure, where each contributes to the bonding arrangement in that molecule. Resonance structures can also be used to suggest subtle features of the electronic structure of functional groups.

Electronegativity and bond polarization. Electronegative elements pull electron density toward themselves. This introduces polarity into bonds, resulting in bond dipoles and molecular dipoles.

Induction. Electronegativity and polarization effects can be felt a few bonds away through an inductive effect.

Polarizability. Polarizability reflects the extent to which electrons can be perturbed from the standard bonding arrangement in response to the approach of another molecule. This effect is important in understanding solvent properties and many reactivity patterns.

Sterics. Two or more groups tend to stay away from each other due to adverse electrostatic repulsion between filled orbitals.

We draw upon each of these notions as necessary to explain various aspects of bonding, reactivity, and structure. We will add solvation effects later in the book. For particular reactions or structures, we may have to refine and/or combine these notions to get the optimal model for the molecule. In some cases we may even need to completely re-think and modify these foundations. However, there is often a tendency in physical organic chemistry to become quite focused on the “exceptions to the rule”. We should keep in mind that the vast majority of organic structures are well described by this simple model.

1.2 A More Modern Theory of Organic Bonding

In Section 1.1 we described the basic bonding patterns of organic molecules and the properties of different types of localized bonds. The concepts introduced were mostly fairly classical valence bond concepts. They have definite predictive power, and that is an important measure of the value of any model. However, the picture we are about to present also has the

same predictive power, but can better explain certain structural issues and experimental observations. In addition, understanding chemical reactivity poses a serious challenge, and this second approach to bonding is very good at predicting the reactivity patterns of organic molecules.

The highest extent of rigor that can be given for bonding is presented in the advanced quantum mechanical analysis that was developed in the first half of the 20th century. Modern calculational methods now provide accurate representations of the molecular orbitals not only of stable molecules, but also of reactive intermediates and even transition states. These powerful computational methods will be described in detail in Chapter 14. Our goal in this chapter is to develop an *understanding* of chemical bonding, and the detailed numerical output of a quantum mechanical calculation is not enough. Here, we will show that certain key concepts and trends that result from the output of such calculations lead to a more rigorous descriptive model of organic bonding than given in the first part of this chapter. Molecular orbital theory (MOT) forms the core of this second model. However, we will see that certain key concepts from the valence bond theory based model for bonding will still be very useful (e.g., sterics, induction, and polarizability), and in fact the distinctions between the two theories are often blurred. Therefore, we again will present a hybrid VBT/MOT model, but now more tilted toward MOT.

Although the starting points for VBT and MOT are different, and the initial mathematics used to explain them are different, both theories can be shown to give similar results when correction factors are added and their respective mathematics are solved completely in a manner consistent with quantum mechanical rules. In addition, straightforward and theoretically justifiable mathematical operations can, in most cases, transform the fully or partially delocalized molecular orbitals we are about to examine into localized orbitals that clearly resemble our VBT notions of discrete and localized bonds. MOT and VBT are not as far apart as they may seem. Hence, one theory is not necessarily *more correct* than the other. Also, there is nothing wrong with a hybrid approach to bonding. Both MOT and VBT are approximations to the “true” answer—a full solution to the Schrödinger equation. A combination of the two is no less approximate, as long as it is thoughtfully applied.

1.2.1 Molecular Orbital Theory

In contrast to VBT, “full-blown” MOT considers the electrons in molecules to occupy molecular orbitals that are formed by linear combinations (addition and subtraction) of *all* the atomic orbitals on *all* the atoms in the structure. In MOT, electrons are not confined to an individual atom plus the bonding region with another atom. Instead, electrons are contained in MOs that are highly delocalized—spread across the entire molecule. MOT does not create discrete and localized bonds between neighboring atoms. An immediate benefit of MOT over VBT is its treatment of conjugated π systems. We don’t need a “patch” like resonance to explain the structure of a carboxylate anion or of benzene; it falls naturally out of the delocalized nature of the MOs. The MO models of simple molecules like ethylene or formaldehyde also lead to bonding concepts that are pervasive in organic chemistry.

On the other hand, the fully delocalized view of MOT that is so useful for certain molecules like benzene is not so useful in other organic molecules. For example, if we want to anticipate the reactivity of 3-heptanone, we really don’t need to consider orbitals that span all seven carbons plus the oxygen. Chemical experience tells us that the action is at the carbonyl, and we want to be able to focus on the π system and lone pairs. In order to do so, we will use an MO approach to establish certain fundamental bonding principles by studying small prototype molecules—ethylene for learning about C=C bonds, formaldehyde for C=O, acetic acid for carboxylic acids, etc. This analysis gives us the MOs for these functional groups. Fortunately, analysis of the computed MOs of complex systems shows that the MOs of the smaller model molecules appear with only minor modifications in the larger molecules. In other words, we can concentrate our analysis on the molecular orbitals of functional groups to understand structure and reactivity. Thus, a key concept we present below is that of **group orbitals**—orbitals that are delocalized only over a defined group of atoms. If we need to understand the molecular orbitals of the entire molecule, we combine these group

orbitals to create molecular orbitals for the entire molecule, and the rules for this orbital mixing procedure are presented in the following discussions.

To create group orbitals or delocalized molecular orbitals, we need to understand how to combine atomic orbitals properly. Therefore, the starting point in developing our second model of organic bonding is a set of rules that lead by inspection to group orbitals and molecular orbitals. This procedure is called qualitative molecular orbital theory (QMOT).

1.2.2 A Method for QMOT

The protocol we will follow was developed by many workers, including Hoffmann and Salem. An especially succinct statement of the procedure embodied in a series of rules was provided by Gimarc, and is presented in Table 1.7. Below we give several examples of using rules 1–13, where rules 14 and 15 become more important in future chapters. Several MOT and quantum mechanical concepts go into the origin of these rules. For example, the orbital mixing follows the well-developed rules of perturbation theory, which we will often discuss (glance back to Section 1.1.7 for our first quick look at mixing). Furthermore, the symmetry of the molecule (see the next Connections highlight) guides the creation of the MOs, a notion that has a quantum mechanical origin discussed in Chapters 14 and 15.

Table 1.7
The Rules of QMOT*

1. Consider valence orbitals only.
2. Form completely delocalized MOs as linear combinations of s and p AOs.
3. MOs must be either symmetric or antisymmetric with respect to the symmetry operations of the molecule.
4. Compose MOs for structures of high symmetry and then produce orbitals for related but less symmetric structures by systematic distortions of the orbitals for higher symmetry.
5. Molecules with similar molecular structures, such as CH_3 and NH_3 , have qualitatively similar MOs, the major difference being the number of valence electrons that occupy the common MO system.
6. The total energy is the sum of the molecular orbital energies of individual valence electrons.
7. If the two highest energy MOs of a given symmetry derive primarily from different kinds of AOs, then mix the two MOs to form hybrid orbitals.
8. When two orbitals interact, the lower energy orbital is stabilized and the higher energy orbital is destabilized. The out-of-phase or antibonding interaction between the two starting orbitals always raises the energy more than the corresponding in-phase or bonding interaction lowers the energy.
9. When two orbitals interact, the lower energy orbital mixes into itself the higher energy one in a bonding way, while the higher energy orbital mixes into itself the lower energy one in an antibonding way.
10. The smaller the initial energy gap between two interacting orbitals, the stronger the mixing interaction.
11. The larger the overlap between interacting orbitals, the larger the interaction.
12. The more electronegative elements have lower energy AOs.
13. A change in the geometry of a molecule will produce a large change in the energy of a particular MO if the geometry change results in changes in AO overlap that are large.
14. The AO coefficients are large in high energy MOs with many nodes or complicated nodal surfaces.
15. Energies of orbitals of the same symmetry classification cannot cross each other. Instead, such orbitals mix and diverge.

*Adapted, with modifications, from Gimarc, B. M. (1979). *Molecular Structure and Bonding: The Qualitative Molecular Orbital Approach*, Academic Press, New York.

Connections

A Brief Look at Symmetry and Symmetry Operations

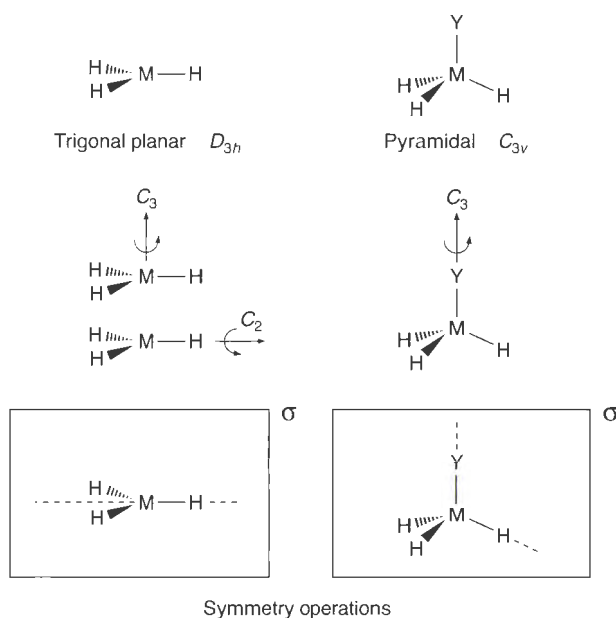
It is clear from Table 1.7 that symmetry plays an important role in QMOT because the concept is used in rules 3, 4, 7, and 15. A full explanation of symmetry operations, point groups, and their relationships to orbitals is beyond the scope of this book (see the reference at the end of this highlight for an excellent discussion). However, we will look at proper and improper rotations in Chapter 6. To understand the QMOT examples discussed below, you should be aware of the symmetries inherent in only two geometries: trigonal planar with all three groups the same, and tetrahedral where one group is different than the other three (also called **pyramidal**). Hence, we discuss these two systems briefly.

A trigonal planar structure with three equivalent ligands belongs to the point group called D_{3h} . The structure possesses a C_3 axis perpendicular to the plane of the molecule that passes through the central atom. It also possesses three C_2 axes perpendicular to the central C_3 axis, one along each M–H bond vector (only one is shown to the right). To possess a C_3 axis and a C_2 axis means that the molecule can be rotated along this axis by 120° and 180° , respectively, returning exactly the same structure with atoms returned to identical positions in space. A D_{3h} structure also possesses three internal mirror planes, called σ planes. They contain the C_3 axis and lie along each M–H bond (only one is shown), where reflection of the atoms through these mirror planes likewise gives back the same structure.

The MH_3Y structure belongs to the point group called C_{3v} . This structure possesses one C_3 axis, no C_2 axes, but does have three σ planes (one defined by each H–M–Y plane; only one is shown).

Rules 3 and 7 address the symmetry properties of the orbitals. With respect to each symmetry operation, an orbital must be symmetric or antisymmetric. For an orbital to be symmetric, it must be unchanged by the sym-

metry operation. To be antisymmetric, all signs (phases) of the orbital must be reversed by the symmetry operation. Rule 4 speaks of “high symmetry” and “less symmetric structures”. While these are not precisely defined terms, their meaning in context is usually clear. Certainly, the D_{3h} structure is of higher symmetry than the C_{3v} . Rule 15 is not one we are going to need until Chapter 15, so it will be addressed at that time. The use of these rules and the notions of symmetry will hopefully become evident with the examples given below, and with the practice given in the end-of-chapter Exercises.



Cotton, F. A. (1971). *Chemical Applications of Group Theory*, 2nd ed., Wiley Interscience, New York.

1.2.3 Methyl in Detail

To illustrate the procedures of QMOT, we will “build” the MOs of planar CH_3 from scratch. We are choosing a reactive species such as this, because we want a simple structure as an easy starting point. To begin with, we will not worry as to whether we have methyl cation, anion, or radical, for, as we will describe later, the orbital diagram is not overly sensitive to such distinctions. For now, let’s just make the MOs of planar CH_3 ; we’ll put the electrons in later. The starting point is planar CH_3 , because pyramidal CH_3 is of lower symmetry (rule 4).

Planar Methyl

Because of rule 1, the orbitals we have to work with for methyl are simple: three hydrogen $1s$ orbitals, a carbon $2s$ orbital, and three carbon $2p$ orbitals. Next, we form the delocalized MOs using the other rules, and those are shown in Figure 1.7. How this is done may at first seem mysterious, but with a little practice it should become clear. For example, using

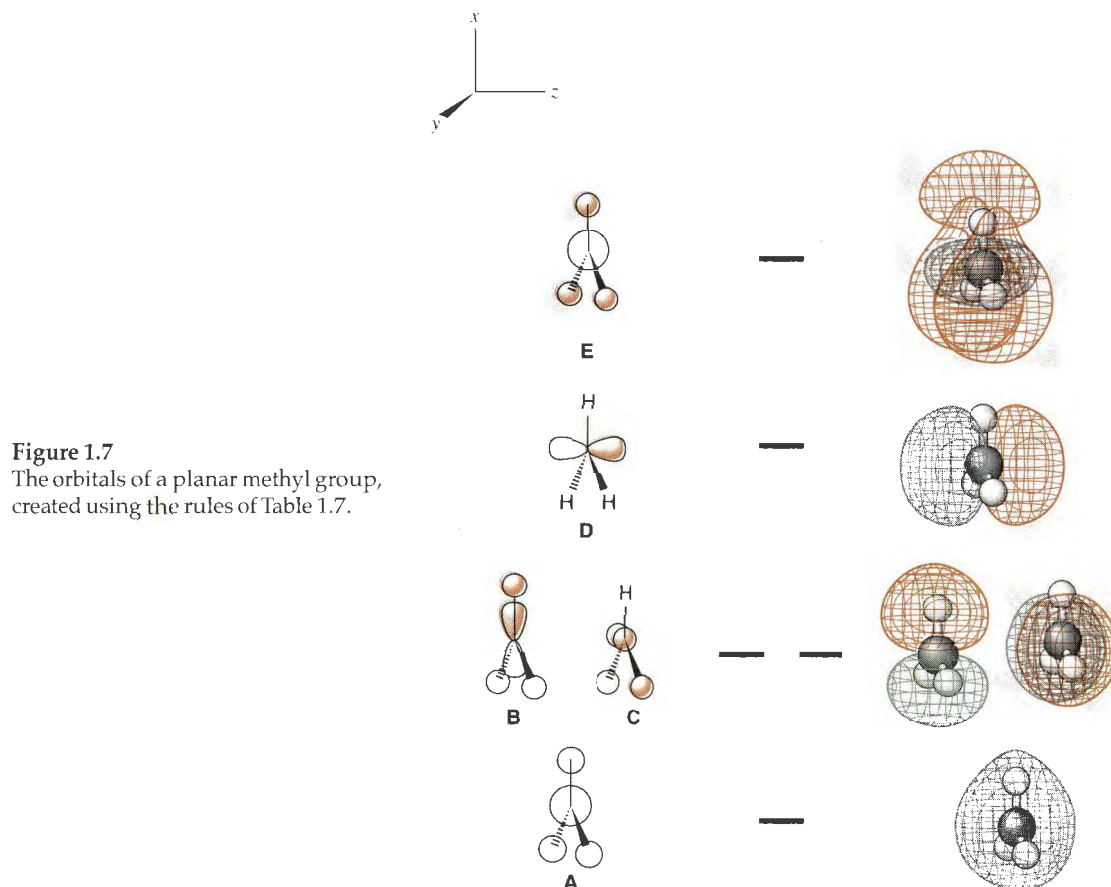


Figure 1.7
The orbitals of a planar methyl group,
created using the rules of Table 1.7.

rules 2 and 3, we mix the carbon $2s$ orbital with the three hydrogen atomic orbitals (AOs) in-phase to produce orbital **A**. Mixing just means adding and subtracting, and we do this visually. Orbital **A** is symmetric with respect to the C_3 axis in the molecule. Our goal here is to focus on the low-lying, bonding MOs, so we mix these s orbitals in-phase, in a bonding manner. If we mix out of phase, we create a high energy, antibonding orbital, shown as **E** in Figure 1.7. Next come the carbon p orbitals. The p_z AO cannot mix with any of the hydrogens, as they all lie on the node of this orbital. We thus have a molecular orbital that is just an atomic p orbital, called a nonbonding orbital, and labeled as **D** in Figure 1.7. The p_x and p_y AOs can mix with the hydrogens to give favorable interaction patterns, as shown in **B** and **C** in the figure. We would call orbitals **A**, **B**, **C**, and **D** the group orbitals for planar methyl.

To create the group orbitals for planar methyl we used a total of seven atomic orbitals. There must be a conservation of the number of orbitals, meaning that the number of molecular orbitals created must equal the number of atomic orbitals we start with. Yet, in Figure 1.7 we only show five orbitals. Orbital **E** and the two MOs that we do not show are called **virtual orbitals**, meaning that they exist but are not typically populated with electrons (see the discussion of adding electrons below). We do not draw two of them because they are not involved in bonding schemes, except with excited electronic states.

We also show the relative energies of these MOs in Figure 1.7, with energy increasing as we go from the bottom of the diagram to the top. Orbitals **A–C** are stabilized by mixing with the H AOs, but **D** is not. The reason **A** lies below **B** and **C** is that carbon $2s$ AOs are lower in energy than $2p$ AOs, and this carries over to the MOs. MOs **B** and **C** are of equal energy—they are degenerate. In this high symmetry system, the x and y axes are equivalent, and so p_x and p_y are degenerate, as are the MOs derived from them. If you are uncomfortable with this symmetry argument, please simply accept this degeneracy as fact for now.

Figure 1.7 also shows the actual MOs of CH_3^+ , obtained from a quantum mechanical calculation. As you can see, our simple qualitative analysis has predicted the essence of the full calculation quite well. You don't actually see the individual s and p atomic orbitals, but you see the results of mixing them. Deciphering the origin of the calculated orbitals is a major goal of our analysis. We want our QMOT model to faithfully capture the essence of the actual computed results, and we will be constantly checking our qualitative reasoning against quantitative calculations to be sure we are getting things right. If the QMOT rules prove effective in reproducing calculated structures, it gives us confidence to use the method to predict other orbitals, too.

The Walsh Diagram: Pyramidal Methyl

Once we have the orbitals for a fragment, we can predict how geometrical distortions will affect the MOs, again using the rules from Table 1.7. We make the MOs of a pyramidal methyl group by distorting the planar system and following the changes as indicated in rule 4. A diagram that follows orbital energies as a function of angular distortions is called a **Walsh diagram**, named after A. D. Walsh, the theoretician who first devised the approach. Figure 1.8 shows the Walsh diagram for CH_3 , along with the results of a calculation on pyramidal methyl.

We begin with the planar system we just developed, and it is reproduced on the left side of Figure 1.8. On pyramidalization, the energy of **A** will change very little. Perhaps a slight

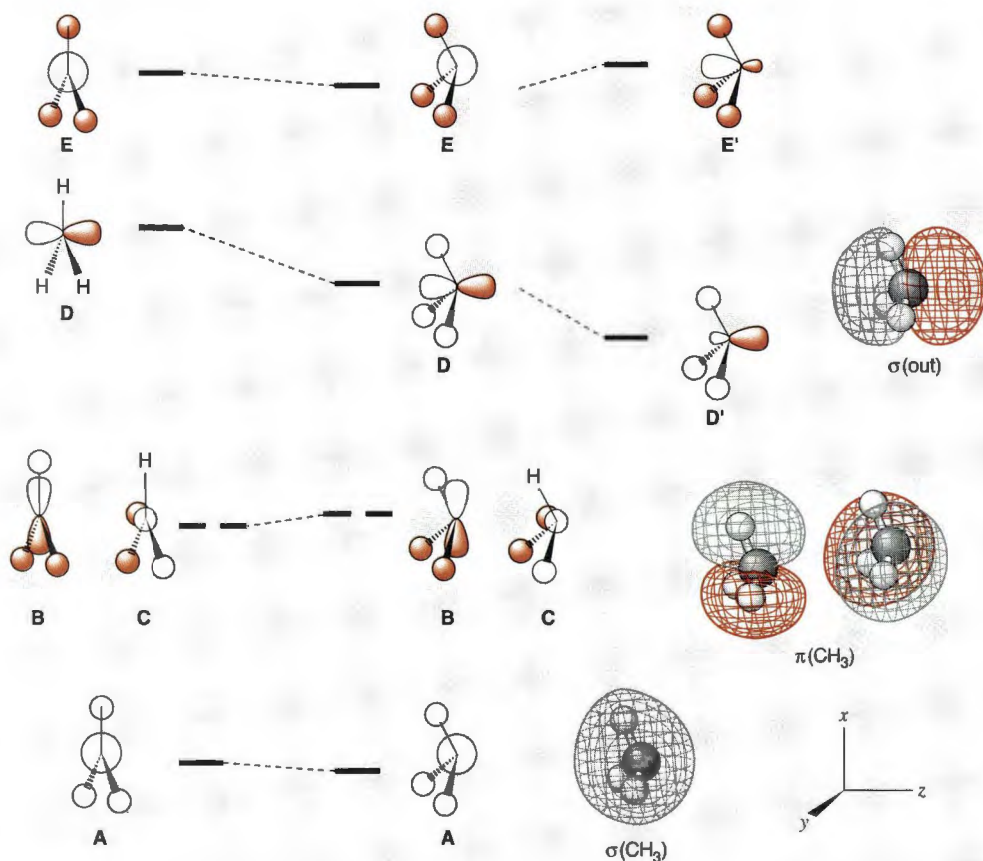


Figure 1.8

The Walsh diagram for CH_3 . Beginning with planar CH_3 on the left and moving right, pyramidalization lowers orbital **A** slightly, raises **B** and **C** more so, and lowers **D** substantially. Also shown is a secondary mixing between **D** and **E** to produce **D'** and **E'**, the final MOs of pyramidal CH_3 . Atomic orbitals are drawn to show phase relationships only; they are not to scale. Also shown are the computed MOs for the system, and the conventional labeling of the group orbitals for methyl.

energy lowering will occur because the hydrogens are closer to each other and can interact favorably. The **B/C** pair will rise in energy on pyramidalization significantly more than **A** is lowered because of the greater directionality of the bonding. In the planar form the p orbitals can point more directly at the hydrogens, maximizing overlap. This overlap diminishes as the system distorts, and this destabilizes the orbitals. Even with distortion, the **B/C** pair of orbitals remains degenerate.

The biggest effect by far of pyramidalization is on orbital **D**. A formerly nonbonding orbital becomes significantly bonding. The hydrogens have moved off the node of the p orbital, allowing favorable bonding interactions to develop. Thus, the energy of **D** is considerably lower in pyramidal vs. planar methane.

Let's look at the bonding in the planar and pyramidal forms. Orbitals **A–C** are strongly C–H bonding, whether in the planar or pyramidal forms. They can contain six electrons. In a VB model, we would want three C–H bonds, each with two electrons, for a total of six C–H bonding electrons. The two models agree. With QMOT, we still have three C–H bonds, described by three occupied MOs that are strongly C–H bonding.

We are not done with the QMOT analysis, because we have not yet considered rule 7. MOs **D** and **E** have the same symmetry, but one is based on a carbon $2p$ orbital and the other on a $2s$ orbital. Thus, rule 7 tells us to mix these two orbitals, and Figure 1.8 shows the result as **D'** and **E'**. Now **D'** looks more like a lone pair orbital, but contrary to simple Lewis diagrams, there is still significant C–H bonding in this lone pair orbital. Note also that **D'** is very much like an sp^n hybrid at carbon; it contains contributions from both the $2s$ and the $2p$, atomic orbitals. Orbital **E'** also looks like a hybrid, but points in the opposite direction from **D'**, just as is expected when one creates sp hybrids (they point in opposite directions on the same atom).

“Group Orbitals” for Pyramidal Methyl

One of the most important concepts in organic chemistry is that of the functional group, a collection of atoms and bonds that shows a consistent reactivity pattern in a wide range of molecular environments. This same notion carries over to electronic structure. Certain collections of atoms contribute a consistent set of orbitals to any molecule in which they are present. To a considerable extent these groups will correspond to the familiar functional groups, but there will be some differences. We will refer to these as **group orbitals**, a collection of partially delocalized orbitals that is consistently associated with a functional group or similar collection of atoms in molecules.

You are already familiar with a few orbitals that can be considered as group orbitals from our first VBT/MOT model of bonding. The localized bonding and antibonding C–C σ and π orbitals given in Section 1.1.7 are group orbitals. We call them group orbitals because we can use these orbitals to describe any σ and π bond, hence describing any alkane C–C bond and any C=C functional group.

As mentioned earlier, one of the goals of introducing a second model of bonding is to define the group orbitals of many types of functional groups, and many will be examined as we go along. However, the analysis just presented actually defined group orbitals also, but for a group not normally considered a functional group—methyl. We are used to thinking of a carbonyl or an olefin as a functional group, but not a methyl or methylene group. However, there are group orbitals for these groups, and we will find they play an important role in understanding many aspects of organic structure and reactivity. We will use these group orbitals to model the bonding in any molecules that contain methyl groups (see Section 1.3).

For convenience, the group orbitals for methyl are given specific descriptive names, as shown in Figure 1.8. We begin with a low-lying, C–H bonding orbital derived from the carbon $2s$ orbital. It is of σ **symmetry**, possessing no nodes along the z axis (the axis that would bond CH_3 to anything else) and is called $\sigma(\text{CH}_3)$. Next comes the C–H bonding orbitals that are derived from carbon $2p$ orbitals. There is a nodal plane that passes through the carbon and contains the z axis in each group orbital of this type, so it is sensible to consider them to be of π **symmetry**. They are called $\pi(\text{CH}_3)$ (a degenerate pair). Next comes another orbital of σ -type symmetry. It points away from the hydrogens, and so it is called $\sigma(\text{out})$.

Putting the Electrons In—The MH_3 System

As stated in rule 5 of Table 1.7, the Walsh diagram we just developed applies not just to CH_3 , but to any MH_3 , where M is a main group element. It is thus the Walsh diagram for BH_3 and NH_3 , too. The only difference is the number of valence electrons. Since B, C, and N have three, four, and five valence electrons, respectively, and the three H's contribute a total of three electrons, we must deposit into Figure 1.8 six, seven, or eight electrons, depending on whether we are considering BH_3 , CH_3 , or NH_3 , respectively. Because neutral CH_3 is a radical, we'll discuss the ramifications of Figure 1.8 on methyl when we look at the geometries of reactive intermediates (see Section 1.4).

Let's focus first on the even electron systems, BH_3 and NH_3 . The six valence electrons of BH_3 occupy MOs **A**–**C** of either the planar or pyramidal form. Now, consider which geometry, planar or pyramidal, the molecule will prefer. For BH_3 , converting the planar form to the pyramidal form will be *destabilizing*. MO **A** is slightly stabilized, but **B** and **C** will be significantly destabilized. Rule 6 states that we will model the energy as the sum of the one-electron energies—that is, as the sum of the orbital energies for occupied orbitals, including a factor of two if the MO is doubly occupied, but only a factor of one for singly occupied MOs. The net effect of pyramidalization is thus destabilizing, and so BH_3 is predicted to be planar. Note that since **D** or **D'** is empty in BH_3 , it has no influence on the geometry.

In contrast, with eight valence electrons, NH_3 is predicted to be pyramidal. Now orbital **D** is occupied in the planar form and **D'** in the pyramidal form, and the substantial stabilization of **D'** associated with pyramidalization outweighs all other considerations. Our simple model has thus made a clear prediction: BH_3 should be planar, while NH_3 should be pyramidal. Moreover, the prediction is correct and is identical to the prediction from VSEPR. However, the MOT model is rooted in the most modern theories of bonding.

1.2.4 The CH_2 Group in Detail

The methyl group was just a start. Let's now use the same procedure for the CH_2 group. Our goals are the same as they were with CH_3 : define the group orbitals and examine how electron population leads to differing structures. Figure 1.9 shows the Walsh diagram for methylene, where we consider the relationship between the linear and bent forms.

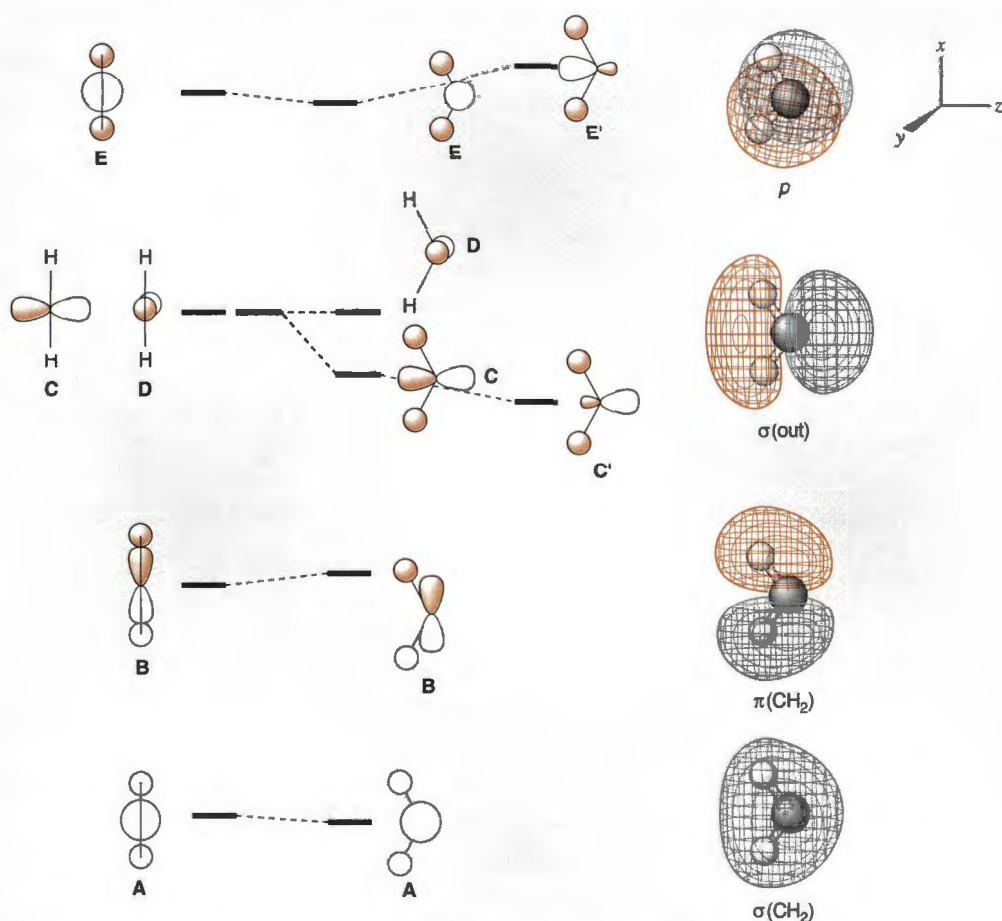
The Walsh Diagram and Group Orbitals

Beginning with the linear structure, we have a low-lying, CH bonding orbital (**A**) derived from the carbon 2s orbital. Now, only one *p* orbital can bond to the hydrogens in the linear form, leaving a degenerate pair of nonbonding *p* orbitals. On going from linear to bent, orbital **A** is slightly stabilized due to the $H\bullet\bullet\bullet H$ overlap. **B** is destabilized more, as C–H overlap drops, just as in the case with CH_3 . However, the dominant change is the drop in **C**, as it goes from nonbonding to bonding. Orbital **D** remains an isolated *p* orbital, so it is nonbonding. As in the CH_3 diagram, we expect a secondary mixing, this time between **C** and **E**, resulting in an important hybrid orbital, **C'**.

The actual MOs of the bent form are also shown in Figure 1.9. These constitute the group orbitals for CH_2 . Similar descriptors for the group orbitals are used as with methyl. We find a $\sigma(CH_2)$, now a single $\pi(CH_2)$, a $\sigma_{out}(CH_2)$, and a lone *p* orbital. Together, the $\sigma(CH_2)$ and the $\pi(CH_2)$ define the C–H bonding orbitals. The $\sigma_{out}(CH_2)$ and the *p* orbital can be used to make bonds to other groups (see Section 1.3).

Putting the Electrons In—The MH_2 System

The diagram given in Figure 1.9 applies to all MH_2 molecules, the most prominent of which is water. Because neutral CH_2 is a carbene, we'll discuss the ramifications of Figure 1.9 on the geometry of CH_2 as a discrete molecule when we analyze reactive intermediates (see Section 1.4). Here, let's briefly consider only H_2O . Water has eight valence electrons, and so both **C'** and **D** are doubly occupied. Thus, water will prefer a bent structure, in order to take advantage of the large stabilization of **C'** that occurs.

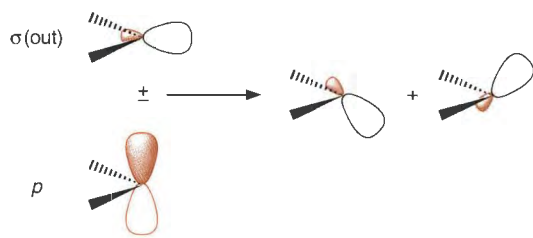
**Figure 1.9**

The Walsh diagram for CH_2 . The linear form is shown on the left, and it is converted to the bent form. Also shown is a secondary mixing between the C and E orbitals to make C' and E'. The computed MOs and the standard group orbital designations are also given.

We all know that water has two lone pairs on the oxygen, but you may be wondering where they are. In this bonding scheme the lone pairs are best thought of as MOs C' and D. In the MO model, the two lone pairs of water (and those of all ethers or other dicoordinate oxygens) are *not* equivalent sp^3 hybrids. One is of σ symmetry (C'), has some O–H bonding character, and lies significantly lower in energy than the other because it has contributions from an oxygen s orbital. The higher lying lone pair is a pure p orbital, and thus has π symmetry. This model is supported by experiment. Water in the gas phase has two ionizations corresponding to the two different oxygen lone pairs, and they are separated by 2.2 eV. In an isolated water molecule the lone pairs are not identical!

What, then, of the often invoked model in which water is treated as if it has a “tetrahedral” structure, with two equivalent lone pairs? As shown in Figure 1.10, the two equivalent lone pairs can be obtained by taking in-phase and out-of-phase combinations of the C' and D orbitals. Is this model also viable for water? For properties such as dipole moment, polarizability, etc., either bonding model is acceptable. That is because these properties depend only on the total electron density of the molecule, which is the same for either model of bonding.

However, the environment that a molecule finds itself in also influences the orbital description. Our analysis to this stage has been for a single isolated MH_2 system. Orbitals C' and D could not mix since they have differing symmetry. However, as soon as another mole-

**Figure 1.10**

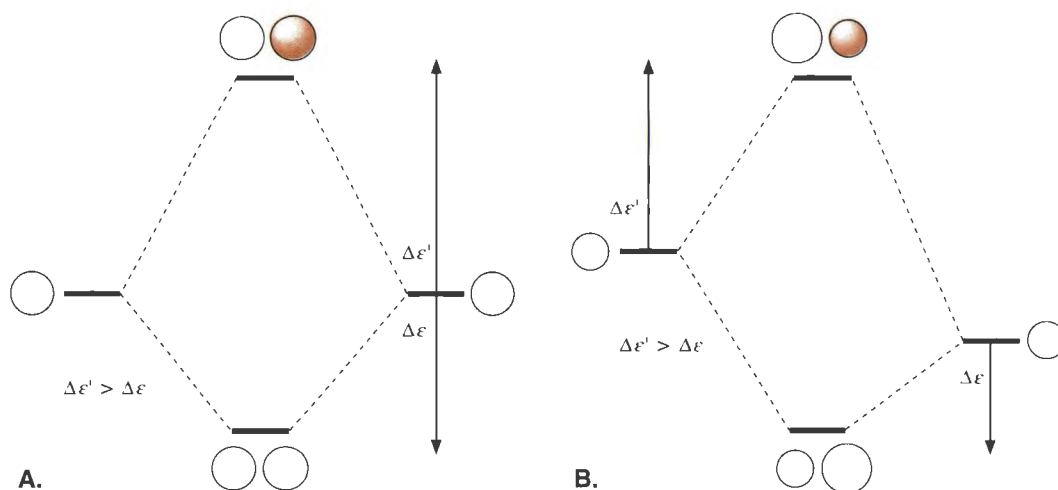
Mixing the two MOs of water that correspond to the lone pairs produces two equivalent lone pair hybrid orbitals.

cule approaches a water molecule, it lowers the symmetry of the system. Now the **C'** and **D** orbitals can mix in MOT and two sp^3 -like lone pair orbitals are obtained (as in Figure 1.10). This would happen in bulk water.

In summary, with QMOT and the Walsh diagrams for CH_3 and CH_2 , we made use of simple notions: s orbitals are lower in energy than p orbitals; strong overlap stabilizes an orbital, and weakening that overlap raises the orbital energy. In the end, though, we mixed atomic orbitals to create delocalized molecular orbitals on the groups. Mixing orbitals is a common tool, and it follows a precise protocol. As mentioned earlier, the QMOT rules were devised in part on a mixing protocol. It is now time to describe that procedure in detail. This will allow us to further develop our repertoire of group orbitals, and to begin the analysis of more standard functional groups, namely alkenes and carbonyls.

1.3 Orbital Mixing—Building Larger Molecules

The essence of orbital mixing is stated in rules 8 and 9 of Table 1.7. The rules are depicted pictorially in Figure 1.11, where we schematically mix orbitals of two separate molecules. You should already be familiar with the idea that mixing two orbitals produces an in-phase, bonding combination, and an out-of-phase, antibonding combination (Figure 1.3 and Section 1.1.7). The former is stabilized (lowered in energy), whereas the latter is destabilized (raised in energy). Figure 1.11 includes some further detail. The figure shows two mixings: Figure 1.11 **A** describes the mixing of degenerate orbitals, and Figure 1.11 **B** shows the mixing of two orbitals that start at different energies. The essential features of the two are the same.

**Figure 1.11**

Orbital mixing. Simple spherical orbitals are shown to illustrate the concept. **A.** The first-order, degenerate mixing. **B.** The second-order, nondegenerate mixing.

A key aspect of orbital mixing is that the antibonding combination is raised in energy more than the bonding combination is lowered in energy. This difference in energy is true for both mixings in Figure 1.11. If both original orbitals are doubly occupied, the resulting two orbitals will also be doubly occupied, and because two electrons are raised in energy more than the other two are lowered, the net interaction is destabilizing. *The mixing of filled orbitals on two separate molecules is always destabilizing.* This four-electron interaction is referred to as **closed shell repulsion**. A different outcome arises if we have only two electrons in the system. The two electrons can come from one in each starting orbital or two in one starting orbital with the other empty. Now the two electrons end up in the lower orbital, and the mixing is stabilizing. *It is always favorable to mix a filled orbital with an empty orbital or to mix two singly occupied orbitals.* These two aspects of orbital mixing are universal, and they will be used throughout this text.

Figure 1.11 **B** illustrates an additional feature of the nondegenerate mixing, which is summarized in rule 9. In this situation, the lower energy, bonding orbital will have a larger contribution from the original orbital that started out lower in energy. The high energy, antibonding orbital will have a larger contribution from the higher-lying original orbital. As shown in Figure 1.11 **B**, mixing produces a **polarization** of the resulting orbitals. This will be a very important feature in subsequent orbital mixings.

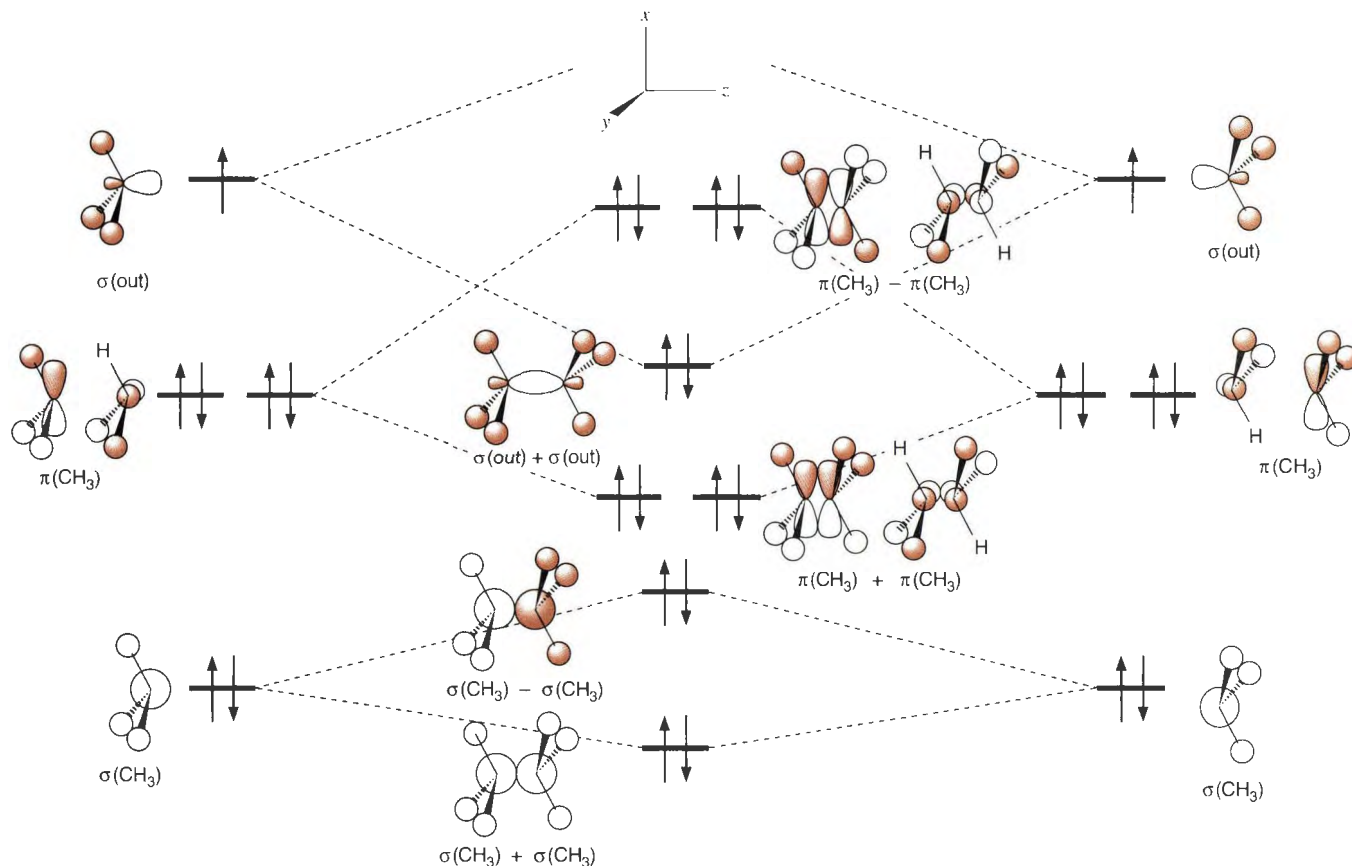
We will see in Chapter 14 that the mixings just described can be treated quantitatively using **perturbation theory**. The mixing of Figure 1.11 **A**, involving degenerate orbitals, is called a **first-order perturbation**, while the nondegenerate mixing of Figure 1.11 **B** is a **second-order perturbation**. With the quantitative tools of perturbation theory, we can provide more detailed analyses of some of the issues addressed here, including, for example, why the orbitals **B** and **C** of Figure 1.7 are degenerate.

In reality, for most situations encountered in organic chemistry, the qualitative descriptions of orbital mixing presented here are adequate. We do need, however, one aspect of perturbation theory that is not evident from the simple analysis given so far. It is generally true that the first-order, degenerate mixing is stronger than the second-order mixing. Also, when comparing second-order mixings, the larger the energy gap between the initial pair of orbitals, the *smaller* the mixing interaction. This statement is known as the **energy gap law**, and it is given as rule 10 in Table 1.7. Note that the terms “extent of mixing”, or “mixing interactions”, refer to the energy changes that occur with mixing. The statement that first-order mixings are stronger than second-order is really just an aspect of the energy gap law, since first-order systems have the smallest possible energy gap: zero. Finally, rule 11 states that the more the interacting fragments overlap, the stronger the mixing interaction. Hence, overlap and energy gap both determine the extent of mixing.

1.3.1 Using Group Orbitals to Make Ethane

In Section 1.2.3 we built the orbitals of CH_3 (actually MH_3) using simple reasoning and symmetry. This construction gets more difficult to do as the molecules get bigger. It is not at all obvious what the orbitals of methylcyclohexane or acrolein should look like. However, there is a way to gain considerable insight into the bonding of larger molecules. The strategy is to build up larger molecules by combining small fragments whose MOs we understand, using the orbital mixing rules we have just developed. We are now in a position to illustrate this strategy by combining two methyl fragments to make ethane.

We want to create an orbital mixing diagram that combines the orbitals of two CH_3 groups to make ethane. We should use the MOs of pyramidal methyl, as this is the geometry appropriate to ethane. The diagram is set up as in Figure 1.12. We need consider only the degenerate (first-order) mixings, because they will be the strongest. The $\sigma(\text{CH}_3)$ and $\pi(\text{CH}_3)$ orbitals are primarily C–H bonding; they do not point out into the C–C bonding region. As such, the overlap in each case should be small, and the mixing interaction energy should be small (rule 11). Thus, these are shown as not especially strong mixings in Figure 1.12.

**Figure 1.12**

The orbital mixing diagram for the formation of ethane from two pyramidal CH_3 groups. The computed MOs are shown in Figure 1.13.

The interesting mixing is of the pair of $\sigma(\text{out})$ orbitals. Because of the hybridization that resulted from rule 7 (Table 1.7) discussed previously, the $\sigma(\text{out})$ orbitals point out into the C–C bonding region. They overlap very well, and so the mixing interaction is quite strong, as shown in the figure. The mixing is strong enough that the $[\sigma(\text{out}) + \sigma(\text{out})]$ MO drops below the out-of-phase combination of the $\pi(\text{CH}_3)$ orbitals. It is difficult to predict how far this orbital will drop and its relative placement with respect to the other orbitals with only the rules of Table 1.7, so this result must just be accepted at this stage of the discussion.

Now we add electrons. As noted above, CH_3 has seven valence electrons. Ethane thus has 14 valence electrons, and as we place them into Figure 1.12, we see that the **highest occupied molecular orbital (HOMO)** of ethane is a degenerate pair of orbitals that can best be described as C–H bonding, but slightly C–C antibonding.

All the combinations derived from $\sigma(\text{CH}_3)$ and $\pi(\text{CH}_3)$ orbitals are occupied. There are six such orbitals with 12 electrons, and ethane has six C–H bonds. Again, the accounting implied by the MO treatment is completely consistent with our simpler views of bonding, consisting of localized C–C and C–H σ bonds. It may at first seem odd that the out-of-phase combinations of the $\sigma(\text{CH}_3)$ and $\pi(\text{CH}_3)$ orbitals are occupied. Shouldn't these be antibonding MOs? They are C–C antibonding, but they are still C–H bonding, and this is enough to make them overall bonding. The $[\sigma(\text{out}) + \sigma(\text{out})]$ MO is the major C–C bonding orbital, and it has two electrons in it.

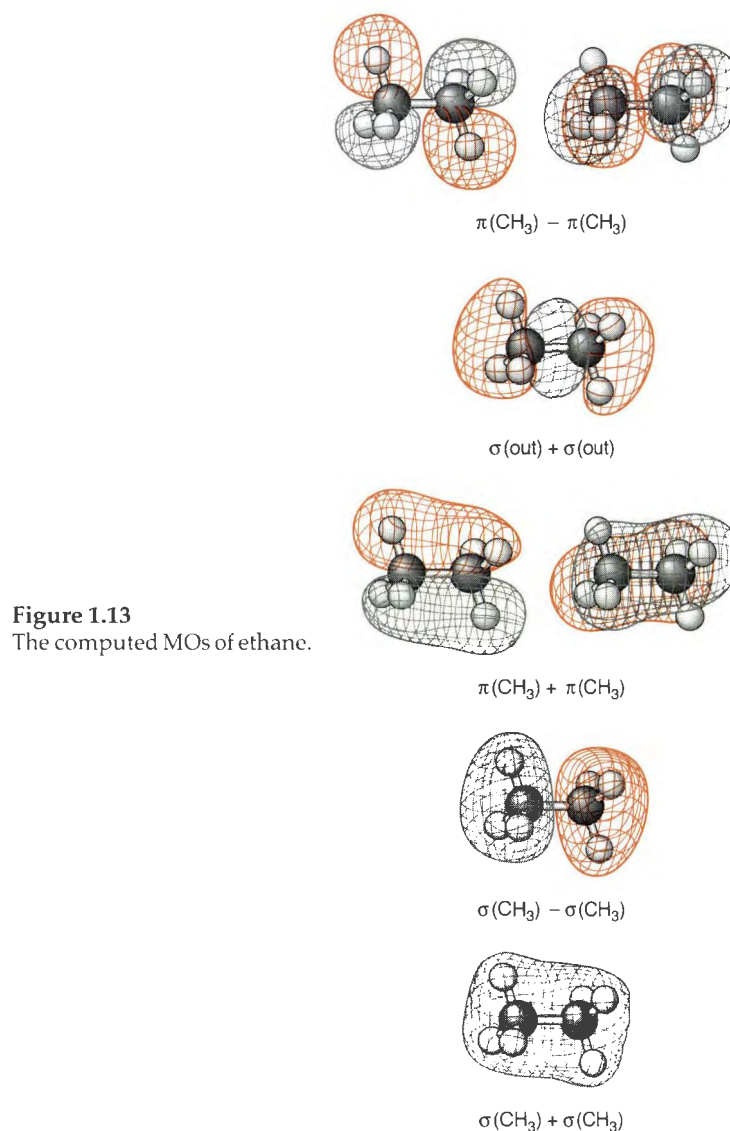


Figure 1.13
The computed MOs of ethane.

For comparison, we show the calculated MOs of ethane in Figure 1.13. Prior to our analysis these might have seemed to be fairly strange orbitals, but now we can see they match up quite nicely with the orbitals we have derived. This highlights the power of group orbitals; their combinations nicely rationalize the origins of orbitals derived from sophisticated calculations. Moreover, experience has shown that the C–C σ bonding orbital of ethane, $[\sigma(\text{out}) + \sigma(\text{out})]$ in Figures 1.12 and 1.13, transfers consistently to all alkane fragments in organic molecules. Thus, whenever we see a C–C single bond in a molecule, we can anticipate an orbital of this type. Likewise, we can anticipate orbitals of π -like character for the C–H bonding interactions.

1.3.2 Using Group Orbitals to Make Ethylene

The standard bonding picture for ethylene is viewed as being made from two sp^2 hybridized carbons, and it contains a C–C double bond comprised of a σ bond and a π bond. In MOT, however, we don't hybridize, and we don't presume bonding arrangements. Instead, we build ethylene with no assumptions by just combining two CH_2 groups. Let's see how well this construction turns out.

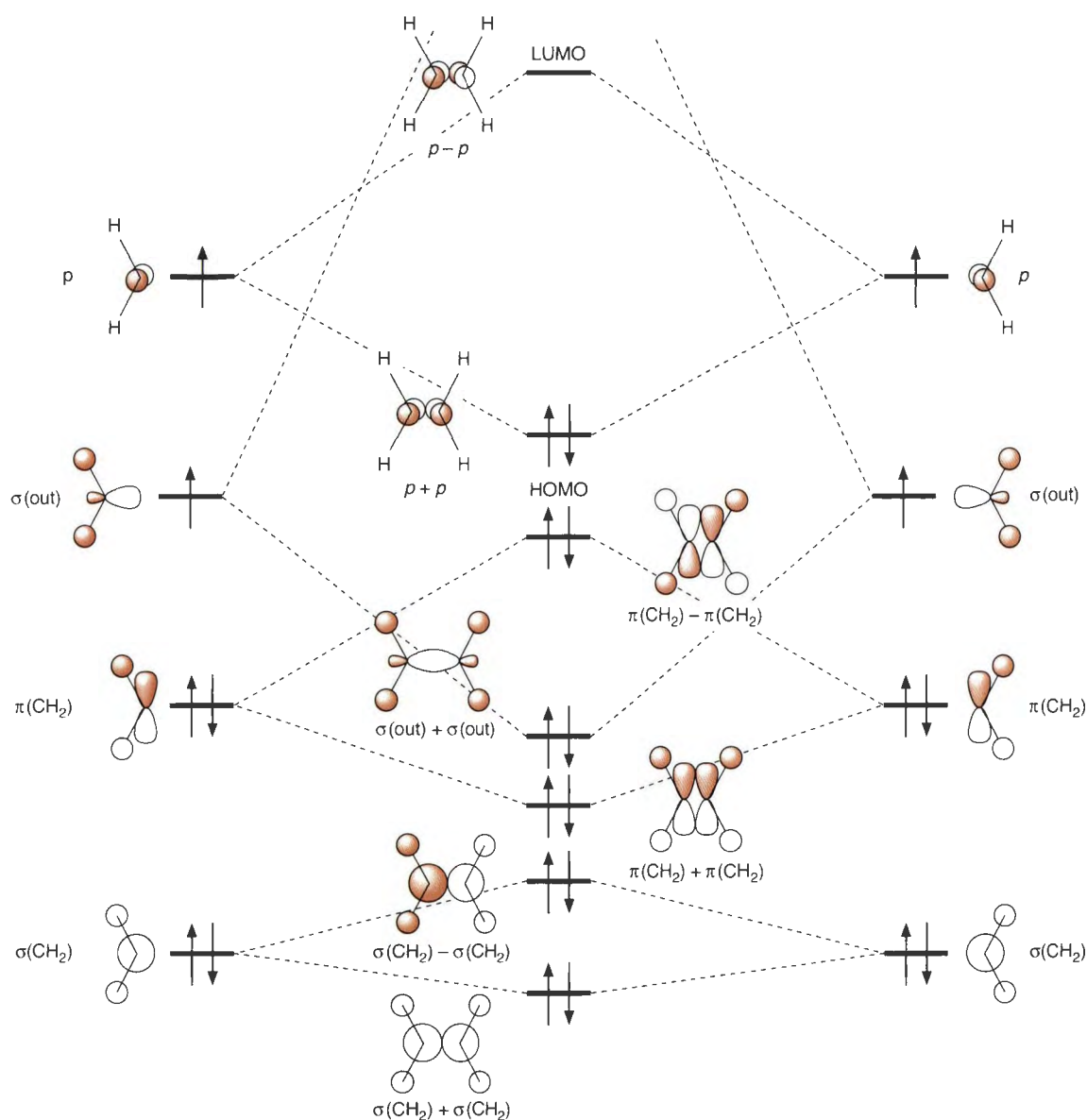


Figure 1.14
Orbital mixing diagram for the formation of ethylene by mixing two CH_2 groups. The computed MOs are shown in Figure 1.15.

The creation of ethylene from two CH_2 groups is shown in Figure 1.14. The actual MOs are shown in Figure 1.15. Hopefully by now the origins of all the MOs are clear. As with ethane, the MOs derived from $\sigma(\text{CH}_2)$ and $\pi(\text{CH}_2)$ make four MOs that are primarily C–H bonding, just as we need for ethylene. The interesting interactions involve $\sigma(\text{out})$ and p .

The hybrid $\sigma(\text{out})$ orbital is strongly directional, pointing along the C–C bond. We expect a strong interaction. The p orbital does not point across the C–C bond. As such, the $(p + p)$ interaction should be weaker than $[\sigma(\text{out}) + \sigma(\text{out})]$, because of poorer overlap. The combination of these interactions produces the C–C double bond of ethylene. Thus, $[\sigma(\text{out}) + \sigma(\text{out})]$ is the major σ bond component of the double bond. The $(p + p)$ mixing produces the π bond of ethylene.

A CH_2 group has six valence electrons, so ethylene has 12. When we place 12 electrons into Figure 1.14, we see that the HOMO is the π orbital. The **lowest unoccupied molecular**

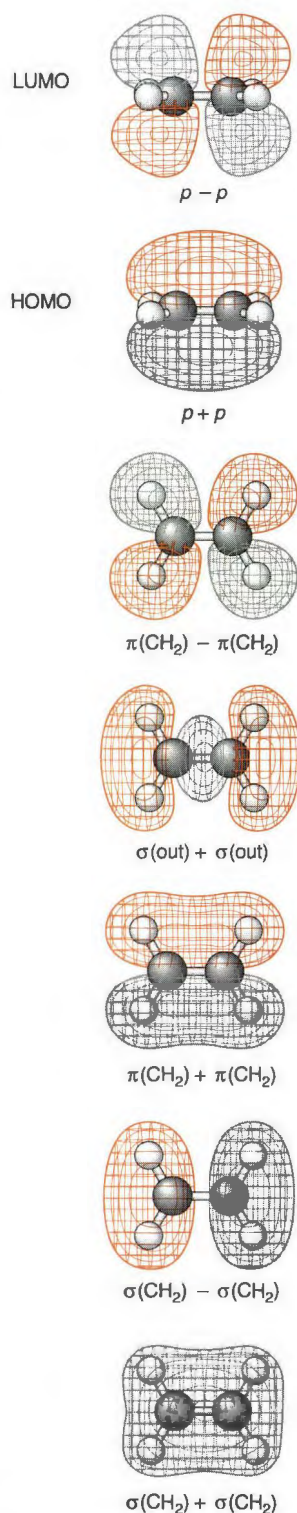


Figure 1.15
The computed MOs of ethylene.
Note the molecule is shown in
different orientations for
different MOs.

orbital (LUMO) is the out-of-phase combination of p orbitals, making it an antibonding π^* orbital. The standard model of bonding in ethylene has been reproduced. We simply follow the rules of orbital mixing, as described in Table 1.7.

1.3.3 The Effects of Heteroatoms—Formaldehyde

In our rules for orbital mixing, rule 5 states that similar molecules have similar MO diagrams. This is essentially true, but there are important differences that we must consider. Formaldehyde and ethylene are **isoelectronic**; they have the same number of valence electrons and the same types of valence orbitals. Thus, we can expect similar MOs for formaldehyde and ethylene, but with some changes (more properly termed “perturbations”) introduced by the oxygen. Experience has shown that the primary consequence of introducing heteroatoms into a hydrocarbon system is to alter orbital energies, as stated in rule 12.

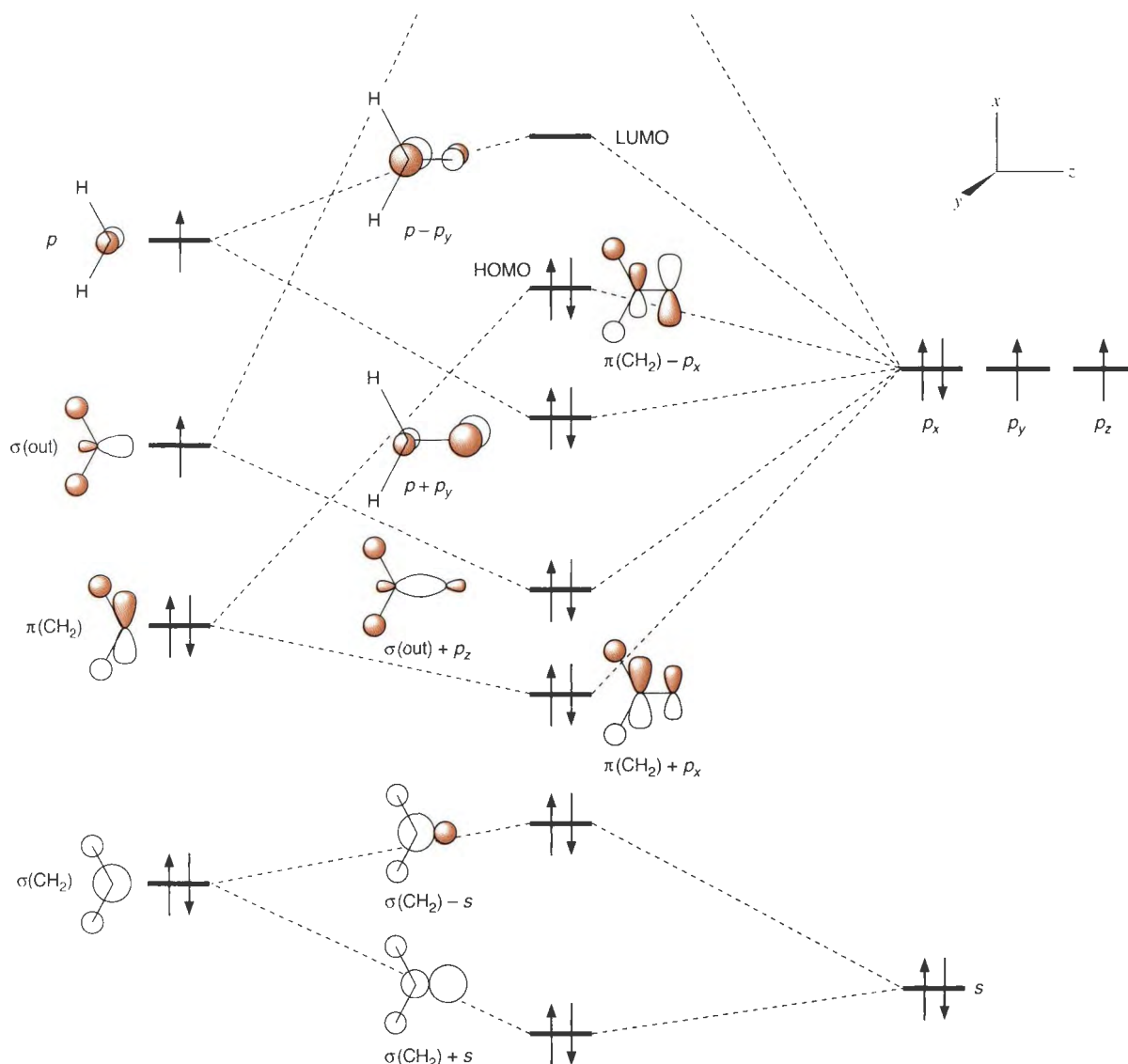
A convenient guideline for orbital energies to be used in mixing diagrams is the set of valence state ionization energies given in Table 1.8. We can see the trends expected from our earlier discussions of electronegativity. The key point is that *electronegative elements have relatively low-lying atomic orbitals*. The larger the ionization energy, the harder it is to remove the electron, so the energy of the orbital is lower. For the particular case at hand, it is clear from Table 1.8 that the atomic orbitals that O contributes to formaldehyde start at much lower energies than those of C or H. As a result, the mixings are no longer degenerate as they were for ethylene, and we must consider the consequences of the second-order perturbation rules for orbital mixing.

The mixing diagram for formaldehyde is shown in Figure 1.16, and the resulting MOs are given in Figure 1.17. The precise energies where we place the oxygen p orbitals are not crucial. However, we must place the oxygen p orbitals below the CH_2 p orbital. Orbitals that are analogous to all the key orbitals of ethylene are created in Figure 1.16, including both the σ and π orbitals of the double bond. Although mixing rule 9 now predicts polarizations in all the orbitals, let’s first focus on the π and π^* orbitals.

Table 1.8
Valence State Ionization Energies*

Orbital		$-\text{H}_{ii}$ (eV)
H	1s	13.6
C	2s	21.4
	2p	11.4
N	2s	26.0
	2p	13.4
O	2s	32.3
	2p	14.8
F	2s	40.0
	2p	18.1
Si	3s	17.3
	3p	9.2
P	3s	18.6
	3p	14
S	3s	20
	3p	11
Cl	3s	30
	3p	15

*Technically, these are not experimentally determined numbers, but rather they are parameters (H_{ii} values) used in a popular computational method termed extended Hückel theory (see Chapter 14, p. 834).

**Figure 1.16**

The orbital mixing diagram of $\text{CH}_2=\text{O}$ from CH_2 plus an oxygen atom. The final MOs are given in Figure 1.17.

Since the building blocks for π and π^* are isolated p orbitals, there is no ambiguity about the relative energies of the two initial orbitals: the oxygen orbital lies below the carbon orbital. As such, there is also no ambiguity about the expected polarizations. The lower energy MO, the π orbital, must be polarized toward the oxygen. Similarly, the higher energy MO, π^* , must be polarized toward carbon. The actual MOs show that this is indeed the case. When you examine the π MO of formaldehyde in Figure 1.17, the expected polarization may not be evident. However, you must remember that the atomic orbitals of electronegative elements are smaller; both the covalent and the van der Waals radii of oxygen are smaller than those of carbon (see above). So, the π MO of formaldehyde does have a larger contribution from the oxygen (numerical rather than graphical descriptions of the MO make this clear). The expected polarization toward carbon is very clear in the π^* MO.

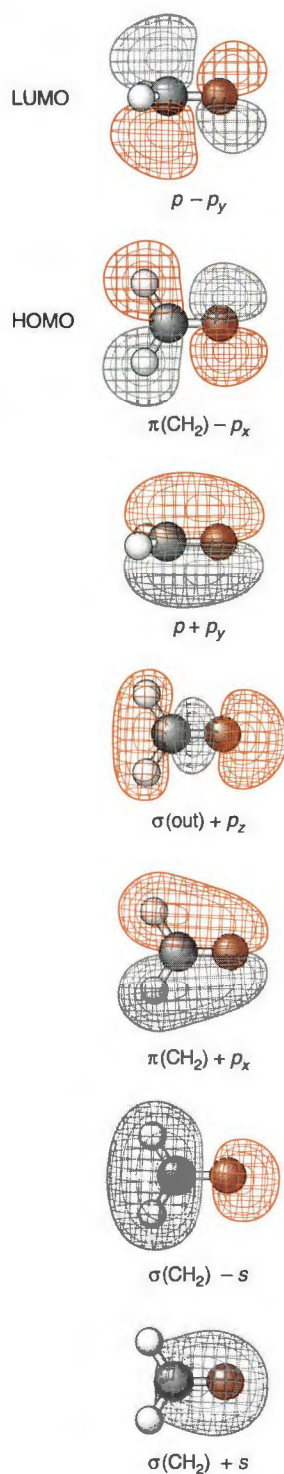
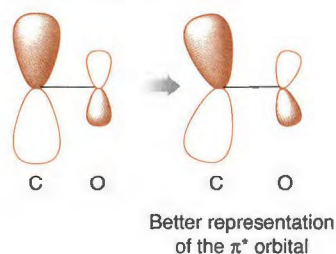


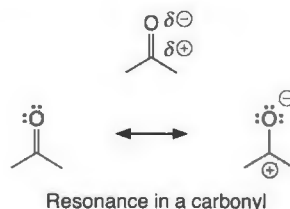
Figure 1.17
The computed MOs of formaldehyde. Note the molecule is shown in different orientations for different MOs.

We still have 12 electrons to allocate. An interesting feature of Figure 1.16 is that when we insert our 12 valence electrons, we find that the LUMO is π^* , as expected. However, the HOMO is not π , but rather a lone pair, which we describe as $\pi(\text{CH}_2)$ minus the oxygen p_x orbital (see Figure 1.16). It is a π -type lone pair with a significant out-of-phase mixture of the $\pi(\text{CH}_2)$ orbital. The second lone pair lies below the HOMO, and is σ -type, formed from mixing $\sigma(\text{out})$ with p_z . Neither of these orbitals are “pure” lone pair orbitals. The higher energy one is significantly C–H bonding, and the lower energy one is significantly C–O bonding. MOT does not always lead to a simple correspondence to our classical views. The situation with formaldehyde is directly analogous to what we saw previously with the water molecule. The canonical (fully delocalized) MOs of oxygen-containing molecules predict the existence of two distinct lone pairs. You can combine these orbitals to produce orbitals that resemble the classic sp^2 hybridization for the oxygen.

Looking ahead, it should be clear that the MO diagram for this prototype carbonyl has significant implications for predicting and rationalizing reactivity patterns. Nucleophiles will interact with the LUMO preferentially at the larger coefficient, the one on carbon. Note that this π^* MO “leans away” from the bonding region, as is typical for antibonding orbitals. The π^* orbital is perpendicular to the plane of the molecule but is slightly tilted toward the CH_2 and O groups. It is better represented in a cartoon as shown below. This “tilting” of the orbital has ramifications for reactivity that we will return to in Chapter 10. The nature of π^* is completely consistent with what we expect for nucleophilic addition to a carbonyl, namely attack at C from a “backside” direction. Also, the polarization of the HOMO toward oxygen has implications for reactivity. For example, we expect protonation on oxygen, not carbon.



As in other cases discussed so far, our QMOT model of a simple carbonyl is completely compatible with more conventional bonding models. In order to explain the polarization we have just discussed, models emphasizing VBT concepts would focus on the greater electronegativity of O vs. C. This electronegativity difference should polarize the bond, as shown below. Alternatively, we could invoke a resonance argument. While a single Lewis structure is certainly a good model for a carbonyl, the resonance interaction shown is not unreasonable. Although it creates charge and disrupts octets, it places a negative charge on a highly electronegative element. Certainly the two forms would not contribute equally to the structure of a carbonyl. The $\text{C}=\text{O}$ form is preferred—perhaps the structure is best thought of as a 4:1 or a 10:1 mixture of the two—the exact details are unimportant. However, even a small contribution from the ionic form has significant implications for carbonyl chemistry. It implies a partial negative charge on O, and a partial positive charge on C. Again, the reactivity patterns of a carbonyl, protonation at O and nucleophilic addition to C, are completely consistent with this model.



The group orbitals of an olefin would be those we derived for ethylene in Figure 1.14. The most important ones will be the π and π^* MOs, designated as the HOMO and LUMO in Figures 1.14 and 1.15. Similarly, the group orbitals of a simple carbonyl compound, an aldehyde or ketone, are those we developed for formaldehyde (Figures 1.16 and 1.17). Now we should consider at least four different MOs: π and π^* along with the two lone pairs.

1.3.4 Making More Complex Alkanes

In alkanes, CH_2 and CH_3 groups will contribute orbitals such as in Figures 1.8 and 1.9. If we consider a linear alkane, $\text{CH}_3(\text{CH}_2)_n\text{CH}_3$, we would describe the bonding as follows. For the terminal methyl groups, the $\sigma(\text{CH}_3)$ and two $\pi(\text{CH}_3)$ orbitals would be used for C–H bonding. The methyl $\sigma(\text{out})$ orbital would be used to make a C–C bond. The resulting orbital would be the C–C bonding orbital discussed for ethane.

The CH_2 group is more challenging to consider because it must make two C–C bonds. To do this, we use the CH_2 $\sigma(\text{out})$ and the p orbitals. You don't have to, but it is convenient to make linear combinations of these two, as we did for water in Figure 1.10 (in fact, our discussion of orbital mixing in water can be extended to any MH_2). This linear combination creates orbitals similar to the sp^3 hybrids available to bond to two other groups. We just mix $\sigma(\text{out})$ and p to give orbitals that point in the correct direction to make CH_2 a tetrahedral center. A linear combination of group orbitals to give orbitals reminiscent of our standard hybridization notions *will always be possible*. The important point is that the orbitals developed by the QMOT approach predict the delocalized orbitals found by high level calculations, where hybridization is not usually the result. We will show experimental evidence in Chapter 14 that supports the fact that hybridization does not actually occur in the standard sp^3 , sp^2 , and sp manner.

Once we see how the C–C bonds are made, we just assume them directly. So, in a linear alkane, we have a backbone of C–C bonding orbitals. The terminal methyl groups make C–C bonds with their respective $\sigma(\text{out})$ orbitals. In addition, each CH_3 group contributes three C–H bonding orbitals, one $\sigma(\text{CH}_3)$ and two $\pi(\text{CH}_3)$, while each CH_2 contributes two C–H bonding orbitals, one $\sigma(\text{CH}_2)$ and one $\pi(\text{CH}_2)$.

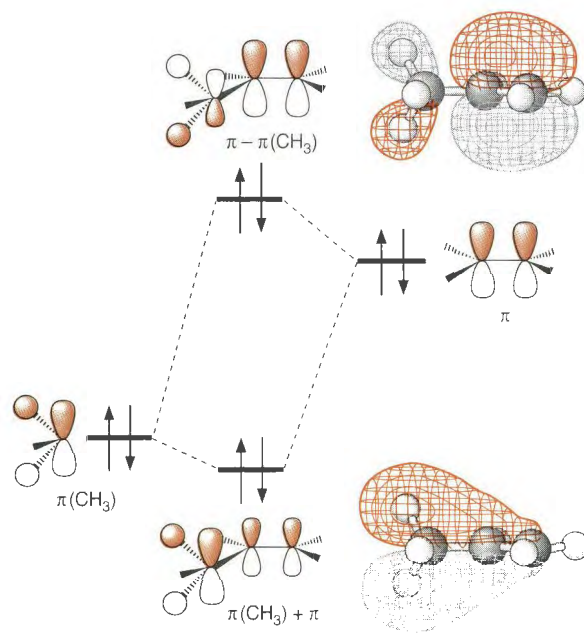
This model of alkane bonding is useful, but admittedly, it really does not produce any new insights compared to the more conventional model emphasizing sp^3 hybridized carbons and simple localized σ bonds. Frankly, if we only ever considered alkanes, there would be no need for CH_2 and CH_3 group orbitals. However, we will now show that in more interesting molecules, the use of group orbitals provides very valuable insights.

1.3.5 Three More Examples of Building Larger Molecules from Group Orbitals

All functional groups have a set of group orbitals associated with them. Appendix 3 shows the full MOs of a number of representative small molecules. These can be considered to be the group orbitals of the analogous functional group in larger molecules. From the basic building blocks we have presented so far, the MOs for these essential functional groups of organic chemistry can be understood. For example, the MOs of a carboxylic acid (Appendix 3) can be developed from an orbital mixing diagram, combining the π MOs of formaldehyde with the appropriate AOs of an oxygen atom. Then, a methyl ester can be “prepared” by mixing in a $\pi(\text{CH}_3)$ orbital. When considering the reactivity of a specific functional group, it will be useful to look at its group orbitals in Appendix 3 as a guideline. Let's do such a mixture for a few additional cases.

Propene

In most instances alkyl groups are spectators in organic reactivity, and the simple VBT model based on hybridization is adequate. However, for a CH_2 or CH_3 that is adjacent to a conventional π system, such as an alkene, a carbocation, or a carbonyl, important interactions between the alkyl group and the π system may occur. In these instances the $\pi(\text{CH}_2)$ or $\pi(\text{CH}_3)$ group orbitals are quite useful.

**Figure 1.18**

An orbital mixing diagram for propene, along with the computed MOs of the molecule.

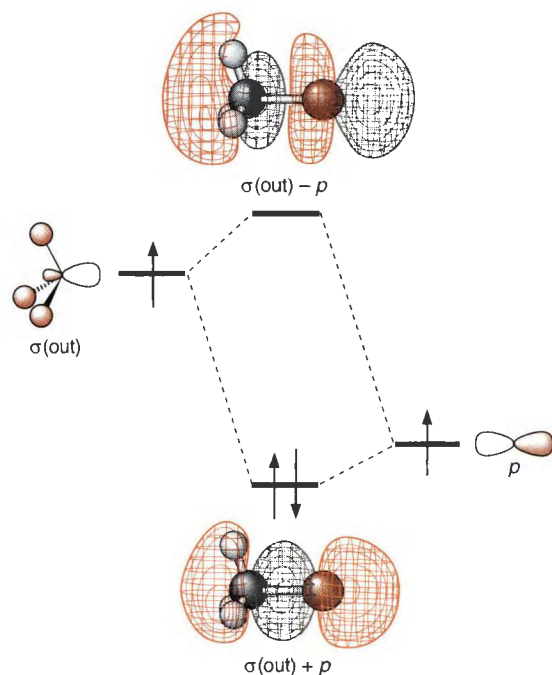
As an example, Figure 1.18 shows an orbital mixing diagram for propene. We are going to focus only on the π system, so we start with the ethylene π MO of Figures 1.14 and 1.15. In conventional views of a methyl group, there really is no way for the CH_3 to interact with the π system of propene. However, it is not just a coincidence that two of the C-H bonding orbitals for CH_3 are called $\pi(\text{CH}_3)$. These group orbitals are of the correct symmetry to interact with conventional π orbitals of the sort seen in alkenes and carbonyls. In our orbital mixing diagram, the two degenerate $\pi(\text{CH}_3)$ group orbitals are low lying in energy; they are C-H σ bonding, and so are lower than a C-C π bond. In any particular conformation of propene, it will always be true that one of the $\pi(\text{CH}_3)$ group orbitals will be able to mix with the ethylene π MO. This mixture is a standard second-order mixing, producing two new propene MOs, as in Figure 1.18. Let's consider the consequences of this mixing.

Compared to ethylene, the π bond of propene is (a) slightly higher in energy, and (b) slightly delocalized onto the methyl group. Note the clear prediction that the olefin π orbital and the $\pi(\text{CH}_3)$ group orbital should be *out of phase* in the " π " orbital, the HOMO of propene. This prediction is confirmed by the actual MO (Figure 1.18). In addition, the mixing is confirmed by experiments. The predicted elevation in orbital energy is fully supported by the fact that the ionization energy of propene (9.73 eV) is measurably lower than that of ethylene (10.51 eV), meaning that the HOMO of propene is higher in energy. The $\pi(\text{CH}_3)$ group orbital has been lowered in energy, a factor not usually of importance in understanding reactivity. An important lesson of this analysis is that there are orbitals of π symmetry in simple alkyl groups such as CH_3 and CH_2 , and these can interact with the more conventional π systems of alkenes, carbonyls, and so on.

The mixing we have just discussed is *not hyperconjugation*, which is the mixing of C-H and C-C bonding orbitals with adjacent empty or partially empty π and p orbitals (see Section 1.4.1). However, a similar mixture can be done with the π^* orbital of the alkene that does represent hyperconjugation, and you are asked to show this in an Exercise at the end of this chapter.

Note that we have mixed filled orbitals in Figure 1.18, which may seem like a violation of the previous statement that such mixings are always destabilizing. Indeed, this interaction *is* destabilizing because four electrons go into the resulting orbitals. However, it is enforced by the close juxtaposition of the $\pi(\text{CH}_3)$ orbital with the π bond due to the presence of the C-C bond between the methyl and the alkene of propene.

There is a distinction in mixing filled orbitals that must be kept straight. If we are considering a possible reaction between two species, and we want to know if favorable or unfavor-

**Figure 1.19**

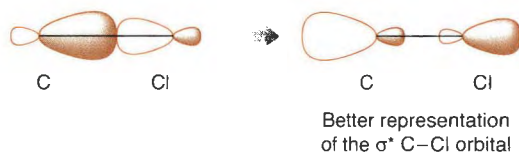
Orbital mixing diagram for CH_3Cl , with computed orbitals shown. $\sigma(\text{out}) + p$ is the bonding orbital. $\sigma(\text{out}) - p$ is the LUMO, σ^* antibonding orbital.

able orbital interactions develop during the reaction, we clearly want to maximize two-electron mixings and minimize four-electron mixings. However, we are doing something different in our analysis of propene. Really, we are performing an after-the-fact analysis of the molecule, trying to understand the results of a full quantum mechanical calculation on the system. The molecule was not *really* formed by mixing fragment orbitals; we have simply found that such analyses lead to the true MOs. For a destabilizing mixing to occur within the molecule, such as that shown in Figure 1.18 for propene, there must be other σ and π bond stabilizing interactions that more than compensate, thus stabilizing the molecule overall.

Methyl Chloride

As another example of the use of orbital mixing strategies for understanding structure and reactivity, we consider methyl chloride as a prototype of an organic molecule singly bonded to an electronegative element. We could construct a complete diagram, using all the CH_3 group orbitals and the AOs of chlorine, but we can also focus only on the C–Cl bond as the most interesting feature of the system.

The C–Cl bond is formed by the mixing of $\sigma(\text{out})$ of the CH_3 with a chlorine p orbital. The mixing diagram is shown in Figure 1.19. Both electronegativity (Table 1.1) and orbital energy arguments (Table 1.8) allow us to place the Cl orbitals below the carbon orbitals. This means that the polarizations should have the bonding combination polarized toward Cl, and the antibonding combination polarized toward C. Actual calculated MOs are shown in Figure 1.19. The most interesting MO is the LUMO. It is the prototype σ^* orbital. It is C–Cl antibonding, in that there is a nodal surface that splits the C–Cl bond in two. A great deal of orbital character lies outside the region between the C and the Cl, and is especially accentuated at the carbon end of the molecule. There is considerable orbital density on the “backside” of the carbon, and hence a cartoon of the σ^* orbital is better represented as shown below rather than as given for generic σ^* orbitals shown in Figure 1.3.



This σ^* orbital leads to predictions about reactivity. In the S_N2 reaction, a nucleophile will react with CH_3Cl . As we will see, nucleophiles have high-lying filled orbitals that can mix with low-lying empty orbitals. The σ^* orbital is the appropriate empty orbital on CH_3Cl , and the fact that it is substantially polarized toward the backside of the carbon atom is nicely consistent with the observed inversion of configuration (backside attack) seen in S_N2 reactions. We can view this as a group orbital for a C–Cl functional group that can be considered present in any alkyl chloride.

Butadiene

If we are going to build up larger π systems than are present in simple alkenes and carbonyls, we again need to linearly combine the appropriate group orbitals. This is done for the π system of butadiene in Figure 1.20, and calculated orbitals are given. The π and π^* orbitals of ethylene are our starting point, and they mix in the manner shown to give four molecular orbitals. These resulting orbitals can be considered to be the π MOs of butadiene, but also the four group orbitals for any diene. Note that the simple schematic mixing of the ethylene π and π^* orbitals does not predict the relative contributions of each p orbital to a given MO, and we will have to wait until Chapter 14 to see why. Yet, the simple mixture gets the nodal properties perfectly correct.

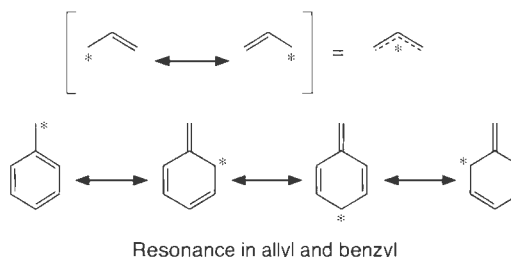
1.3.6 Group Orbitals of Representative π Systems: Benzene, Benzyl, and Allyl

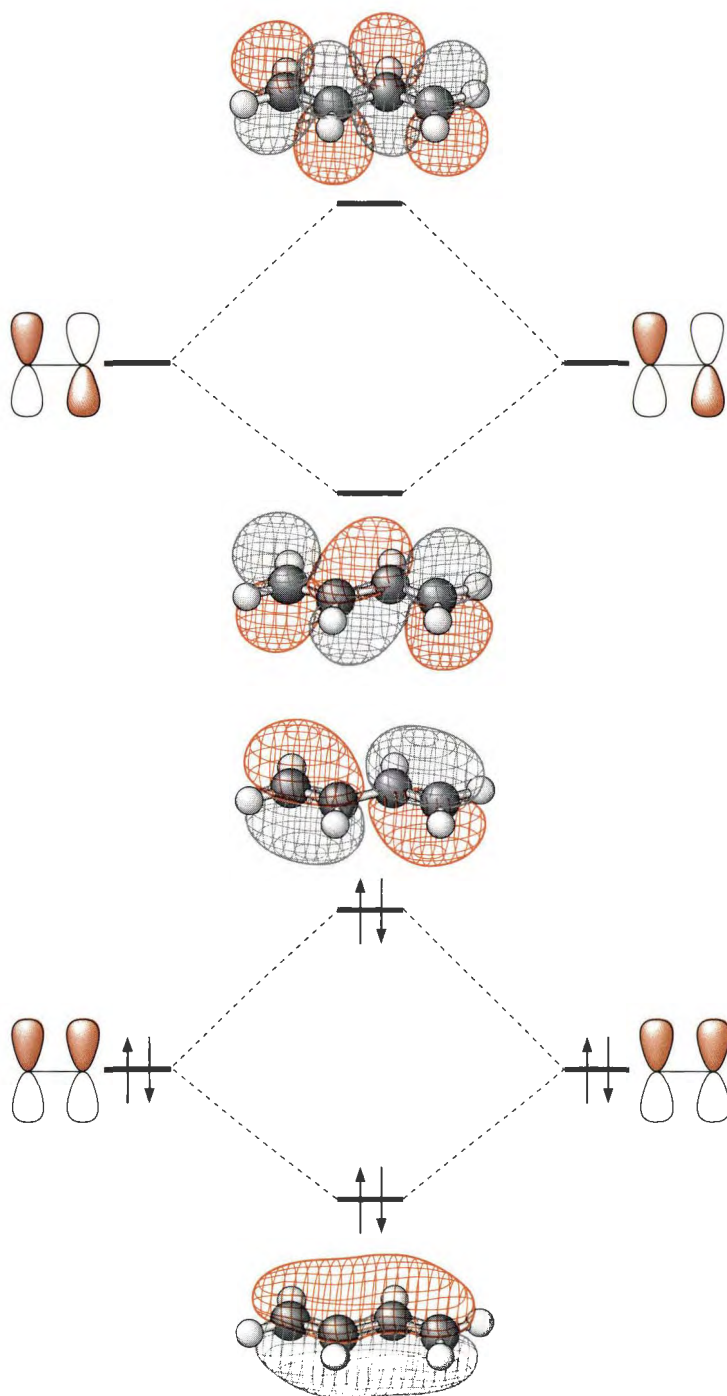
We have focused our discussion to this point on group orbitals for small alkyl fragments, σ systems, and a few simple π systems. Yet, as shown in Appendix 3, a significant fraction of important functional groups in organic chemistry have complex π systems. To build many complex π systems by combining small π systems, we need to know the group orbitals of some representative small π systems other than just ethylene. We now give the MOs of three other essential building blocks of organic chemistry: benzene, benzyl, and allyl. In evaluating organic structure and reactivity, these units show up repeatedly, and so it is worthwhile to present them here.

Figure 1.21 presents the MOs of benzene, both in symbolic form and as produced by an accurate quantum mechanical calculation. Also shown is the “HOMO” of benzyl—that is, the singly occupied orbital of benzyl radical, the empty orbital of benzyl cation, and the doubly occupied HOMO of benzyl anion. To a good approximation, the MO has the same form for all three structures. Here, we show the orbitals from the top, so that the p orbitals that make up the molecular orbitals appear only as spheres.

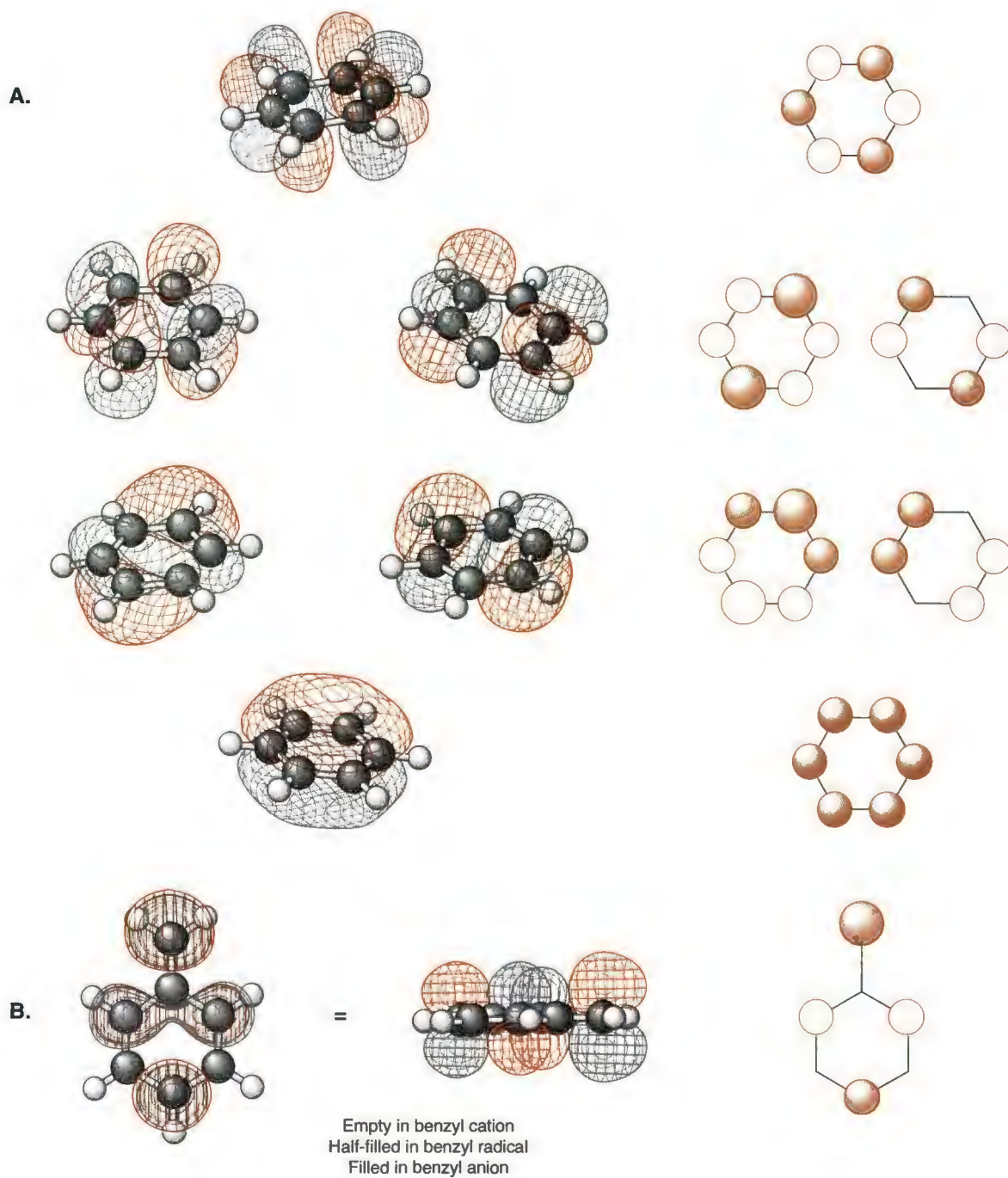
Figure 1.22 shows the group MOs of allyl. The most distinctive feature is that the middle MO has a node through the central carbon, and so is formed from only C1 and C3. This is the MO that is empty in allyl cation, singly occupied in allyl radical, and doubly occupied (HOMO) in allyl anion.

The HOMOs of allyl and benzyl are completely consistent with resonance models for the two (shown below). Allyl resonance places the charge ($^+ = +$, $-$, or $\bullet$) only on C1 and C3, where the orbital has finite coefficients. In benzyl, there is activity only at the benzylic carbon and the ortho and para positions of the ring. Once again, we emphasize that the MOT approach does not require the “patch” fix of resonance to get the π properties correct.



**Figure 1.20**

The mixing diagram for creating the π MOs of butadiene from ethylene group orbitals and the calculated orbitals for the π system of butadiene.

**Figure 1.21**

A. The computed MOs of benzene and conventional representations.

B. Two views of the key MO of a benzyl group and the conventional representation.

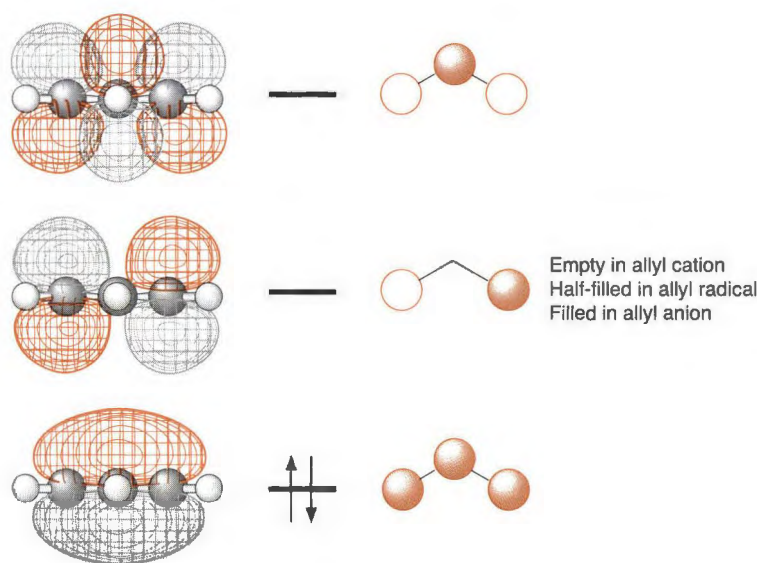


Figure 1.22
The MOs of an allyl fragment.

It is an interesting exercise to form the MOs of allyl using an orbital mixing strategy, and such an analysis is shown in Figure 1.23. Here, we show the p orbitals viewed from the top, so all we see is a circle. For allyl, it is simplest to first mix C1 and C3. We can form the in-phase and out-of-phase combinations of p orbitals, but to a good approximation, these nonadjacent orbitals do not overlap significantly, so there is no energy split. Now we bring in C2. It can mix only with the in-phase combination of C1 and C3, because it lies on the node of the out-of-phase combination. The new mixing produces two new MOs, while the C1/C3 out-of-phase combination comes across unchanged. The result is the allyl pattern of Figure 1.22.

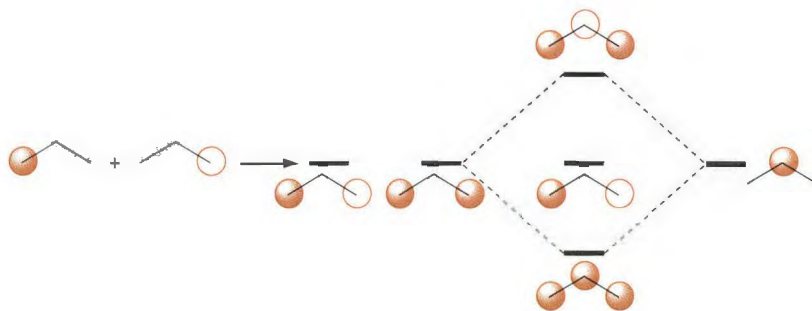
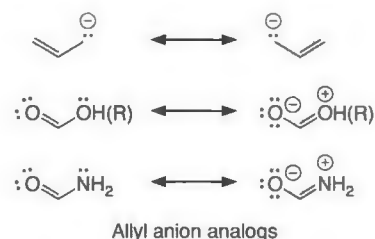


Figure 1.23
An orbital mixing approach to forming the MOs of an allyl fragment.

1.3.7 Understanding Common Functional Groups as Perturbations of Allyl

One of the most useful insights to be derived from the group orbitals of allyl is a picture of the group orbitals for the carboxylic acid, ester, and amide functional groups. These functional groups have π systems that can be viewed as isoelectronic with allyl anion. The resonance structures shown in the margin highlight this concept. Appendix 3 shows the actual π systems for these functional groups, where it is clear that the allyl nature is retained. The three π MOs of the functional groups are polarized in various ways due to the addition of the heteroatoms, yet the general nodal properties of allyl are retained.



1.3.8 The Three Center–Two Electron Bond

Starting from the classic picture of bonds between adjacent atoms, we often need other concepts to fully explain the bonding in a given molecule. As explained, resonance is such a concept, required to explain the bonding in molecules with delocalized π systems. Another bonding situation that requires a special treatment for VBT, but not with MOT, is the three center–two electron bond. As the name implies, the three center–two electron bond is associated with electron-deficient systems, those for which there are not enough valence electrons to make conventional two center–two electron bonds among all the atoms. The boranes and related structures use the three center–two electron bond extensively, creating highly bridged structures. In organic chemistry, the most common electron-deficient species are carbocations. The three center–two electron bond also figures prominently in carbocation chemistry, so we describe it briefly here.

The starting point for understanding the three center–two electron bond is the chemistry of boron-containing compounds. Simple boron compounds are isoelectronic with carbocations: BH_3 has the same number of valence electrons and orbitals as CH_3^+ . We already noted that a QMOT treatment of the two structures produces qualitatively similar results, so BH_3 has the same planar structure as CH_3^+ . As you may recall from introductory chemistry, though, BH_3 in the gas phase dimerizes to diborane, B_2H_6 , and this molecule has the unconventional structure shown in Figure 1.24 A. The molecule has 12 valence electrons, eight of which are used to make the four conventional B–H bonds. The remaining four valence electrons are partitioned, two apiece, to the two B–H–B bridging systems, forming a pair of three center–two electron bonds.

The stability of this well-established bonding pattern can be understood from simple MO arguments. The B–H–B arrangement in diborane is another three-orbital mixing problem, like allyl. In fact, the mixing diagram we developed for allyl translates directly to the

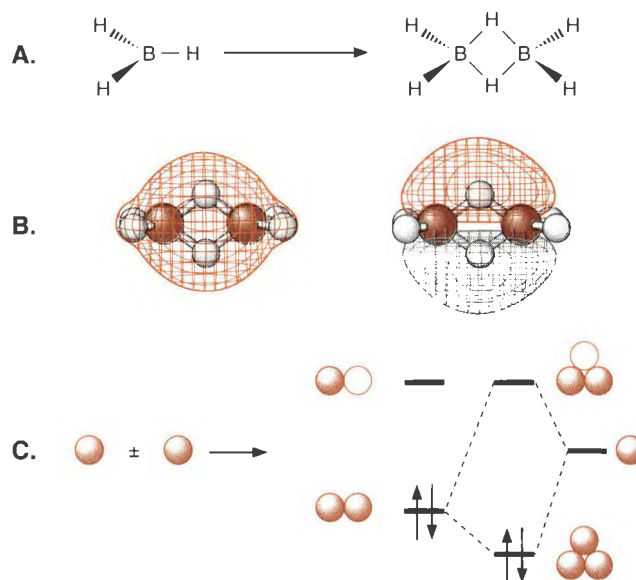


Figure 1.24

Three center–two electron bonds. **A.** Borane dimerizes to produce diborane. **B.** The calculated MOs primarily responsible for the bonding of the bridging hydrogens in diborane. Since there are two B–H–B bridges, there are two bonding orbitals. Shown are the in-phase (left) and out-of-phase (right) combinations of the fully symmetric B–H–B orbital combinations. **C.** A qualitative picture for how a single three center–two electron bond is constructed. Only the lowest of the three combinations, the fully in-phase MO, is occupied. The two MOs of part **B** are in-phase and out-of-phase combinations of this MO, because diborane has two three center–two electron bonds.

three center–two electron bond. In Figure 1.23, the open and filled circles were meant to be the “tops” of p orbitals, showing their phase relationships. However, they could just as easily be s orbitals or sp^3 hybrids. A recurring theme of QMOT, which we have already seen when discussing CH_2 and CH_3 , is that the distinction between σ and π systems is not as clear-cut as introductory texts would suggest, and so we should not be surprised that the MO patterns in a σ system such as we are developing here are topologically identical to those of an analogous π system. Although formally σ bonding is involved, the mixing for the B–H–B system (Figure 1.24 C) still produces three molecular orbitals in a manner analogous to Figure 1.23. Since only one of the three orbitals is highly stabilized, the optimal arrangement involves only two electrons. It is a three-atom system involving only one occupied MO, a **three center–two electron bond**.

Note that it is generally better to make two, conventional, two center–two electron bonds rather than a single three center–two electron bond. However, boranes are electron deficient; there just are not enough valence electrons to make the requisite number of conventional bonds. The system does the best it can by forming a pair of three center–two electron bonds.

There is absolutely no doubt that three center–two electron bonding is an important component of the electronic structures of a wide range of boron-containing compounds. Given the isoelectronic relationship to CH_3^+ , it should come as no surprise that similar bonding should arise in carbocations, and indeed it does (see the next section). There was initially considerable resistance to this notion; the tetravalent carbon was sacrosanct. However, it is now clear that three center–two electron bonds do contribute to many carbocation structures. The three center–two electron bond allows a wide range of exotic looking, highly-bridged structures, often referred to as non-classical carbocations (see Chapters 2, 11, and 14 for discussions). While a detailed description of the bonding in some of the more elaborate systems may seem complicated, at its core is the special stability of the three center–two electron bond.

1.3.9 Summary of the Concepts Involved in Our Second Model of Bonding

The more modern theory of bonding extends the notion of bonding and antibonding orbitals that describe localized σ and π bonds to increasingly delocalized orbitals. All organic functional groups can be envisioned as consisting of a set of delocalized orbitals that are combined to build up the whole molecule. These “group orbitals” will be very useful in future analyses, and as described in this chapter, they nicely predict the structures of stable molecules and reactive intermediates. The new concepts that were introduced in developing this second model of bonding are summarized as follows:

QMOT. A set of rules for predicting how orbitals will mix to make delocalized molecular orbitals for molecular groups or molecules.

Orbital mixing. MOs associated with particular bonds can be formed by mixing orbitals from the bonding partners according to the rules of QMOT. When such mixing occurs, the destabilization of the higher orbital is greater than the stabilization of the lower orbital, and in the case of the nondegenerate second-order mixing, predictable orbital polarizations result. This mixing can be between atomic orbitals on adjacent atoms, or between group orbitals on adjacent groups.

Transferable, partially localized MOs—group orbitals. Functional groups such as olefins, ketones, amides, aromatics, etc., contribute a set of transferable orbitals to a molecule. For example, a simple olefin always has π and π^* orbitals. Even CH_3 and CH_2 groups contribute partially delocalized molecular orbitals to molecules, and importantly, these include $\pi(\text{CH}_3)$ and $\pi(\text{CH}_2)$ group orbitals that can interact strongly with conventional π systems.

1.4 Bonding and Structures of Reactive Intermediates

In this chapter we have focused on the molecular and electronic structures of organic molecules and on the theories of bonding that explain them. A major goal of physical organic chemistry, however, is to explain reactivity, and a key concept in reaching that goal is the reactive intermediate. A **reactive intermediate** in organic chemistry is typically some form of carbon that either does not have the requisite four bonds, or has less than an octet of electrons, or is highly strained (see Chapter 2). Heteroatom analogs of these structures are also important. For example, carbanions do not have four bonds, while radicals, carbocations, and carbenes are also lacking an octet. Rather than disperse our discussion of the structures of these prototype reactive intermediates throughout the text, we group them here. This allows us to directly connect the structures to the theoretical models of bonding we have developed and to compare and contrast the properties of the basic classes of reactive intermediates. Our notions of bonding should be applicable to both stable structures and reactive structures. The structures of stable organic molecules are mostly straightforward. However, interesting issues arise in describing reactive intermediates that will allow us to further develop many of the models described thus far in this chapter. This further development includes the QMOT model given in Sections 1.2 and 1.3, but also other concepts such as electronegativity and resonance. This is the first of two preliminary descriptions of reactive intermediates. In the following chapter, which emphasizes thermodynamic stabilities, we will evaluate quantitatively the features that stabilize or destabilize reactive intermediates. We begin our analysis of reactive intermediates here by showing how the Walsh diagrams we have already developed provide a solid foundation for understanding the structural features of reactive intermediates.

We generated the group orbitals of CH_2 and CH_3 so we could use them as fragments to build up and understand larger molecules. However, each of these is also a well-defined species in its own right, and more importantly, they are the fundamental structures for a range of reactive intermediates we will see throughout this textbook. Here we will consider the extent to which the simple Walsh diagrams can help us anticipate the fundamental features of reactive intermediates that will allow us to understand their reactivities. A key issue for these structures is whether the planar or pyramidal (linear or bent) form is preferred. A basic assumption of the group orbital concept is that the essential features of an orbital diagram will not change as we alter atom types or the number of electrons. Thus, the basic orbital diagrams for the prototype structures (methyl cation, methyl radical, and methyl anion) are all the same to a good approximation. The primary difference is just the number of electrons we put into the diagram of Figure 1.8. We will see that this distinction is enough to yield very different properties for the three types of reactive intermediates.

1.4.1 Carbocations

From the standpoint of molecular and electronic structure, carbocations present a much greater variety and more significant conceptual challenges than carbanions, radicals, or carbenes. Many aspects of carbocation chemistry have been at times quite controversial. Here, we describe the bonding models for basic carbocation structures. First, however, we must explain some nomenclature.

A **carbocation** is a structure with a positive charge that is associated primarily with a carbon center or, in the case of delocalized systems, a collection of carbons. Heteroatom substitution is allowed, and it will inevitably lead to some positive charge being on the heteroatoms. There are two types of carbocations—carbenium ions and carbonium ions. **Carbenium ions** are trivalent species with a formula of R_3C^+ . **Carbonium ions** encompass two related types of structures. First are pentavalent species of the general formula R_5C^+ . While not common in solution phase chemistry, such pentavalent carbons are common in the gas phase. Also referred to as carbonium ions are carbocations that have an important contribution from three center–two electron bonding. The nomenclature will become more clear as we present examples below.

This distinction between the two types of carbocations was championed by Olah, and is consistent with other nomenclature. Just as hydronium and ammonium ions represent protonated forms of water and ammonia, carbonium ions can be considered to contain protonated forms of methane and other typical, tetravalent carbon compounds. Likewise, carbenium ions can be thought of as protonated carbenes ($\text{:CH}_2 + \text{H}^+$ gives CH_3^+), consistent with the nomenclature system. In the older literature, only the term carbonium ion is used in connection with what is more precisely called a carbenium ion. We feel the more modern nomenclature has value, so we use it in this book.

Carbenium Ions

The prototype carbenium ion is methyl cation, CH_3^+ . It has six valence electrons (one from each of three H's and three from C, given the + charge). These six electrons will fill the lowest three MOs of Figure 1.8, with the p_z orbital remaining empty. Since pyramidalization destabilizes the B/C pair in Figure 1.8 more than it stabilizes the single A MO, CH_3^+ will prefer a planar structure. The lowering of the D orbital is inconsequential, because there are no electrons in it. The preference for planarity is therefore substantial. Calculations suggest that a CH_3^+ that is pyramidalized to have tetrahedral angles is less stable than the planar form by about 30 kcal/mol. We will see experimental evidence in support of this in Chapter 2. This is completely consistent with the classic view of sp^2 hybridization and what we know about the planar structure of CH_3^+ .

This very simple analysis predicts the correct geometry, but as we now describe, there are a lot of complications to carbenium structures that require more complex pictures of bonding than is apparent from this simple analysis, and even just the substitution of an alkyl group will start to perturb the analysis.

Alkyl substitution stabilizes carbocations, and we will see in Chapter 2 that the order of stabilities for carbenium ions is $3^\circ > 2^\circ > 1^\circ > \text{methyl}$. Why should this be so? The reason is an orbital mixing phenomenon. The difference between methyl and ethyl cations is the CH_3 group, and as we have emphasized in this chapter, there are two orbitals on a methyl group with π symmetry. Just as with propene (Figure 1.18), in any given conformation of the ethyl cation one of these $\pi(\text{CH}_3)$ orbitals will be of the right symmetry to mix with this empty p orbital. The difference between this situation and the mixing in propene is that the $\pi(\text{CH}_3)$ is filled and the p orbital is empty, and it is always highly favorable to mix a filled orbital with an empty orbital.

The orbital mixing diagram for ethyl carbenium is shown in Figure 1.25 A, along with the resulting HOMO and LUMO. The empty orbital (LUMO) of ethyl carbenium is delocalized onto the methyl group. As such, we expect a significant amount of the positive charge on the CH_3 of ethyl cation, and high-level calculations suggest a charge of roughly +0.17 associated with the CH_3 . Also, there should be geometry changes. Examination of the filled orbital shows a sort of “ π bond” in ethyl carbenium, and we might expect this to lead to a shortening of the C–C bond. Indeed, calculations predict a bond length of 1.425 Å, significantly shorter than a typical bond between an sp^3 and an sp^2 carbon (1.51 Å; Table 1.4). Support for this bond shortening comes from the crystal structure of the *t*-butyl cation [$(\text{CH}_3)_3\text{C}^+$], a result made possible by the development of stable ion media (see Chapter 2). The C–C bonds in this cation are 1.442 Å long, again shorter than a typical sp^3 – sp^2 bond. We would also expect a lengthening of the C–H bonds of the CH_3 , due to the removal of electrons from the bonding $\pi(\text{CH}_3)$ molecular orbital, and this too is borne out by calculations. Another interesting effect is the distortion of the methyl C–H bond that aligns with the empty orbital. As shown in Figure 1.25 A, this bond “leans” toward the empty orbital, reducing the C–C–H angle to 95° .

The orbital mixing interaction we have just described is simply the MOT version of what in VBT terms would be called **hyperconjugation**. Hyperconjugation is often depicted with structures such as that shown in the margin, where a neighboring bond formed from a carbon sp^3 hybrid leans toward the carbocation center. Figure 1.25, on the other hand, gives a more sophisticated picture. The essence of hyperconjugation can also be viewed with the

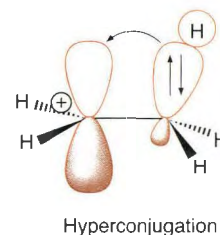
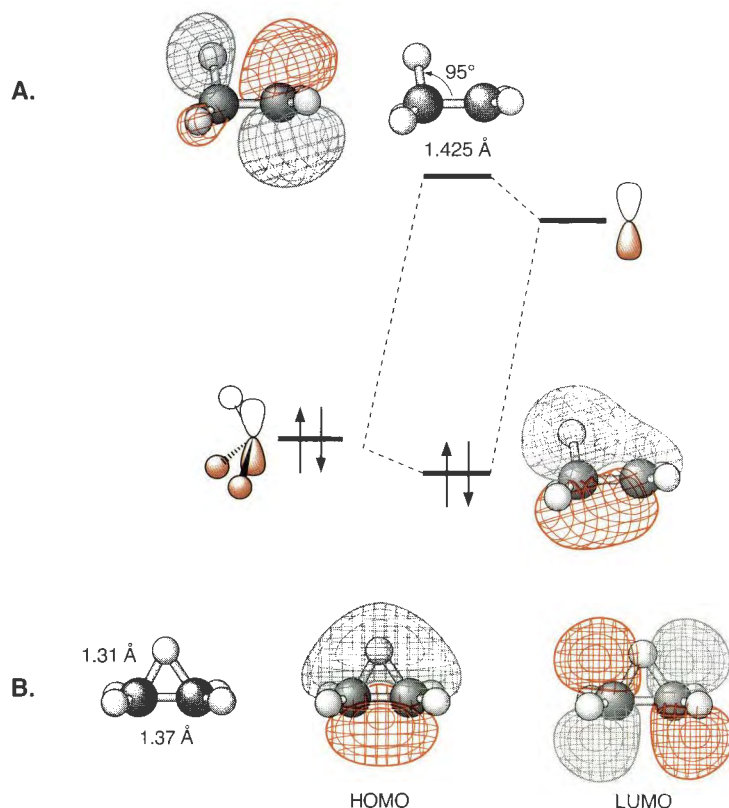
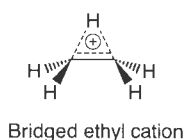
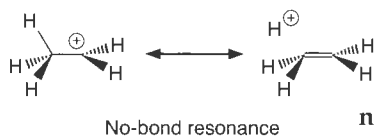


Figure 1.25

C_2H_5^+ . **A.** The ethyl carbenium ion. Shown are a structure for the ion and an orbital mixing diagram between an empty p orbital and a $\pi(\text{CH}_3)$ orbital, to produce the HOMO and LUMO of the ion.

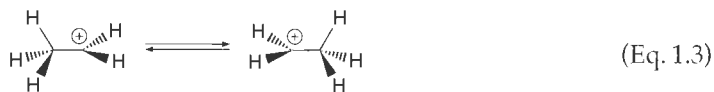
B. The bridged ethyl carbonium ion. The structure is shown, along with the HOMO and the LUMO.



no-bond resonance form shown in the margin, and it successfully recapitulates all the major features of the MOT analysis.

Interplay with Carbonium Ions

The structure we have discussed for the ethyl cation is appealing and makes sense. In addition, as shown in Eq. 1.3, it is well-established that there is a very rapid scrambling of the five hydrogens of the ethyl cation both in solution and in the gas phase (we will discuss these rearrangements in Chapter 11). The scrambling of hydrogens in ethyl cation suggests some role for the bridged structure shown in the margin. The issue is whether this symmetrical, bridged structure should be considered as a transition state or an intermediate. The bridged species is now considered a carbonium ion, because it has a hypervalent hydrogen. The bonding situation is directly analogous to that of diborane discussed previously. The C–H–C unit involves a three center–two electron bond.



The structure of the bridged ion (be it transition state or intermediate) is intriguing, and a calculated structure is shown in Figure 1.25 **B**. To a good approximation, it is simply ethylene that has been protonated directly in the middle of the π bond. As expected, the C–C bond has elongated (1.37 Å vs. 1.31–1.34 Å for a typical olefin; see Table 1.4), and the bridging C–H bonds are quite long. The HOMO and LUMO of the cation look very much like the HOMO and LUMO of ethylene. At the same time, they are not really very different from the corresponding orbitals of the “conventional” hyperconjugated ethyl cation in Figure 1.25 **A**.

Our discussion of ethyl cation illustrates many issues that are universal in carbocation chemistry. As already mentioned, carbenium ions are especially prone to rearrangement. Often, several different but similar structures can equilibrate rapidly via such rearrange-

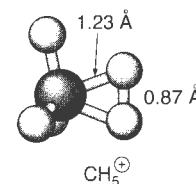
ments. This rapid equilibration is a manifestation of the fact that the potential energy surfaces of carbocations are often very flat, involving several related structures that are similar in energy and that interconvert via processes with very small barriers. This is a challenging situation. Many of the most contentious debates in physical organic chemistry have boiled down to efforts to sort out very subtle structural and energetic differences on a given carbocation potential energy surface. If we look at the structures and MOs of the two forms of ethyl cation shown in Figure 1.25, the differences are certainly small. In addition, we might expect that such situations would be strongly influenced by context effects, such as solvent, counterion, temperature, exact substitution pattern, and so on.

In the specific case of ethyl cation, high-level theory actually favors the bridged structure as the lowest in energy by several kcal/mol. In fact, many calculations indicate that the classical non-bridged structure is best thought of as a transition state for the hydrogen scrambling reaction. However, in other cases the balance may be tilted in the other direction to favor the carbenium ion. For example, the *t*-butyl cation $[(\text{CH}_3)_3\text{C}^+]$ is a classical hyperconjugated carbenium ion. It is planar at the cationic carbon with 120° bond angles. Our goal at this point is simply to alert you to the potential for such complications, and to keep the big picture in mind; carbocation potential energy surfaces are often flat, with several possible isomers that differ very little in structure and energy.

Allyl and benzyl cations are the prototype delocalized carbenium ions (look at Figures 1.21 B and 1.22). Conventional π delocalization does not convert a carbenium ion to a carbanion in this case. No hypervalent atoms are involved in any of the resonance structures for allyl or benzyl. The same orbitals that carry the negative charge in allyl and benzyl anion carry the positive charge in allyl and benzyl cation.

Carbonium Ions

The prototype carbonium ion, CH_5^+ , is readily generated in a mass spectrometer. Also, with very strong acids like FSO_3H , protonated alkanes can be generated, making CH_5^+ an important model compound. MH_5 compounds are well known, the prototype being the trigonal bipyramidal PH_5 . However, CH_5^+ does not adopt such a highly symmetric geometry. A representative structure is shown in the margin. The easiest way to think about this structure is as a pyramidal CH_3^+ bonded to a molecule of H_2 . This makes a three center–two electron bond of the sort we have described for BH_3 . The CH_3^+ contributes an empty σ_{out} orbital that interacts with the filled σ bonding orbital of H_2 , making the three center–two electron system. The H_2 fragment has a slightly elongated H–H bond distance of $0.87 \text{ \AA}$ (vs. $0.746 \text{ \AA}$ for the isolated molecule), while the C–H bonds involving these hydrogens are substantially elongated. The positive charge is distributed around the entire structure. The molecular structure given for CH_5^+ is reasonable for discussion purposes, but as the next Going Deeper highlight describes, it does not represent the true “nature of the beast”.



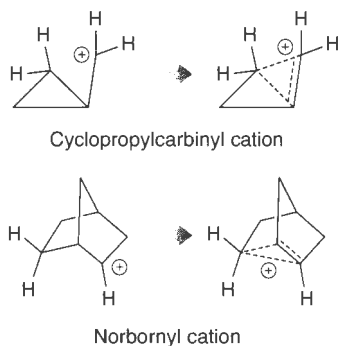
Going Deeper

CH_5^+ —Not Really a Well-Defined Structure

We have just described the structure given for CH_5^+ as “representative” because the situation is actually fairly complicated. The CH_5^+ ion undergoes a rapid scrambling process in which all five hydrogens become equivalent. Estimates are that the barrier to this process is less than 1 kcal/mol, making CH_5^+ a highly fluxional molecule. The potential energy surface for this carbocation is extremely flat. In fact, some very high-level calculations have led to the conclusion that there is effectively *no barrier* to the fluxional process, in which case the notion of “molecular structure” is poorly defined. This small ion may be a quantum mechanical species, for which the classical model of a

rigid geometrical object is simply not applicable. Not surprisingly, spectroscopic characterization of such a structure is challenging. However, a very sophisticated gas phase infrared (IR) spectroscopy study reported in 1999 found a complex collection of lines that have been assigned to CH_5^+ . The complexity of the spectrum and its dissimilarity to spectra of other simple molecules like CH_3^+ or CH_4 seems consistent with the non-standard description of CH_5^+ . However, full assignment of the lines was not possible. Stay tuned.

White, E. T., Tang, J., and Oka, T. “ CH_5^+ : The Infrared Spectrum.” *Science*, 284, 135 (1999). Marx, D., and Parrinello, M. “Structural Quantum Effects and 3-Center 2-Electron Bonding in CH_5^+ .” *Nature*, 375, 216 (1995).



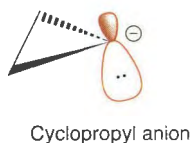
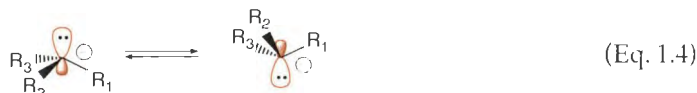
As discussed, an R_5C^+ species can be considered to be a bridging structure with a hypervalent carbon. The bridged picture of ethyl cation has a hypervalent hydrogen. Many other carbocations are also bridged species best defined as carbonium ions. Some of these compounds are called **non-classical cations**. These are defined as containing bridged structures with three center–two electron bonds and a hypervalent atom. Although ethyl cation fits this description, the term is normally associated with structures where a C–C σ bond bridges to create a hypervalent carbon. Two classic examples are cyclopropylcarbinyl cation and norbornyl cation, shown in the margin. We will discuss these structures much more in Chapters 2, 11, and 14.

1.4.2 Carbanions

Now let's consider the prototypical carbanion, CH_3^- , with a total of eight valence electrons. The **D** orbital of Figure 1.8 is now filled, and the stabilization it gains by pyramidalization dominates the Walsh diagram. This stabilization is enhanced by further mixing to produce **D'**. Thus, simple carbanions should be pyramidal, again consistent with experimental observations.

As noted previously, a basic tenet of the group orbital concept is that the MOs of MH_3 are essentially the same, regardless of the identity of M, the only difference being the number of valence electrons. We have already noted the isoelectronic relationships between BH_3 and CH_3^+ , and between NH_3 and CH_3^- , and so, again, Figure 1.8 predicts that NH_3 should be pyramidal and BH_3 should be planar, as is observed experimentally.

The prototype structure of a simple, pyramidal carbanion has a lone pair of electrons in the σ_{out} orbital of Figure 1.8. An equivalent way to think about the structure is as an sp^3 hybrid, placing the lone pair in an sp^3 -like orbital. In most cases the barrier to inversion at a carbanion center is small, roughly 1–2 kcal/mol for simple systems. Thus, on most reaction time scales, carbanions behave as if they were effectively planar (Eq. 1.4).



Several factors can significantly raise the **inversion barrier** at a carbanion center. One is incorporation into a small ring. Consider the inversion of the cyclopropyl anion. In the pyramidal ground state, the lone pair occupies a hybridized orbital and the carbanion center would be roughly sp^3 hybridized. In the planar transition state for inversion, the lone pair is in a p orbital, and the carbanion center is essentially sp^2 hybridized. In both the ground and transition states, the C–C bonds must be bent to accommodate the 60° bond angles of the cyclopropane ring. This is more difficult when a carbon is sp^2 hybridized (favoring 120° angles) than when the carbon is sp^3 hybridized (favoring 109.5° angles). This effect raises the energy of the planar transition state for inversion in the small ring relative to an open system. As a result, cyclopropyl anions often react as if they were stably pyramidal.

Electronegative substituents attached to the anionic center can also substantially raise the inversion barrier. This electronegativity effect is general. While ammonia has an inversion barrier of ~ 5 kcal/mol, NF_3 has a barrier of ~ 50 kcal/mol. This high barrier can be understood by recalling that electronegative substituents prefer to bond to orbitals with greater amounts of p character, because p orbitals in general are easier to withdraw electrons from than s orbitals. Thus, electronegative substituents preferentially stabilize the sp^3 ground state over the sp^2 transition state, raising the inversion barrier.

An alternative rationalization can be developed by referring to the CH_3 Walsh diagram. On going from NH_3 to NF_3 we replace H by the highly electronegative F. As noted in the discussion of formaldehyde (Section 1.3.3), electronegative atoms lower the energies of *all* the MOs to which they contribute. Returning to the Walsh diagram (Figure 1.8), the effect of fluorine on the planar structure is to lower all the orbital energies except **D**, which has no contribution from attached fluorines. However, with the pyramidal structure, all the orbitals are lowered in energy, including **D**, because they all have contributions from the fluorines. This stabilizes more electrons in the pyramidal structure of an anion because the **D** orbitals are populated, and thus raises the inversion barrier.

Connections

Pyramidal Inversion: NH_3 vs. PH_3

The pyramidal inversion barrier, E_{inv} , is a crucial property of any pyramidal tricoordinate molecule. As discussed previously, NH_3 is pyramidal, but the barrier is so low (~ 5 kcal/mol) that amines act planar at room temperature. PH_3 , however, has a very high E_{inv} , on the order of 35 kcal/mol, so phosphines are stable pyramids at conventional temperatures.

To rationalize this difference between NH_3 and PH_3 , we focus on the key interaction between **D** and **E** to generate **D'** and **E'** (Figure 1.8). This greatly stabilizes **D'** and is a key factor favoring the pyramidal form.

The magnitude of the stabilization increases as the initial energy gap between **D** and **E** decreases. Two factors make this gap *smaller* for PH_3 than for NH_3 . First, P is less electronegative than N, so **D** will be higher-lying in PH_3 , thus diminishing the gap. Second, the $2s$ and $2p$ orbitals in the first row of the periodic table have similar sizes, but beyond the first row the s orbitals are significantly more contracted than the p orbitals. Consequently, at distances where the phosphorus $3p$ orbitals interact strongly with the H $1s$ orbitals, the P $3s$ orbitals interact relatively weakly, making **E** less antibonding in PH_3 than in NH_3 , and further diminishing the initial **D/E** gap.

Substituents that stabilize a carbanion by π delocalization will favor the planar structure. This preference is because the interaction with the π substituent will be greatest in this form, where we have a pure p orbital rather than a $\sigma(\text{out})$ orbital to participate in the delocalization. Given the small intrinsic preference for a pyramidal structure, we would expect the delocalization (resonance) effect to win out, making the anionic carbon planar. Examples of such substituents that will lead to a planar neighboring anionic carbon are cyano, nitro, and carbonyl. Resonance structures make this argument clear.

Allyl and benzyl anions are also planar. The HOMOs of these anions are shown in Figures 1.21 **B** and 1.22, respectively, and we expect significant negative charge only on those atoms that have significant contribution in the HOMO.

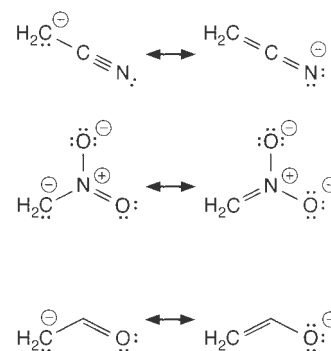
1.4.3 Radicals

Methyl radical has seven valence electrons. The crucial orbital that is stabilized by pyramidalization, σ_{out} , is only singly occupied (Figure 1.8), and the stabilization it provides is to some extent offset by the destabilization of other, doubly occupied orbitals. It is difficult to predict the geometry of this species based on the Walsh diagram, and it is best to say that no obvious preference for planar or pyramidal geometry can be predicted. This, too, is “consistent” with experiment, as simple radicals show only a very weak preference for the planar structure, and simple substitution can produce pyramidal radicals. The net result is that the parent methyl radical, $\text{CH}_3^\bullet$, is planar, but the energy cost for distorting away from planarity is small.

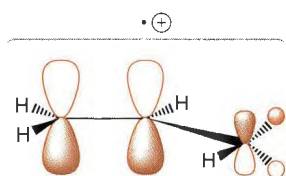
In fact, extensive study has revealed that the intrinsic preference for planarity in methyl radical is so small that essentially all other localized radicals are not planar. Two factors favor pyramidalization in radicals. The first is an electronic effect of the sort discussed previously for anions. $\text{CF}_3^\bullet$ is very strongly pyramidal for the same reason as discussed above for NF_3 . Electronegative substituents prefer bonding to an sp^3 hybrid rather than an sp^2 , and this preference completely shifts the balance toward pyramidalization in the case of the radical system. The second reason that radicals are pyramidal is a conformational effect, and this is discussed in Chapter 2.

Just as with allyl and benzyl cations and anions, allyl and benzyl radicals are stabilized by delocalization. The crucial molecular orbital, the **singly occupied molecular orbital (SOMO)**, is essentially the same as the MO that is doubly occupied for the respective anions (see Figures 1.21 **B** and 1.22).

There are also reactive intermediates known as **radical cations**. The geometries of such species can also be understood using our notions of bonding. One example is the one-electron oxidation of an alkene, where the electron is removed from the π orbital. For all alkenes besides ethylene (see Chapter 2 for a discussion of the radical cation of ethylene), oxidation retains a planar structure in the alkene. However, the mixing of the alkyl group's π -like group orbitals with the now singly occupied π orbital becomes even more pro-



Stabilized carbanions



Radical cation of propene

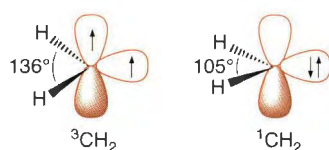
nounced than we saw for propene (Figure 1.18). Therefore, the SOMO is even more delocalized onto the neighboring R group, as shown in the margin for the propene radical cation, where a bracket is used to denote one electron and a formal positive charge in the MO.

1.4.4 Carbenes

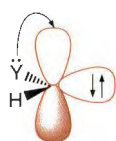
Lastly, let's consider carbenes, neutral :CR_2 species, the prototype of which is the molecule methylene, :CH_2 . For the reactive intermediates we have considered so far—cations, anions, and radicals—the Walsh diagram for CH_3 provides the starting point for the discussion. For carbenes, the relevant electronic structure questions can be considered by referring to the Walsh diagram for CH_2 in Figure 1.9. We begin with the linear form on the left side of Figure 1.9. Methylene has six valence electrons, and we can place two electrons each in the **A** and **B** MOs, leaving two valence electrons to occupy the degenerate **C/D** pair. This leads to a unique feature of carbenes vs. the other reactive intermediates we have studied. Each carbene is in fact two reactive intermediates, differentiated by the spin state of the system. If the spins are aligned, then the spin state of the system is $S = \frac{1}{2} + \frac{1}{2} = 1$, and the multiplicity is $m_s = 2S + 1 = 3$. This is a **triplet** state. If we have opposing spins, $S = \frac{1}{2} + (-\frac{1}{2}) = 0$ and $m_s = 1$. This is a **singlet** state. These two states are expected to have substantially different molecular and electronic structures and to show distinct reactivities. The Walsh diagram of Figure 1.9 provides an excellent starting point for considering these issues.

Hund's rule predicts the high-spin, triplet state should be preferred at the linear geometry, and indeed it is. As the H–C–H angle contracts, a gap opens up between **C** and **D**. If we keep one electron in each MO, this distortion should be mildly stabilizing. With small bending, there is no large benefit to pairing two electrons into **C**, and the triplet is still preferred. However, when the angle becomes small enough, the lower energy of orbital **C** will overcome the electron repulsion energy, and a singlet with both electrons in **C** will become the ground state.

It is impossible to unambiguously predict the absolute ground state of methylene with simple models such as Walsh diagrams. As we will see in Chapter 14, understanding and predicting spin preferences requires more advanced treatments of electronic structure than we are providing here. However, some predictions are still possible. For example, the triplet state should have a wider H–C–H angle than the singlet. This is indeed the case; the angle is 136° for the triplet and 105° for the singlet. Experimentally it turns out that the triplet is the global ground state in methylene, by approximately 9 kcal/mol. All simple dialkyl carbenes have triplet ground states.



Carbene electron configurations



Resonance in singlet carbenes

The Walsh diagram provides a satisfying analysis of the electronic structure of carbenes, and the essential features of the system can be summarized by the simple representations shown in the margin. The triplet has two electrons in two very different orbitals, what we have called p and $\sigma(\text{out})$. We might expect its reactivity to be similar to that of radicals, and indeed this is the case. The singlet state is quite different. It contains a lone pair of electrons in an MO [$\sigma(\text{out})$] that is reminiscent of an sp^2 hybrid. It also contains an empty p orbital, just like a simple carbenium ion. Its reactivity patterns should be quite different from radicals, and we will see that they are (Chapter 10).

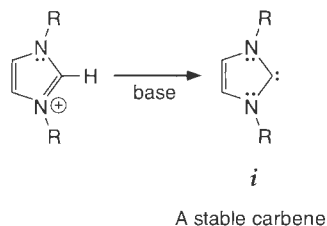
While simple carbenes have a triplet ground state, appropriate substituents can reverse this preference, such that some substituted carbenes show a large energetic preference for the singlet. The bonding and structural model we have developed provides excellent guidance as to how we might create a carbene that has a singlet ground state. The most effective way is to interact with the empty **D** orbital of the singlet, as shown in the margin. Carbenes with lone-pair donating substituents such as N, O, and halogens can have singlet ground states because of such an interaction (see the next Going Deeper highlight for an example). The effect can be quite large; in difluorocarbene the singlet lies below the triplet by ~ 50 kcal/mol! Detailed theoretical studies reveal a linear correlation between the singlet–triplet gap and the electron pair donating ability of the attached substituent(s). As is typical, a resonance model also nicely rationalizes the stabilizing effect of donating substituents on carbenes.

Going Deeper

Stable Carbenes

We generally think of carbenes as extremely reactive species, and for the most part they are. In recent years, however, clever application of the concepts described here has led to some remarkable new carbenes. The breakthrough occurred in 1991 when Arduengo and co-workers at DuPont reported the synthesis and *isolation* of carbene *i*. Two factors contribute to the stability of this type of carbene. First is the steric bulk of the R group. The first example had R = adamantyl, a large, aliphatic ring system. Later examples included heavily substituted aromatics as the R group. The second effect is electronic. Two potent electron donors are attached to the carbene center. As just discussed, these should greatly stabilize the singlet state, and indeed these types of carbenes have a singlet ground state. Remarkably, these molecules can be crystallized, and an x-ray structure reveals an N–C–N angle of 102° at the carbene center, in excellent agreement with the expectation for a singlet carbene. Samples of *i* are stable for years, as long as they are protected from air. Many deriva-

tives have been made, and extensive physical characterization has provided detailed insights into carbene electronic structure. These stable carbenes are not solely theoretical curiosities. They have recently found use as ligands for an important class of ruthenium-based olefin metathesis catalysts that have profoundly influenced synthetic organic chemistry (see Chapter 12). Thus, research into basic reactive intermediates can lead to fundamental insights and useful new materials.



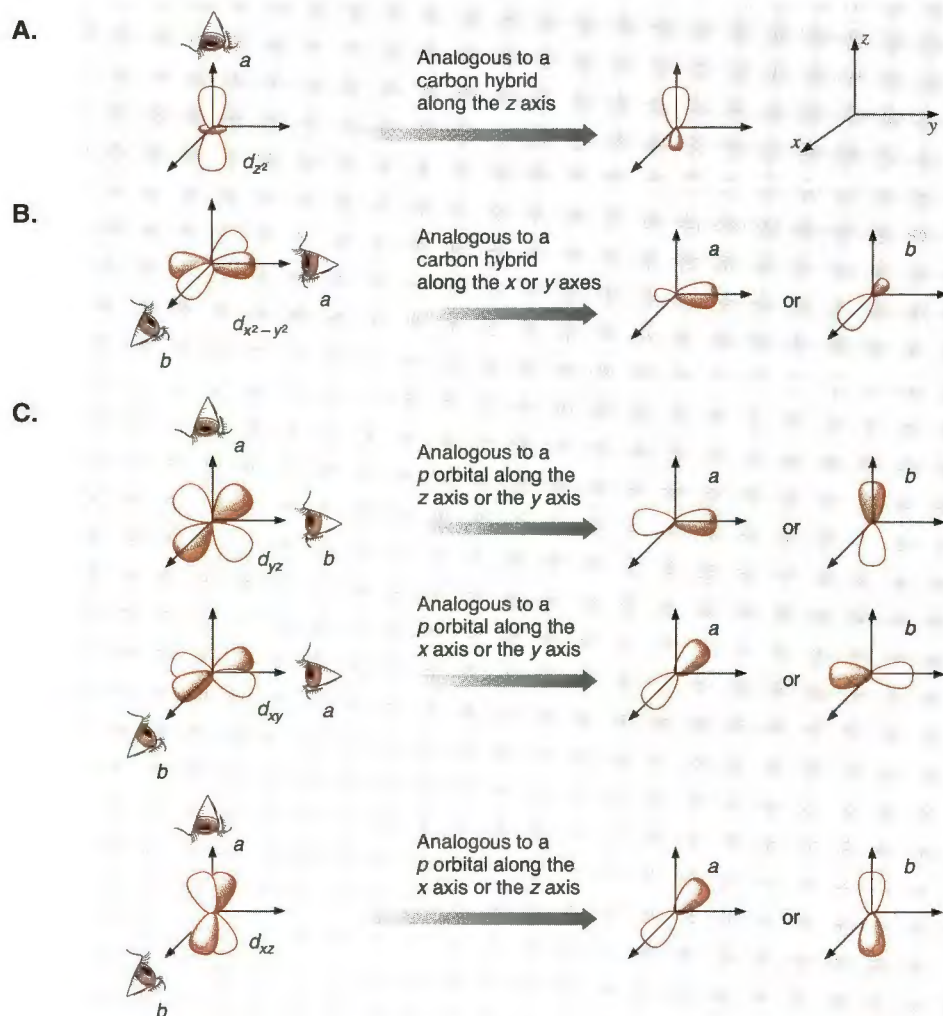
Arduengo, A. J., III. "Looking for Stable Carbenes." *Acc. Chem. Res.*, **32**, 913–921 (1999).

1.5 A Very Quick Look at Organometallic and Inorganic Bonding

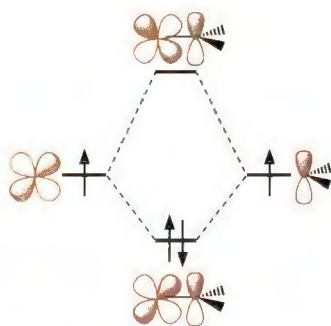
One theme of this textbook is to consistently tie organic chemistry to organometallic chemistry, which is just one of the current chemical subdisciplines where the tools of physical organic chemistry are often applied. In this regard, it is useful in this chapter to develop a simple bonding model for organometallic and inorganic complexes, and not just look at organic bonding. Here we examine a model analogous to the first VBT/MOT model of organic bonding given in Section 1.1. We will leave an examination of structure in organometallic systems to Chapter 12, and we will examine more complex MOT ideas about bonding in metal-containing systems in Chapter 14.

The same kind of localized and discrete σ and π bonds often associated with organic compounds can be assigned to the bonding in organometallic compounds. One simple concept for visualizing bonding in metal-containing systems is to make direct analogies to such bonding in organic systems. For example, let's examine the shapes and nodal properties of the d orbitals shown in Figure 1.26. First, notice that the d_{z^2} orbital is directional along the z axis, as is a hybrid orbital along an axis, and is therefore "sigma-like". In fact, we refer to it as having **sigma symmetry** along the z axis. Likewise, the $d_{x^2-y^2}$ orbital is aligned along the x and y axes, and is therefore considered to have sigma symmetry when viewed down these axes. Hence, when forming discrete localized bonds to ligands, the d_{z^2} orbital and the $d_{x^2-y^2}$ orbital can make sigma bonds that are placed along the z or the x and y axes, respectively. We essentially use these orbitals just as we use hybrid or $\sigma(\text{out})$ orbitals on carbon.

Now let's examine the analogies that can be drawn between the rest of the d orbitals and p orbitals. When citing down the x axis, the d_{xz} and d_{xy} orbitals look like two orthogonal p orbitals. Likewise, citing down the y axis, both the d_{yz} and d_{xy} look like p orbitals. Therefore, these orbitals are considered to have **π symmetry**. When creating double bonds to ligands,

**Figure 1.26**

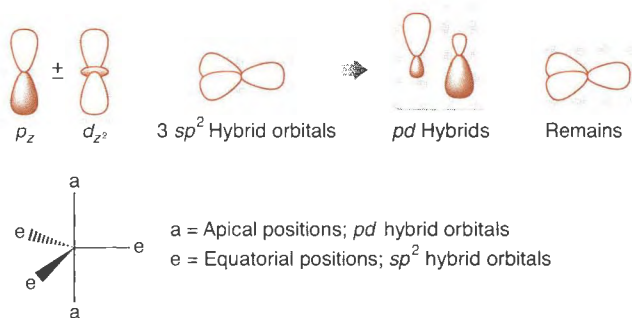
Schematic representations of d orbitals and their analogous carbon orbitals. **A.** Looking along the positive z axis (designated as an eyeball), the carbon hybrid along the z axis appears the same as a d_{z^2} orbital. They are referred to as having σ symmetry, analogous to σ (out) orbitals or the standard hybrid orbitals for making σ bonds to organic groups. **B.** Looking along the positive x or y axes, the $d_{x^2-y^2}$ orbital looks like a carbon hybrid. **C.** The d_{yz} , d_{xy} , and d_{xz} orbitals appear as p orbitals, depending upon the line of sight. They are referred to as having π symmetry. The a and b descriptors on each orbital refer to the line of sight for the corresponding eyeballs.

**Figure 1.27**

Orbital mixing diagram for the combination of a π -symmetry d orbital with a carbon p orbital to create a metal-carbon π bond.

these orbitals on the metals are used exactly as we use p orbitals on carbons. This combination is very clear in Figure 1.27, where an orbital mixing diagram is given for a π bond between a metal and a carbon, as would be present in an organometallic complex that has an $M=CR_2$ functional group.

Lastly, there is even a hybrid orbital approach to modeling the bonding in organometallic and inorganic complexes. Using the same concepts presented in Section 1.1.6 for the linear combinations of s and p orbitals to create hybrids, we can mix d orbitals with the s and p orbitals. For example, if you start with an sp^2 hybridization state, and mix the remaining p_z orbital with the d_{z^2} orbital, a hybridization state known as dsp^3 is obtained. As shown in Figure 1.28, the bonds to the apical positions are pd hybrid orbitals, while the equatorial positions are sp^2 hybrid orbitals. Such hybridization is appropriate for trigonal bipyramidal complexes.

**Figure 1.28**

Hybridization for trigonal bipyramidal geometries starts with the central atom sp^2 hybridized. The orbitals aligned along the equatorial positions remain as sp^2 hybrids. However, the orbitals along apical positions are pd hybrids.

The mixing of an s , three p 's, and the d_{z^2} and $d_{x^2-y^2}$ orbitals leads to d^2sp^3 hybridization, which is appropriate for octahedral complexes. For almost all bonding geometries in inorganic and organometallic chemistry, hybrid orbitals are useful, and this very brief introduction to bonding using metals will be enough to take us a long way in understanding structure and reactivity.

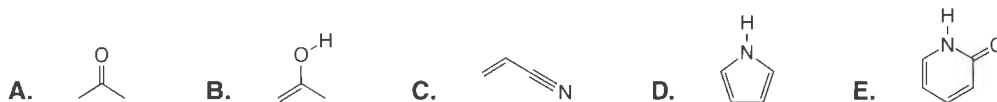
Summary and Outlook

This chapter represents just a beginning. It is only a first look into bonding, with hints at structure and reactivity. We presented two models for bonding, a classical one and a more modern approach, and showed that they can be used to understand stable organic structures and reactive intermediates. Hopefully this chapter has refreshed your memory, and has sparked an interest in you to learn how these notions of bonding can be put to use. We do exactly that in the next several chapters where the focus is more upon structure. Furthermore, after analyzing structure, we can take the bonding concepts and look at reactivity. Structure and reactivity actually take up approximately two-thirds of this book, and not until we have to look at some very specialized reactions (pericyclic and photochemical) will we need to develop a more sophisticated theory of bonding. With our current qualitative models we can go a long way in our analysis of topics in organic chemistry.

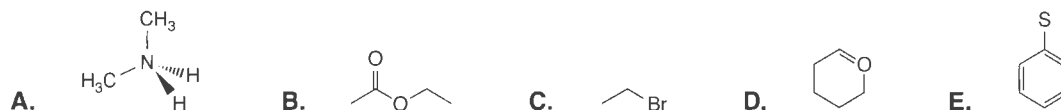
For many students and professors, this chapter may completely suffice as a review of bonding, because it is sufficient for almost all organic transformations. For other students and professors, however, it may be desirable to now go directly to Chapter 14, where the concepts introduced herein are discussed more quantitatively and modern methods in computational electronic structure theory are covered. This is a decision to be made on an individual basis. However, it should be appreciated that to the best of the authors' abilities, we took the topics of this chapter only to a depth that is routinely used when thinking about organic structure and bonding by a non-expert in quantitative methods. Our intention is for Chapter 14 to stand alone, so it can be covered at any point during the course to learn more advanced concepts and quantitative methods.

Exercises

1. State the hybridization of the non-hydrogen atoms in the following structures.

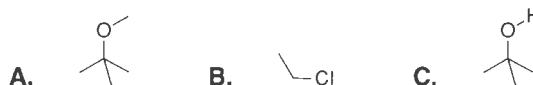


2. Assuming that each atom in the following structures has an octet of electrons, identify which compounds have an atom that has a formal charge. Identify what and where that charge is. In the compounds without formal charges, identify any significant bond polarizations by writing δ^+ and δ^- near the appropriate atoms.



3. For the bond polarizations in Exercise 2, draw a dipole arrow for the polarized bonds.

4. Draw the σ and σ^* molecular orbitals for discrete and localized bonds formed between hybridized carbons and heteroatoms (a heteroatom is any atom besides C and H), as well as heteroatoms and hydrogen, in the following structures. Indicate any polarization these bonds may have in your diagram by drawing the shapes of the discrete bonding and antibonding orbitals in a manner indicative of this polarization.



5. Draw the π and π^* molecular orbitals for the discrete and localized π bonds in the following structures. Indicate any polarization these bonds may have in your diagram through the shapes of the discrete bonding and antibonding orbitals.



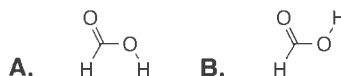
6. Show any plausible resonance structures for the molecules given in Exercises 1 and 5. If there are no plausible resonance structures, indicate this (note that the molecule corresponding to letter A is the same in each problem).

7. Discuss the hybridization in the C–C and C–H bonds of cyclopropane, given that the H–C–H bond angle is 118° .

8. Carbon tetrachloride has neither a dipole moment nor a quadrupole moment, but it does have an octopole moment. Provide a simple description of this moment.

9. Formaldehyde has a fairly large dipole moment of 2.33 D, but CO has a small dipole moment of 0.11 D. Use resonance and electronegativity arguments to explain these results.

10. Consider bond dipoles to predict which conformer of formic acid should have the higher dipole moment, A or B.



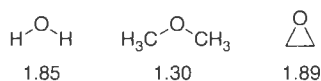
11. Pauling proposed the following correlation between electronegativity difference and percent ionic character in a bond: $\text{Ionic character} = 100 \times [1 - e^{-(X_A - X_B)^2 / 4}]$ where X_A is the electronegativity of A. Calculate the percent ionic character in HF, HCl, HBr, and HI. The most polar bond for the elements in Table 1.1 would be in KF. What is its percent ionic character?

12. Which should have a larger dipole moment, formic acid (the higher dipole conformer of Exercise 10) or formamide?

13. The electronegativities of the $-\text{C}\equiv\text{CH}$ and $-\text{C}\equiv\text{N}$ groups are the same (Table 1.1), but the dipole moments of $\text{CH}_3\text{C}\equiv\text{CH}$ and $\text{CH}_3\text{C}\equiv\text{N}$ are 0.78 D and 3.92 D, respectively. Explain why the dipole moments are so different.

14. Draw a molecule that has no dipole moment, but has a quadrupole moment with the topology of a d_{xy} orbital, rather than the d_{z^2} topology.

15. Predict the hybridization state, sp^n , for the carbon hybrid orbitals used for the C–H bonds of CH_3Cl (H–C–H angle of 110.5°), and the C–H bonds of ethylene (H–C–H angle of 117.3°).
16. Rationalize the differences between the following dipole moments for water, dimethyl ether, and ethylene oxide (oxirane):



17. Convince yourself that the C–H bonds of cyclohexane would add up to give a quadrupole moment, as they do in benzene, if carbon were significantly more electronegative than hydrogen. Contrast this quadrupole moment to that in benzene.
18. Geometries for small molecules such as methyl fluoride can be determined from microwave spectroscopy, whereas various forms of diffraction—x-ray, electron, or neutron diffraction—can be applied to larger molecules. Shown below are bond lengths and angles for various methyl halides determined by various methods. Rationalize the trend in the C–X bond lengths (angstroms). Also rationalize the H–C–H and H–C–X bond angles using VSEPR. Does an argument based upon electronegativities also explain the trends in bond angles?

Molecule	C–H bond length	C–X bond length	H–C–H bond angle	H–C–X bond angle
CH_3F	1.09	1.385	110.2	108.2
CH_3Cl	1.096	1.781	110.52	108.0
CH_3Br	1.11	1.939	111.12	107.14
CH_3I	1.096	2.139	111.5	106.58

19. The calculated H–C–H bond angle in methyl radical is 120° , the calculated H–C–F angle in CH_2F radical is 115° , and the calculated F–C–F angle in CHF_2 is 112° . Rationalize why the structures become more pyramidal with increasing fluorine substitution.
20. Describe and draw (using cartoons) how the three group orbitals for a pyramidal methyl group (Figure 1.8) can be linearly combined to give orbitals similar to the individual σ bonds formed from sp^3 hybridized carbon atoms and hydrogen 1s orbitals.
21. Draw mixing diagrams using $\sigma(\text{out})$ orbitals that compare the formation of the C–C σ bond in ethane, created by the approach of two pyramidal methyl groups, to the formation of the C–F σ bond in CH_3F . Use the same energy axis for both plots. Highlight the differences and explain how this diagram would rationalize reactivity differences in alkanes and alkyl fluorides with nucleophiles.
22. Rationalize the differences between the π orbitals of butadiene and acrolein (the MOs are shown in Appendix 3). The results of Figure 1.20 should be a useful starting point.
23. Sketch the highest occupied MO of propene and compare and contrast this MO with the analogous MO of acetaldehyde ($\text{CH}_3\text{CH}=\text{O}$). Consider the methyl C–H bonding in your answer.
24. Sketch the π^* MO of propene. Consider the methyl C–H bonding in your answer.
25. The H–P–H bond angle in PH_3 is nearly 90° , whereas that for NH_3 is only slightly contracted from the idealized 109.5° . Explain this difference using a VSEPR argument, and then use a hybridization/electronegativity argument.
26. Draw an orbital mixing diagram for a metal–methyl single bond using the d_{z^2} orbital on the metal and a $\sigma(\text{out})$ orbital on the methyl group.
27. Sketch the π and lone pair group orbitals for the following organometallic group, called a Fischer carbene. (Hint: Use the group orbitals of formic acid as your guide.)



28. Use an orbital mixing diagram to rationalize the nature of the HOMO of methylamine shown in Appendix 3.
29. Starting with two allyl groups, use orbital mixing to predict all the orbitals for the π system of hexatriene.
30. Create the group orbitals for an MH system, where M is a second row element such as C or N.
31. Use the appropriate group orbitals and the QMOT rules in Table 1.7 to create the molecular orbitals of protonated formaldehyde ($\text{CH}_2=\text{OH}^+$), starting with methylene and OH.

32. Sketch the π orbitals for allyl anion and for an enolate anion, and discuss the differences.
33. Combine the lone pair orbitals on formaldehyde to achieve what resemble the classic sp^2 hybrid lone pairs.
34. Draw the virtual orbitals that were not shown in Figure 1.8 for the planar and pyramidal forms of CH_3 and in Figure 1.9 for the linear and bent forms of CH_2 .
35. If there are two types of ligands bound to a trigonal bipyramidal metal atom, will the more electronegative ligands prefer the apical or equatorial positions? Explain your answer.

Further Reading

Much of the material in this chapter is review, and if you are not familiar with the material in Section 1.1, you should consult your introductory texts in general chemistry, organic chemistry, and / or physical chemistry. Selected references to more advanced topics are given below.

Qualitative Molecular Orbital Theory, Perturbation Theory, and Group Orbitals

- Albright, T. A., Burdett, J. K., and Whangbo, M.-H. (1985). *Orbital Interactions in Chemistry*, John Wiley & Sons, New York.
- Borden, W. T. (1975). *Modern Molecular Orbital Theory for Organic Chemists*, Prentice-Hall, Englewood Cliffs, NJ.
- Gimarc, B. M. (1979). *Molecular Structure and Bonding: The Qualitative Molecular Orbital Approach*, Academic Press, New York.
- Jorgensen, W. L., and Salem, L. (1973). *The Organic Chemist's Book of Orbitals*, Academic Press, New York. (A classic book that is unfortunately out of print—get one if you can!)

Valence Bond Theory, Hybridization, Electronegativity, and Resonance

- Pauling, L. (1960). *The Nature of the Chemical Bond and the Structure of Molecules and Crystals; an Introduction to Modern Structural Chemistry*, 3d ed., Cornell University Press, Ithaca, NY.
- Wheland, G. W. (1995). *Resonance Theory in Organic Chemistry*, John Wiley & Sons, New York.
- Epiotis, N. D. (1983). *Unified Valence Bond Theory of Electronic Structure*, Springer-Verlag, Berlin.
- Bodrowicz, F. W., and Goddard, W. A., III in *Modern Theoretical Chemistry, Methods of Electronic Structure Theory*, H. F. Schaefer III (ed.), Plenum Press, New York, 1977, Vol. 3, Chapter 4.

A Useful Compilation of Standard Bond Lengths

- Allen, F. H., Kennard, O., Watson, D. G., Brammer, L., Orpen, A. G., and Taylor, R. "Tables of Bond Lengths Determined by X-ray and Neutron Diffraction. Part 1. Bond Lengths in Organic Compounds." *J. Chem. Soc., Perkin Trans. II*, S1–S19 (1997).